

Chapter 20

System Registers (SYSREG)

20.1 Overview

ESP32-P4 includes various features that can be configured through dedicated registers. This chapter mainly enumerates such registers. For detailed introduction about a specific feature, please go to respective chapters.

20.2 Function Description

20.2.1 HP System Registers

20.2.1.1 External Memory Encryption/Decryption Configuration

Register [HP_SYSTEM_CRYPTO_CTRL_REG](#) configures encryption and decryption options of the external memory (external Flash and RAM). For details, please refer to Chapter [32 External Memory Encryption and Decryption \(XTS_AES\)](#).

20.2.1.2 HP Peripherals Clock Configuration and Power Control

The following registers are used to control the power signals of memory in corresponding peripherals.

- RSA: [HP_SYSTEM_HP_RSA_PD_CTRL_REG](#)
- ECC: [HP_SYSTEM_HP_ECC_PD_CTRL_REG](#)
- RSA: [HP_SYSTEM_HP_UART_PD_CTRL_REG](#)

The following fields in register [HP_SYSTEM_HP_PERI_MEM_CLK_FORCE_ON_REG](#) control the clock gating signals of memory in corresponding peripherals.

- GDMA: [HP_SYSTEM_GDMA_MEM_CLK_FORCE_ON](#)
- BitScrambler
 - TX memory: [HP_SYSTEM_BITSCRAMBLER_TX_MEM_CLK_FORCE_ON](#)
 - RX memory: [HP_SYSTEM_BITSCRAMBLER_RX_MEM_CLK_FORCE_ON](#)
- RMT: [HP_SYSTEM_RMT_MEM_CLK_FORCE_ON](#)

20.2.1.3 HP Cache Clock and Reset Configuration

ESP32-P4 allows software to directly control the clock and reset signals of HP cache.

The following fields in register [HP_SYSTEM_HP_CACHE_CLK_CONFIG_REG](#) control the clock signals of different HP cache. Writing 1 to respective fields enables the clock for respective cache.

- L1 icache 0: [HP_SYSTEM_L1_IO_CACHE_CLK_ON](#)
- L1 icache 1: [HP_SYSTEM_L1_I1_CACHE_CLK_ON](#)
- L1 dcache: [HP_SYSTEM_L1_D_CACHE_CLK_ON](#)
- L2 cache: [HP_SYSTEM_L2_CACHE_CLK_ON](#)

The following fields in register [HP_SYSTEM_CACHE_RESET_CONFIG_REG](#) control the reset signals of different HP cache. Writing 1 and then 0 to respective fields resets respective cache.

- L1 icache 0: [HP_SYSTEM_L1_IO_CACHE_RESET](#)
- L1 icache 1: [HP_SYSTEM_L1_I1_CACHE_RESET](#)
- L1 dcache: [HP_SYSTEM_L1_D_CACHE_RESET](#)

20.2.1.4 HP SPM and HP L2MEM Clock Configuration

The following register fields control the clock signals of HP [SPM](#) (Scratchpad Memory) and HP [L2MEM](#). Writing 1 to respective register fields forcibly enable the clock of respective memory.

- HP SPM: [HP_SYSTEM_HP_SPM_CLK_FORCE_ON](#)
- L2MEM: [HP_SYSTEM_L2_MEM_CLK_FORCE_ON](#)
- L2ROM: [HP_SYSTEM_L2_ROM_CLK_FORCE_ON](#)

20.2.1.5 HP SPM Parity Check Configuration

ESP32-P4's has implemented Parity Check in HP SPM to detect one-bit memory flip error.

This feature can be configured and controlled via the following registers:

- [HP_SYSTEM_HP_SPM_PARITY_CHECK_CTRL_REG](#): write 1 to enable the parity check in HP SPM.
- [HP_SYSTEM_HP_SPM_INIT_REG](#): configures the initialization of HP SPM before enabling the parity check.
- Interrupt ([HP_SPM_PARITY_ERR_INT](#)) related registers:
 - [HP_SYSTEM_HP_SPM_INT_RAW_REG](#): the raw interrupt status of [HP_SPM_PARITY_ERR_INT](#)
 - [HP_SYSTEM_HP_SPM_INT_ST_REG](#): the masked interrupt status of [HP_SPM_PARITY_ERR_INT](#)
 - [HP_SYSTEM_HP_SPM_INT_ENA_REG](#): write 1 to enable [HP_SPM_PARITY_ERR_INT](#)
 - [HP_SYSTEM_HP_SPM_INT_CLR_REG](#): write 1 to clear [HP_SPM_PARITY_ERR_INT](#)
- [HP_SYSTEM_HP_SPM_PARITY_INT_RECORD_REG](#): records the address when [HP_SPM_PARITY_ERR_INT](#) occurs.
- [HP_SYSTEM_HP_SPM_ERR_RESP_CTRL_REG](#): write 1 so HP SPM reports parity error to the HP CPU and triggers exception.

For detailed configuration steps, please refer to Chapter [7 System and Memory](#).

20.2.1.6 HP L2MEM ECC Check Configuration

ESP32-P4's has implemented the Error Correction Code (ECC) check in HP L2MEM (including L2RAM and cache) to correct one-bit memory flip error.

This feature can be configured and controlled via the following registers:

- [HP_SYSTEM_L2_MEM_L2_RAM_ECC_REG](#) and [HP_SYSTEM_L2_MEM_L2_CACHE_ECC_REG](#): write 1 to enable the ECC check for L2MEM and cache respectively.
- [HP_SYSTEM_L2_MEM_REFRESH_REG](#): configures the initialization of HP L2MEM before enabling the ECC check.
- Interrupt ([L2_MEM_ECC_ERR_INT](#)) related registers:
 - [HP_SYSTEM_L2_MEM_INT_RAW_REG](#): the raw interrupt status of [L2_MEM_ECC_ERR_INT](#)
 - [HP_SYSTEM_L2_MEM_INT_ST_REG](#): the masked interrupt status of [L2_MEM_ECC_ERR_INT](#)
 - [HP_SYSTEM_L2_MEM_INT_ENA_REG](#): write 1 to enable [L2_MEM_ECC_ERR_INT](#)
 - [HP_SYSTEM_L2_MEM_INT_CLR_REG](#): write 1 to clear [L2_MEM_ECC_ERR_INT](#)
- [HP_SYSTEM_L2_MEM_INT_RECORD1_REG](#): records information when [L2_MEM_ECC_ERR_INT](#) occurs.
- [HP_SYSTEM_L2_MEM_ERR_RESP_CTRL_REG](#): write 1 so HP L2MEM reports ECC error to HP CPU and triggers exception.

For detailed configuration steps, please refer to Chapter [7 System and Memory](#).

20.2.1.7 BitScrambler Configuration

ESP32-P4 has some peripherals that support DMA. These peripherals sometimes need BitScramblers to change the format of data before communicating with DMA.

Configure [HP_SYSTEM_BITSCRAMBLER_PERI_SEL_REG](#) to attach such a peripheral to the RX and TX channels of bitscramblers respectively.

For detailed configuration steps, please refer to Chapter [59 BitScrambler](#).

For a list of DMA-capable peripherals, please refer to Chapter [10 Reset and Clock](#).

20.2.1.8 Ethernet MAC Control

ESP32-P4 can send and receive data via Ethernet MAC (Media Access Controller) according to the IEEE 802.3 standard.

The control and configuration registers are:

- [HP_SYSTEM_GMAC_CTRL0_REG](#)
- [HP_SYSTEM_GMAC_CTRL1_REG](#)
- [HP_SYSTEM_GMAC_CTRL2_REG](#)

For details, please refer to Chapter [52 Ethernet Media Access Controller \(EMAC\)](#).

20.2.1.9 USB OTG 2.0 Control

ESP32-P4 features a USB 2.0 High-Speed On-The-Go peripheral (OTG_HS). To control it, use [HP_SYSTEM_USBOTG20_CTRL_REG](#).

For details, please refer to Chapter [49 USB 2.0 High-Speed OTG](#).

20.2.1.10 CPU Control and Record

ESP32-P4 configures CPU debug features and stores its status information via the following system registers.

- [HP_SYSTEM_CPU_CORESTALLED_ST_REG](#): records the core stalled status of CPU0 and CPU1.
- [HP_SYSTEM_CORE_DEBUG_RUNSTALL_CONF_REG](#): write 1 to enable debug runstall feature between HP CPU and LP CPU.
- [HP_SYSTEM_HP_CORE_DMACTIVE_LPCORE_REG](#): represents the dmaactive value of LP CPU's debug module.

For details, please refer to Chapter [1 High-Performance CPU](#).

20.2.1.11 HP GPIO Hold Control

Each GPIO pin of ESP32-P4 has an individual hold function that is controlled by the following registers:

- [HP_SYSTEM_GPIO_O_HOLD_CTRL0_REG](#): controls the hold signal of HP GPIO16 ~ HP GPIO47
- [HP_SYSTEM_GPIO_O_HOLD_CTRL1_REG](#): controls the hold signal of GPIO48 ~ HP GPIO56
- [HP_SYSTEM_GPIO_DED_HOLD_CTRL_REG](#): the lower 6 bits control the hold signal of flash pads, and the higher 20 bits control the hold signal of 16 line PSRAM pad

For details, please refer to Chapter [9 GPIO Matrix and IO MUX](#).

20.2.1.12 HP GPIO Output Control

Each GPIO pin of ESP32-P4 can be configured to directly output the desired value via the following registers (hysteresis):

- [HP_SYSTEM_GPIO_O_HYS_CTRL0_REG](#): configures whether or not to enable the hysteresis feature of HP GPIO16 ~ HP GPIO47.
- [HP_SYSTEM_GPIO_O_HYS_CTRL1_REG](#): configures whether or not to enable the hysteresis feature of HP GPIO48 ~ HP GPIO56.

For details, please refer to Section [9.10](#) in Chapter [9 GPIO Matrix and IO MUX](#).

20.2.1.13 Illegal Access and Unauthorized Access

ESP32-P4's buses return error response and record transfer address, transfer direction, and master id when illegal access and unauthorized access occur.

- Illegal access: access to unmapped address space
- Unauthorized access: access to mapped address space without permission

Such information will be logged in the following registers:

- HP CPU and other AHB masters ([ICM_CPU_ADDRHOLE_INT](#))
 - [HP_SYSTEM_ICM_CPU_ADDRHOLE_ADDR_REG](#)
 - [HP_SYSTEM_ICM_CPU_ADDRHOLE_INFO_REG](#)
- AXI masters ([ICM_SYS_ADDRHOLE_INT](#))
 - [HP_SYSTEM_ICM_SYS_ADDRHOLE_ADDR_REG](#)
 - [HP_SYSTEM_ICM_SYS_ADDRHOLE_INFO_REG](#)

For detailed information on address space mapping, please refer to Chapter 7 *System and Memory*.

For detailed information on access permission management, please refer to Chapter 19 *Permission Control (PMS)*.

20.2.1.14 HP CPU Bus Timeout Protection

ESP32-P4 supports Timeout Protection on HP CPU's IBUS, DBUS and AHB bus.

When a transfer is initiated, the counter inside the Timeout Protection module increments by one on each clock cycle.

- If the accumulated value remains below the timeout threshold and the slave device responds to the transfer, the internal counter is cleared.
- If the accumulated value exceeds the timeout threshold and the slave device has not responded, the Timeout Protection module takes control of the bus to return an error response to the HP CPU. At the same time, an interrupt is generated.

Related control and configuration registers are described below.

HP CPU IBUS Timeout Protection

Configure [HP_SYSTEM_HP_CORE_IBUS_TIMEOUT_REG](#) to enable timeout protection and set the timeout threshold for HP CPU accessing HP SPM, L2MEM and EXT MEM by IBUS.

When a timeout occurs, the [HP_COREx_IBUS_TIMEOUT_INT](#) interrupt will be asserted.

HP CPU DBUS Timeout Protection

Configure [HP_SYSTEM_HP_CORE_DBUS_TIMEOUT_REG](#) to enable timeout protection and set the timeout threshold for HP CPU accessing HP SPM, L2MEM and EXT MEM by DBUS.

When a timeout occurs, the [HP_COREx_DBUS_TIMEOUT_INT](#) interrupt will be asserted.

HP CPU AHB BUS Timeout Protection

Configure [HP_SYSTEM_HP_CORE_AHB_TIMEOUT_REG](#) to enable timeout protection and set the timeout threshold for HP CPU accessing LP peripherals, HP peripherals registers by AHB bus.

When a timeout occurs, the [HP_COREx_AHB_TIMEOUT_INT](#) interrupt will be asserted.

20.2.1.15 AXI Matrix

ESP32-P4's AXI Matrix connects multiple AXI-compliant master and slave devices, enabling efficient communication.

- Masters such as all AHB masters (master port1 through AHB2AXI bridge), HP CPU cache (master port2), DW GDMA_0 (master port6), DW GDMA_1 (master port7), AXI GDMA (master port9), 2DDMA (master port11), H264_0 (master port12), and H264_1 (master port13)
- Slaves such as L2MEM (slave port2), Flash (slave port3), PSRAM (slave port4), DSI Host (slave port6) and CSI Host(slave port7).

All AXI master ports except master port 2 in the AXI matrix support the QoS (Quality of Service) bandwidth regulator. The AXI matrix also includes a deadlock timeout protection mechanism, slave arbiter priority configuration for slave ports, and QoS arbiter priority configuration for master ports.

Deadlock Timeout Protection

ESP32-P4's Deadlock Timeout Protection is always enabled. Configure the timeout threshold for all AXI masters in the AXI matrix that access AXI Slaves such as Flash, PSRAM, DSI HOST, and CSI HOST via [HP_SYSTEM_ICM_DLOCK_TIMEOUT_REG](#).

When a timeout occurs, AXI Matrix will record transfer direction, AXI ID, master ID and slave ID in [HP_SYSTEM_ICM_DLOCK_STATUS_REG](#), and assert [ICM_DLOCK_INTR](#) interrupt at the same time.

Slave Arbiter Priority

Configure the arbitration priority for slaves in the AXI Matrix to arbitrate reading data and responses via [HP_SYSTEM_ICM_SLV_ARB_PRIORITY_REG](#).

The higher the value, the higher the priority.

Master QoS Arbiter Priority

Configure the QoS (Quality of Service) values for masters in the AXI matrix to arbitrate the read address channel and write address channel via [HP_SYSTEM_ICM_MST_ARQOS_REGO_REG](#) and [HP_SYSTEM_ICM_MST_AWQOS_REGO_REG](#), respectively.

A higher QoS value indicates a higher priority during arbitration.

20.2.1.16 Post Write

The following registers controls the Post Write mode of different components to speed up the write access.

- [HP_SYSTEM_PERI1_APB_POSTW_EN_REG](#): enables the Post Write mode for APB2APB bridge between peripherals.
- [HP_SYSTEM_APB_SYNC_POSTW_EN_REG](#): enables the Post Write mode for APB2APB bridge in DSI, CSI and ISP.
- [HP_SYSTEM_CPU_ICM_H2X_POST_WR_EN](#): enables the Post Write mode for AHB2AXI bridge from AHB masters to AXI slaves.

20.2.2 LP System Registers

20.2.2.1 LP Timer Stall Control

The [LP_SYSTEM_SYSTIMER_STALL_SEL](#) field allows selection of the systimer_stall signal source for the LP timer, choosing between HP CPU0 and HP CPU1:

- 0: from CPU0
- 1: from CPU1

20.2.2.2 Focused Ion Beam (FIB) Control

The [LP_SYSTEM_LP_FIB_SEL](#) field allows selection of the control source for the brownout detector and the super watchdog, choosing between the analog system or software register.

- Bit-1: selects the control source for the brownout detector.
 - 0: controlled by software register
 - 1: controlled by the analog system.
- Bit-2: selects the control source for the super watchdog.
 - 0: controlled by software register
 - 1: controlled by the analog system.

Configure [LP_SYSTEM_DIG_FIB](#) to enable the brownout detector or the super watchdog via software register.

- Bit-1: write 1 to enable the brownout detector.
- Bit-2: write 1 to enable the super watchdog.

Read [LP_SYSTEM_ANA_FIB](#) to check the FIB state of brownout detector or the super watchdog via analog system (FIB state).

- Bit-1: brownout detector
- Bit-2: super watchdog

20.2.2.3 Boot Mode

Configure [LP_SYSTEM_FORCE_DOWNLOAD_BOOT](#) to forcibly switch the chip from SPI Boot mode to Joint Download Boot mode.

20.2.2.4 LP CPU Control

Configure the LP CPU via [LP_SYSTEM_SYS_CTRL_REG](#). In particular:

- [LP_SYSTEM_LP_CPU_BOOT_ADDR](#): configures the boot address of the LP CPU.
- [LP_SYSTEM_LP_CPU_EXC_PC_REG](#): represents the exception PC address of the LP CPU.
- [LP_SYSTEM_LP_CPU_DBG_PC_REG](#): represents the current PC address of the LP CPU.
- [LP_SYSTEM_LP_CORE_ERR_RESP_DIS](#): configures whether or not to disable error response for respective LP CPU buses.

- Bit-0: IBUS
- Bit-1: DBUS
- Bit-2: AHB bus
- [LP_SYSTEM_LP_CORE_ETM_WAKEUP_FLAG](#): indicates the status of an ETM task triggered by a wakeup event of the LP CPU.
- [LP_SYSTEM_LP_CORE_ETM_WAKEUP_FLAG_CLR](#): write 1 to clear [LP_SYSTEM_LP_CORE_ETM_WAKEUP_FLAG](#).

20.2.2.5 Reset Control

ESP32-P4 supports for types of resets that occur at different levels. Such reset can be controlled by some system registers introduced in the following sections.

For a full list of all the reset types and reset sources, please refer to Chapter [10 Reset and Clock](#).

Analog and System Reset Source Bypass Control

[LP_SYSTEM_ANA_RST_BYPASS](#): configures whether to bypass the following reset sources:

- Brown-out detector
- Super watchdog
- Power glitch detector

[LP_SYSTEM_SYS_RST_BYPASS](#): configures whether to bypass the following reset sources:

- Software
- HP Watchdog timer
- LP Watchdog timer
- eFuse

Power-On Reset Bypass Control

[LP_SYSTEM_HP_PO_RSTN_BYPASS_CTRL](#): configures whether or not to bypass the following reset sources for the HP power domain during the Power-On reset.

- Bit-0: eFuse
- Bit-1: LP watchdog timer
- Bit-2: USB (JTAG)
- Bit-3: USB (UART)
- Bit-4: Unused
- Bit-5: Unused
- Bit-6: HP watchdog timer
- Bit-7: Software

[LP_SYSTEM_HP_PO_CNNT_RSTN_BYPASS_CTRL](#): configures whether or not to bypass the following reset sources for the HP CNNT power domain during the Power-On reset.

- Bit-0: eFuse
- Bit-1: LP watchdog timer
- Bit-2: USB (JTAG)
- Bit-3: USB (UART)
- Bit-4: Unused
- Bit-5: Unused
- Bit-6: HP watchdog timer
- Bit-7: Software

For details about ESP32-P4's power domains, please refer to Chapter 14 [Low-Power Management](#).

Software System Reset

Configure [LP_SYSTEM_SYS_SW_RST](#) to trigger a software system reset.

IOMUX Reset

Configure [LP_SYSTEM_IO_MUX_RESET_DISABLE](#) to disable the HP IOMUX reset.

20.2.2.6 Wakeup Control

These registers control the behavior of some PMU wakeup sources.

[LP_SYSTEM_USBOTG20_IN_SUSPEND](#): configures whether or not to enable wakeup signals from USB OTG2.0.

[LP_SYSTEM_USBOTG20_WAKEUP_CLR](#): write 1 to clear wakeup signals from USB OTG2.0 to PMU.

20.2.2.7 LPROM and LP SPM Clock Control

[LP_SYSTEM_LP_SPM_ROM_CLK_FORCE_ON](#): set to force on LP SPM_ROM clock if LP CPU clock is on.

[LP_SYSTEM_LP_SPM_RAM_CLK_FORCE_ON](#): set to force on LP SPM_RAM clock.

20.2.2.8 Analog Voltage Comparator

Analog voltage comparator x ($x = 0, 1$) are configured by the following register fields:

- [LP_SYSTEM_XPD_COMP \$x\$](#) ($x = 0, 1$): configures whether or not to enable analog voltage comparator x .
- [LP_SYSTEM_MODE_COMP \$x\$](#) ($x = 0, 1$): configures the comparison mode:
 - 0: comparing the main voltage with the external reference voltage
 - 1: comparing the main voltage with the internal reference voltage
- [LP_SYSTEM_DREF_COMP \$x\$](#) ($x = 0, 1$): configures the internal reference voltage.

- The adjustable voltage range is 0 to $(0.7 * VDDPST_6)$ V.
- The adjustable step is $0.1 * VDDPST_6$ V.

For details, please refer to Chapter [63 Analog Voltage Comparator](#).

20.2.2.9 Illegal Access and Unauthorized Access

ESP32-P4's buses return error response and record transfer address, transfer direction, and master id when illegal access and unauthorized access occur.

- Illegal access: access to unmapped address space
- Unauthorized access: access to mapped address space without permission

Such information will be logged in the following registers:

- For LP CPU's access by IBUS and DBUS ([IDBUS_ADDRHOLE_INTR](#))
 - [LP_SYSTEM_IDBUS_ADDRHOLE_ADDR_REG](#)
 - [LP_SYSTEM_IDBUS_ADDRHOLE_INFO_REG](#)
- For LP CPU's access by AHB or an access from HP CPU ([LP_ADDRHOLE_INTR](#)):
 - [LP_SYSTEM_LP_ADDRHOLE_ADDR_REG](#)
 - [LP_SYSTEM_LP_ADDRHOLE_INFO_REG](#)

For detailed information on address space mapping, please refer to Chapter [7 System and Memory](#).

For detailed information on access permission management, please refer to Chapter [19 Permission Control \(PMS\)](#).

20.2.2.10 LP Bus Timeout Protection

ESP32-P4 supports Timeout Protection on LP CPU's IBUS, DBUS and AHB bus.

When a transfer is initiated, the counter inside the Timeout Protection module increments by one on each clock cycle.

- If the accumulated value remains below the timeout threshold and the slave device responds to the transfer, the internal counter is cleared.
- If the accumulated value exceeds the timeout threshold and the slave device has not responded, the Timeout Protection module takes control of the bus to return an error response to the LP CPU. At the same time, an interrupt is generated.

Related control and configuration registers are described below.

LP CPU IBUS Timeout Protection

Configure [LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_REG](#) to enable timeout protection and set the timeout threshold for LP CPU accessing LPRom and LP SPM by IBUS. When a timeout occurs, the [LP_CORE_IBUS_TIMEOUT_INTR](#) interrupt will be asserted.

LP CPU DBUS Timeout Protection

Configure [LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_REG](#) to enable timeout protection and set the timeout threshold for LP CPU accessing LPROM and LP SPM by DBUS.

When a timeout occurs, the [LP_CORE_DBUS_TIMEOUT_INTR](#) interrupt will be asserted.

LP CPU AHB Bus Timeout Protection

Configure [LP_SYSTEM_LP_CORE_AHB_TIMEOUT_REG](#) to enable timeout protection and set the timeout threshold for LP CPU accessing LP peripherals, HP peripherals and CPU peripherals by AHB bus.

When a timeout occurs, the [LP_CORE_AHB_TIMEOUT_INTR](#) interrupt will be asserted.

20.2.2.11 RNG Control

Registers [LP_SYSTEM_RNG_DATA_REG](#) and [LP_SYSTEM_RNG_CFG_REG](#) are used to configure and control the random number generator.

20.3 Interrupt

ESP32-P4's System Registers can generate the following interrupt signal(s) that will be sent to the [Interrupt Matrix](#).

- [HP_SYSREG_INTR](#)
- [SYS_ICM_INTR](#)
- [LP_SYSREG_INTR](#)

There are several internal interrupt sources from System Registers that can generate the above interrupt signal(s). The interrupt sources from System Registers are listed with their trigger conditions and the resulted interrupt signal(s) in Table 20.3-1.

Table 20.3-1. System Registers Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
CPU_INT_FROM_CPU0	Software interrupt. Triggered by writing 1 to HP_SYSTEM_HP_CPU_INT_FROM_CPU_0_REG	HP_SYSREG_INTR
CPU_INT_FROM_CPU1	Software interrupt. Triggered by writing 1 to HP_SYSTEM_HP_CPU_INT_FROM_CPU_1_REG	
CPU_INT_FROM_CPU2	Software interrupt. Triggered by writing 1 to HP_SYSTEM_HP_CPU_INT_FROM_CPU_2_REG	
CPU_INT_FROM_CPU3	Software interrupt. Triggered by writing 1 to HP_SYSTEM_HP_CPU_INT_FROM_CPU_3_REG	
L2_MEM_ERR_RESP_INT	Triggered when an error response occurs in L2MEM	
L2_MEM_EXCEED_ADDR_INT	Triggered when L2MEM exceeds its address range	
L2_MEM_ECC_ERR_INT	Triggered when a one-bit flip error is detected and corrected by ECC in L2MEM	
HP_SPM_PARITY_ERR_INT	Triggered when a parity error occurs in HP SPM	

CPU_ICM_H2X_BRESP_ERR_INT	Triggered when a BRESP error occurs in Post Write mode in AHB2AXI	
HP_CORE1_DBUS_TIMEOUT_INT	Triggered upon HP CPU1 DBUS timeout	
HP_CORE0_DBUS_TIMEOUT_INT	Triggered upon HP CPU0 DBUS timeout	
HP_CORE1_IBUS_TIMEOUT_INT	Triggered upon HP CPU1 IBUS timeout	
HP_CORE0_IBUS_TIMEOUT_INT	Triggered upon HP CPU0 IBUS timeout	
HP_CORE1_AHB_TIMEOUT_INT	Triggered upon HP CPU1 AHB bus timeout	
HP_CORE0_AHB_TIMEOUT_INT	Triggered upon HP CPU0 AHB bus timeout	
ICM_CPU_ADDRHOLE_INT	Triggered when illegal/unauthorized access occurs in HP CPU and AHB matrix	SYS_ICM_INTR
ICM_SYS_ADDRHOLE_INT	Triggered when illegal/unauthorized access occurs in HP AXI matrix	
ICM_DLOCK_INT	Triggered when any AXI transfer in the AXI matrix times out	LP_SYSREG_INTR
SLOW_CLK_TICK_INT	ETM tick event interrupt. The interrupt source is periodically triggered at the frequency of LP_SLOW_CLK when the LP CPU is not sleeping. This event source can be disabled via the LP_CLKRST_ETM_EVENT_TICK_EN.	
ETM_TASK_ULP_INT	From ULP_TASK_INT_CPU. Ref to 13 Event Task Matrix (ETM) for details	
LP_CORE_DBUS_TIMEOUT_INT	Triggered upon LP CPU DBUS timeout	
LP_CORE_IBUS_TIMEOUT_INT	Triggered upon LP CPU IBUS timeout	
LP_CORE_AHB_TIMEOUT_INT	Triggered upon LP CPU AHB timeout	
IDBUS_ADDRHOLE_INT	Triggered when illegal/unauthorized access occurs on LP CPU IBUS or DBUS	
LP_ADDRHOLE_INT	Triggered when illegal/unauthorized access occurs in LP AHB matrix	

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

20.4 Register Summary

20.4.1 HP Register Summary

The addresses in this section are relative to HP System Registers base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
Version Register			
HP_SYSTEM_VER_DATE_REG	Version control	0x0000	R/W
HP_SYSTEM_CLK_EN_REG	Clock gating control	0x0004	R/W
Interrupt Register			
HP_SYSTEM_HP_CPU_INT_FROM_CPU_0_REG	Software interrupt 0	0x0010	R/W
HP_SYSTEM_HP_CPU_INT_FROM_CPU_1_REG	Software interrupt 1	0x0014	R/W
HP_SYSTEM_HP_CPU_INT_FROM_CPU_2_REG	Software interrupt 2	0x0018	R/W
HP_SYSTEM_HP_CPU_INT_FROM_CPU_3_REG	Software interrupt 3	0x001C	R/W
HP_SYSTEM_HP_SPM_INT_RAW_REG	HP SPM interrupt raw	0x00FC	R/WTC /SS
HP_SYSTEM_HP_SPM_INT_ST_REG	Masked HP SPM interrupt status	0x0100	RO
HP_SYSTEM_HP_SPM_INT_ENA_REG	HP SPM interrupt enable	0x0104	R/W
HP_SYSTEM_HP_SPM_INT_CLR_REG	HP SPM interrupt clear	0x0108	WT
HP_SYSTEM_HP_CORE_TIMEOUT_INT_RAW_REG	HP CPU timeout interrupt raw	0x01A8	R/WTC /SS
HP_SYSTEM_HP_CORE_TIMEOUT_INT_ST_REG	HP CPU timeout masked interrupt	0x01AC	RO
HP_SYSTEM_HP_CORE_TIMEOUT_INT_ENA_REG	HP CPU timeout interrupt enable	0x01B0	R/W
HP_SYSTEM_HP_CORE_TIMEOUT_INT_CLR_REG	HP CPU timeout interrupt clear	0x01B4	WT
HP_SYSTEM_HP_CACHE_CLK_CONFIG_REG	Cache clock configuration	0x0020	R/W
HP_SYSTEM_CACHE_RESET_CONFIG_REG	Cache reset configuration	0x0024	R/W
HP_SYSTEM_HP_SPM_RAM_PWR_CTRL0_REG	HP SPM clock force on	0x003C	R/W
HP_SYSTEM_L2_ROM_PWR_CTRL0_REG	L2ROM clock force on	0x0040	R/W
HP_SYSTEM_HP_L2_MEM_RAM_PWR_CTRL0_REG	L2MEM clock force on	0x0060	R/W
HP_SYSTEM_CPU_CORESTALLED_ST_REG	HP CPU stall status	0x0064	RO
HP_SYSTEM_CRYPTO_CTRL_REG	Encryption control	0x0070	R/W
HP_SYSTEM_GPIO_O_HOLD_CTRL0_REG	HP GPIO hold control 0	0x0074	R/W
HP_SYSTEM_GPIO_O_HOLD_CTRL1_REG	HP GPIO hold control 1	0x0078	R/W
HP_SYSTEM_L2_MEM_INT_RAW_REG	HP L2MEM interrupt raw	0x009C	R/WTC /SS
HP_SYSTEM_L2_MEM_INT_ST_REG	Masked HP L2MEM interrupt status	0x00A0	RO
HP_SYSTEM_L2_MEM_INT_ENA_REG	HP L2MEM interrupt enable	0x00A4	R/W
HP_SYSTEM_L2_MEM_INT_CLR_REG	HP L2MEM interrupt clear	0x00A8	WT
HP_SYSTEM_AHB2AXI_BRESP_ERR_INT_RAW_REG	AHB to AXI BRESP error interrupt raw	0x0188	R/WTC /SS
HP_SYSTEM_AHB2AXI_BRESP_ERR_INT_ST_REG	AHB to AXI BRESP error interrupt status	0x018C	RO

Name	Description	Address	Access
HP_SYSTEM_AHB2AXI_BRESP_ERR_INT_ENA_REG	AHB to AXI BRESP error interrupt enable	0x0190	R/W
HP_SYSTEM_AHB2AXI_BRESP_ERR_INT_CLR_REG	AHB to AXI BRESP error interrupt clear	0x0194	WT
HP_SYSTEM_L2_MEM_L2_RAM_ECC_REG	HP L2MEM ECC enable	0x00AC	R/W
HP_SYSTEM_L2_MEM_INT_RECORD0_REG	HP L2MEM interrupt record 0	0x00B0	RO
HP_SYSTEM_L2_MEM_INT_RECORD1_REG	HP L2MEM interrupt record 1	0x00B4	RO
HP_SYSTEM_L2_MEM_L2_CACHE_ECC_REG	L2 cache ECC enable	0x00C4	R/W
HP_SYSTEM_HP_L1CACHE_BUS0_ID_REG	L1 cache bus 0 ID	0x00C8	R/W
HP_SYSTEM_HP_L1CACHE_BUS1_ID_REG	L1 cache bus 1 ID	0x00CC	R/W
HP_SYSTEM_GPIO_DED_HOLD_CTRL_REG	HP GPIO hold control	0x00F0	R/W
HP_SYSTEM_HP_USB2OTG_MEM_CTRL_REG	MEM in USB OTG FS clock force on	0x00F8	R/W
HP_SYSTEM_HP_SPM_PARITY_INT_RECORD_REG	HP SPM parity interrupt status	0x010C	RO
HP_SYSTEM_HP_L1_CACHE_PWR_CTRL_REG	HP L1 cache memory force on	0x0110	R/W
HP_SYSTEM_HP_L2_CACHE_PWR_CTRL_REG	HP L2 cache memory force on	0x0114	R/W
HP_SYSTEM_HP_CPU_WAITI_CONF_REG	CPU_WAITI configuration	0x0118	R/W
HP_SYSTEM_CORE_DEBUG_RUNSTALL_CONF_REG	CPU debug runstall configuration	0x011C	R/W
HP_SYSTEM_HP_CORE_AHB_TIMEOUT_REG	HP CPU AHB bus timeout	0x0120	R/W
HP_SYSTEM_HP_CORE_IBUS_TIMEOUT_REG	HP CPU IBUS timeout	0x0124	R/W
HP_SYSTEM_HP_CORE_DBUS_TIMEOUT_REG	HP CPU DBUS timeout	0x0128	R/W
HP_SYSTEM_HP_ICM_CPU_H2X_CFG_REG	AHB to AXI bridge configuration	0x0138	varies
HP_SYSTEM_PERI1_APB_POSTW_EN_REG	HP PERI1 APB post write enable	0x013C	R/W
HP_SYSTEM_BITSCRAMBLER_PERI_SEL_REG	BitScrambler peripherals select	0x0140	R/W
HP_SYSTEM_APB_SYNC_POSTW_EN_REG	HP PREIO APB post write enable	0x0144	R/W
HP_SYSTEM_HP_RSA_PD_CTRL_REG	RSA memory PD control	0x01D0	R/W
HP_SYSTEM_HP_ECC_PD_CTRL_REG	ECC memory PD control	0x01D4	R/W
HP_SYSTEM_HP_RNG_CFG_REG	RNG configuration	0x01D8	varies
HP_SYSTEM_HP_UART_PD_CTRL_REG	Memory PD control	0x01DC	R/W
HP_SYSTEM_HP_PERI_MEM_CLK_FORCE_ON_REG	HP PERI memory clock force on	0x01E0	R/W
HP_SYSTEM_GDMA_CTRL_REG	DW-GDMA control	0x0148	R/W
HP_SYSTEM_GMAC_CTRL0_REG	GMAC control 0	0x014C	varies
HP_SYSTEM_GMAC_CTRL1_REG	GMAC control 1	0x0150	RO
HP_SYSTEM_GMAC_CTRL2_REG	GMAC control 2	0x0154	RO
HP_SYSTEM_USBOTG20_CTRL_REG	USB OTGHS control	0x015C	varies
HP_SYSTEM_HP_SPM_ERR_RESP_CTRL_REG	HP SPM error response enable	0x0160	R/W
HP_SYSTEM_L2_MEM_REFRESH_REG	HP SPM refresh	0x0164	varies
HP_SYSTEM_HP_SPM_INIT_REG	HP SPM initialization	0x0168	varies
HP_SYSTEM_HP_SPM_PARITY_CHECK_CTRL_REG	HP SPM parity check enable	0x016C	R/W
HP_SYSTEM_L2_MEM_ERR_RESP_CTRL_REG	HP L2MEM error response	0x0198	R/W
HP_SYSTEM_L2_MEM_AHB_BUFFER_CTRL_REG	HP L2MEM AHB buffer enable	0x019C	R/W
HP_SYSTEM_HP_CORE_DMACTIVE_LPCORE_REG	HP CPU debug module active	0x01A0	RO
Control Registers			
HP_SYSTEM_HP_CORE_ERR_RESP_DIS_REG	HP CPU error response masked	0x01A4	R/W
HP_SYSTEM_GPIO_O_HYS_CTRL0_REG	HP GPIO hold control 0	0x01C0	R/W

Name	Description	Address	Access
HP_SYSTEM_GPIO_O_HYS_CTRL1_REG	HP GPIO hold control 1	0x01C4	R/W

20.4.2 ICM Register Summary

The addresses in this section are relative to relative to AXI ICM base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
Version Register			
HP_SYSTEM_ICM_VER_DATE_REG	ICM version control	0x0000	R/W
Clock Gating Register			
HP_SYSTEM_ICM_CLK_EN_REG	Clock gating control	0x0004	R/W
Status Register			
HP_SYSTEM_ICM_DLOCK_STATUS_REG	ICM deadlock status	0x0008	RO
HP_SYSTEM_ICM_SYS_ADDRHOLE_ADDR_REG	System address hole address	0x0038	RO
HP_SYSTEM_ICM_SYS_ADDRHOLE_INFO_REG	System address hole information	0x003C	RO
HP_SYSTEM_ICM_CPU_ADDRHOLE_ADDR_REG	CPU address hole address	0x0040	RO
HP_SYSTEM_ICM_CPU_ADDRHOLE_INFO_REG	CPU address hole information	0x0044	RO
Interrupt Register			
HP_SYSTEM_ICM_INT_RAW_REG	ICM interrupt raw status	0x000C	R/WTC /SS
HP_SYSTEM_ICM_INT_ST_REG	ICM interrupt status	0x0010	RO
HP_SYSTEM_ICM_INT_ENA_REG	ICM interrupt enable	0x0014	R/W
HP_SYSTEM_ICM_INT_CLR_REG	ICM interrupt clear	0x0018	WT
Configuration Register			
HP_SYSTEM_ICM_SLV_ARB_PRIORITY_REG	Slave arbitration priority configuration	0x0024	R/W
HP_SYSTEM_ICM_MST_ARQOS_REGO_REG	ARQoS configuration	0x0028	R/W
HP_SYSTEM_ICM_MST_AWQOS_REGO_REG	AWQoS configuration	0x0030	R/W
HP_SYSTEM_ICM_DLOCK_TIMEOUT_REG	ICM bus timeout configuration	0x0048	R/W

20.4.3 LP Register Summary

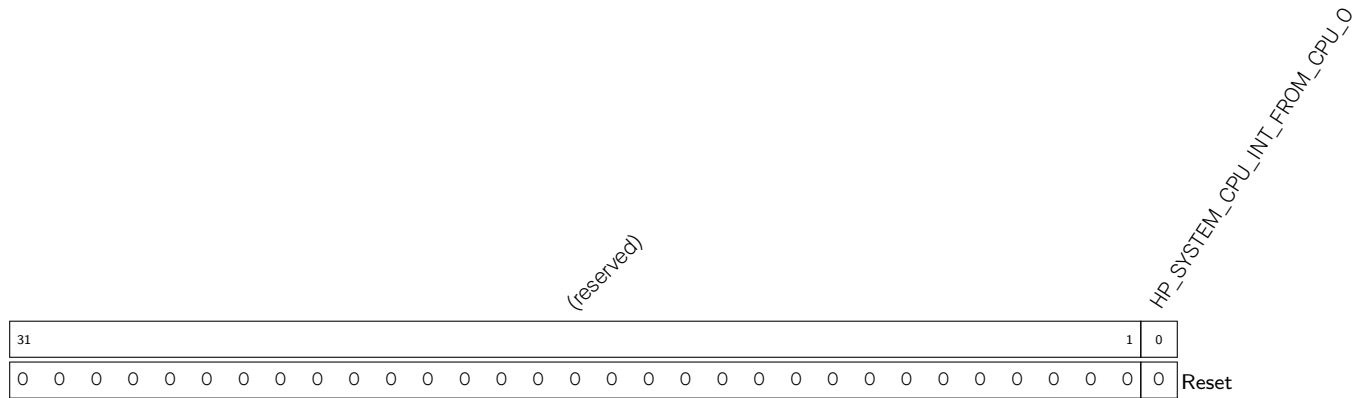
The addresses in this section are relative to LP System Registers base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
Configuration & Control Registers			
LP_SYSTEM_LP_SYS_VER_DATE_REG	Version control	0x0000	R/W
LP_SYSTEM_LP_SYS_CTRL_REG	System control	0x0008	varies
LP_SYSTEM_LP_CLK_CTRL_REG	Clock control	0x000C	R/W
LP_SYSTEM_LP_RST_CTRL_REG	Reset control	0x0010	R/W

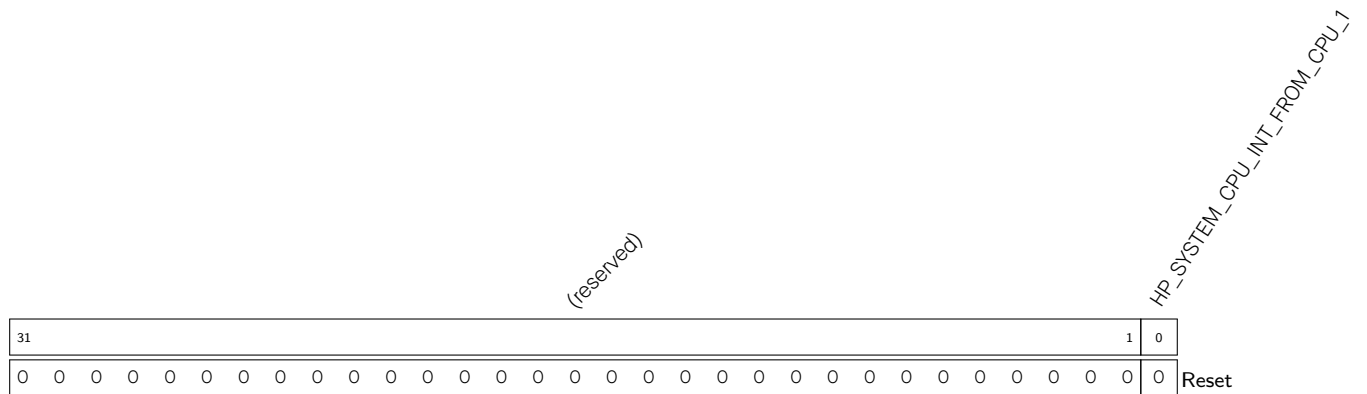
Name	Description	Address	Access
LP_SYSTEM_LP_CORE_BOOT_ADDR_REG	CPU boot address	0x0018	R/W
LP_SYSTEM_LP_SPM_PWR_CTRL_REG	LPROM and LP SPM clock control	0x0024	R/W
LP_SYSTEM_USB_CTRL_REG	USB control	0x0100	varies
LP_SYSTEM_PAD_COMPO_REG	Pad compensation 0	0x0148	R/W
LP_SYSTEM_PAD_COMP1_REG	Pad compensation 1	0x014C	R/W
LP_SYSTEM_HP_MEM_AUX_CTRL_REG	High-performance memory auxiliary control	0x0180	R/W
LP_SYSTEM_LP_MEM_AUX_CTRL_REG	Low-power memory auxiliary control	0x0184	R/W
LP_SYSTEM_HP_ROM_AUX_CTRL_REG	High-performance ROM auxiliary control	0x0188	R/W
LP_SYSTEM_LP_ROM_AUX_CTRL_REG	Low-power ROM auxiliary control	0x018C	R/W
LP_SYSTEM_HP_POR_RST_BYPASS_CTRL_REG	Power-on reset control	0x01A0	R/W
LP_SYSTEM_LP_CORE_AHB_TIMEOUT_REG	CPU AHB bus timeout	0x01B0	R/W
LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_REG	CPU IBUS timeout	0x01B4	R/W
LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_REG	CPU DBUS timeout	0x01B8	R/W
LP_SYSTEM_LP_CORE_ERR_RESP_DIS_REG	CPU error response control	0x01BC	R/W
LP_SYSTEM_RNG_CFG_REG	RNG configuration	0x01C0	varies
Status Registers			
LP_SYSTEM_LP_ADDRHOLE_ADDR_REG	AHB bus address hole	0x0168	RO
LP_SYSTEM_LP_ADDRHOLE_INFO_REG	AHB bus address hole info	0x016C	RO
LP_SYSTEM_LP_CPU_DBG_PC_REG	CPU debug PC	0x0190	RO
LP_SYSTEM_LP_CPU_EXC_PC_REG	CPU exception PC	0x0194	RO
LP_SYSTEM_IDBUS_ADDRHOLE_ADDR_REG	CPU IBUS and DBUS address hole	0x0198	RO
LP_SYSTEM_IDBUS_ADDRHOLE_INFO_REG	CPU IBUS and DBUS address hole info	0x019C	RO
LP_SYSTEM_RNG_DATA_REG	RNG result	0x01A4	RO
Interrupt Registers			
LP_SYSTEM_INT_RAW_REG	Raw interrupt	0x0170	R/SS /WTC
LP_SYSTEM_INT_ST_REG	Masked interrupt	0x0174	RO
LP_SYSTEM_INT_ENA_REG	Interrupt enable	0x0178	R/W
LP_SYSTEM_INT_CLR_REG	Interrupt clear	0x017C	WT

Register 20.3. HP_SYSTEM_HP_CPU_INT_FROM_CPU_0_REG (0x0010)



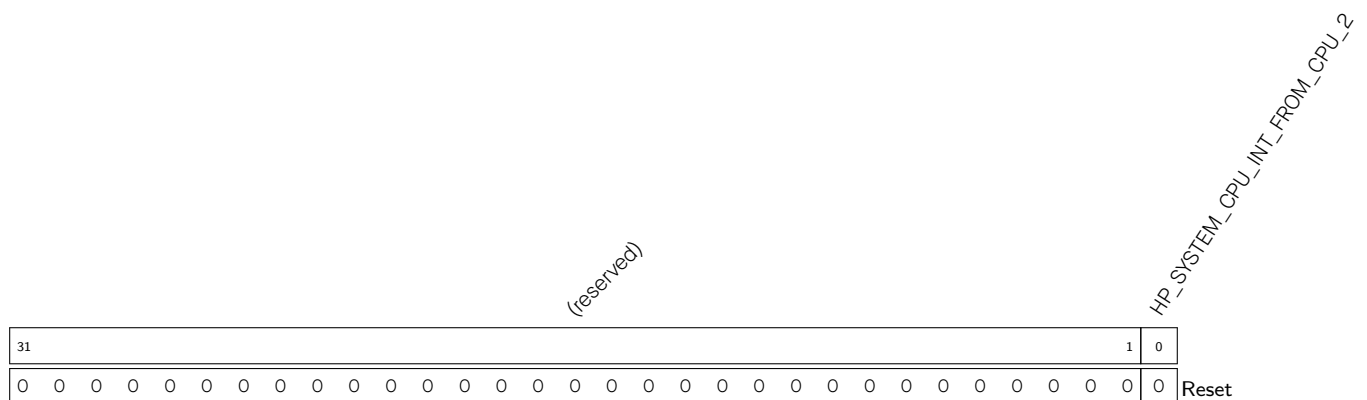
HP_SYSTEM_CPU_INT_FROM_CPU_0 Set this bit to generate CPU_INT_FROM_CPU0. (R/W)

Register 20.4. HP_SYSTEM_HP_CPU_INT_FROM_CPU_1_REG (0x0014)



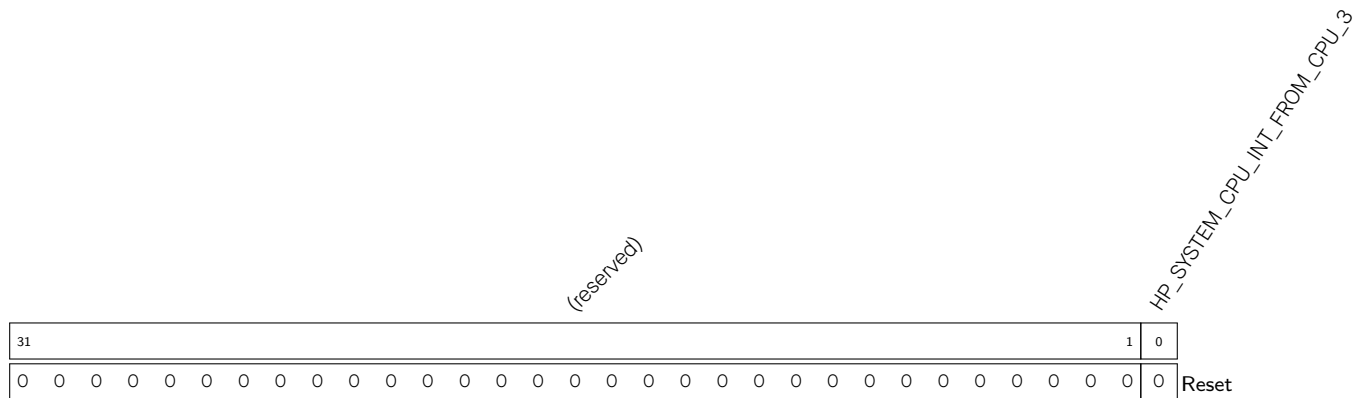
HP_SYSTEM_CPU_INT_FROM_CPU_1 Set this bit to generate CPU_INT_FROM_CPU1. (R/W)

Register 20.5. HP_SYSTEM_HP_CPU_INT_FROM_CPU_2_REG (0x0018)



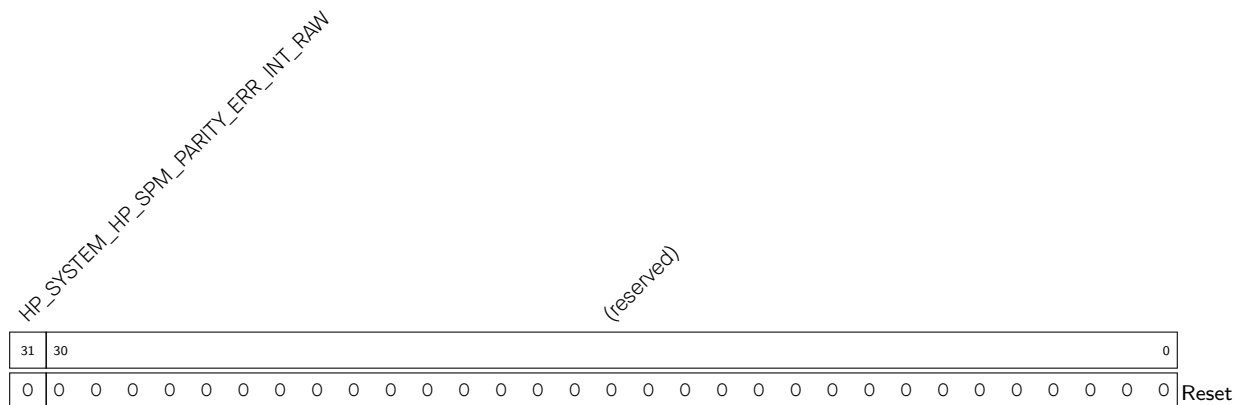
HP_SYSTEM_CPU_INT_FROM_CPU_2 Set this bit to generate CPU_INT_FROM_CPU2. (R/W)

Register 20.6. HP_SYSTEM_HP_CPU_INT_FROM_CPU_3_REG (0x001C)



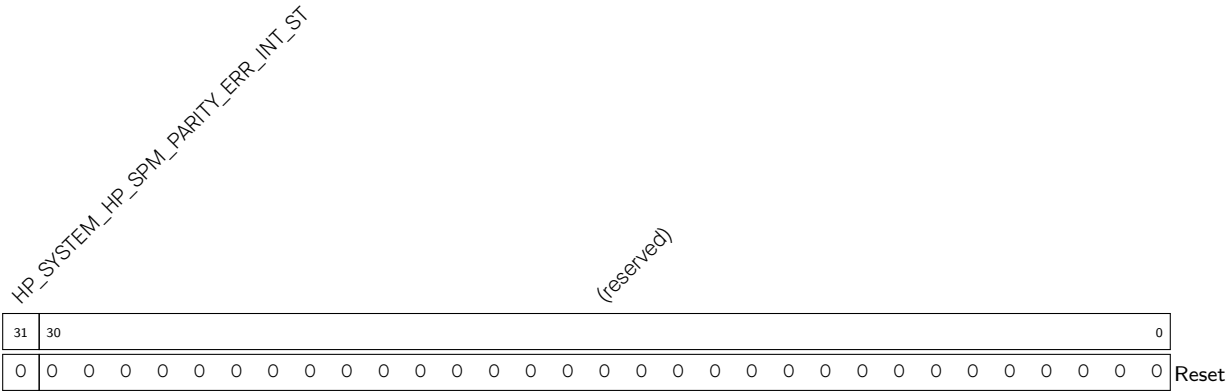
HP_SYSTEM_CPU_INT_FROM_CPU_3 Set this bit to generate CPU_INT_FROM_CPU3. (R/W)

Register 20.7. HP_SYSTEM_HP_SPM_INT_RAW_REG (0x00FC)



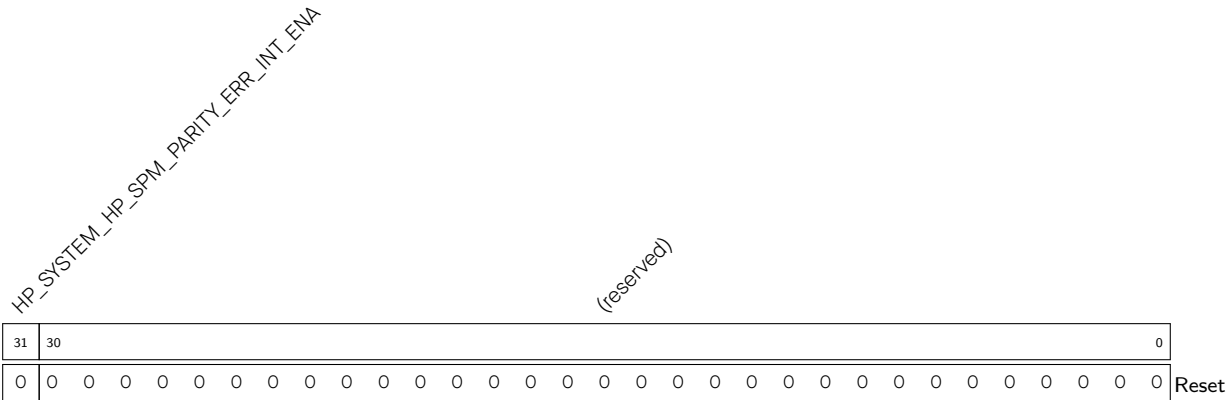
HP_SYSTEM_HP_SPM_PARITY_ERR_INT_RAW The raw interrupt status of [HP_SPM_PARITY_ERR_INT](#), triggered when a parity error occurs. (R/WTC/SS)

Register 20.8. HP_SYSTEM_HP_SPM_INT_ST_REG (0x0100)



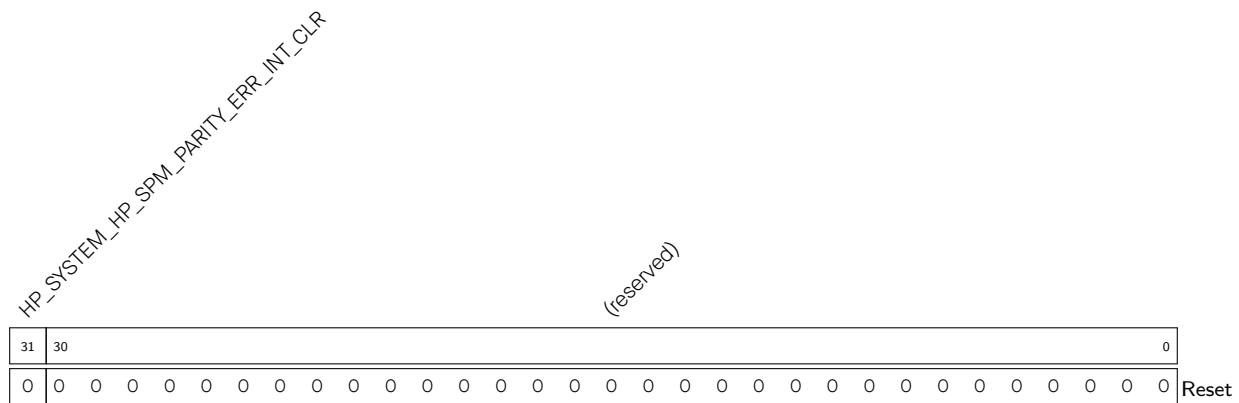
HP_SYSTEM_HP_SPM_PARITY_ERR_INT_ST The masked interrupt status of [HP_SPM_PARITY_ERR_INT](#) interrupt. (RO)

Register 20.9. HP_SYSTEM_HP_SPM_INT_ENA_REG (0x0104)



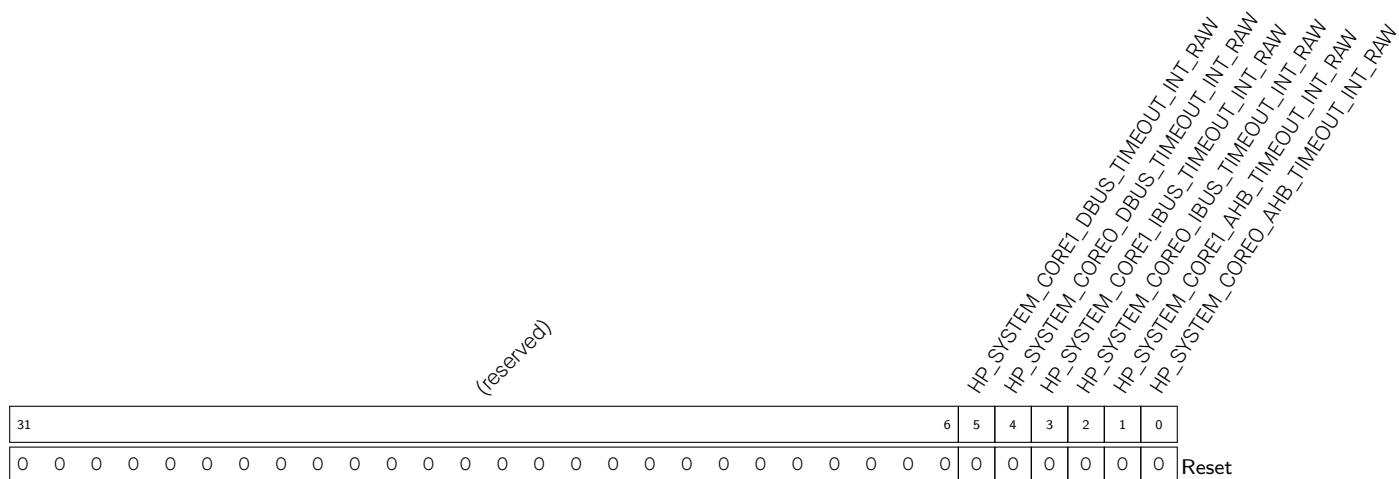
HP_SYSTEM_HP_SPM_PARITY_ERR_INT_ENA Write 1 to enable [HP_SPM_PARITY_ERR_INT](#) interrupt. (R/W)

Register 20.10. HP_SYSTEM_HP_SPM_INT_CLR_REG (0x0108)



HP_SYSTEM_HP_SPM_PARITY_ERR_INT_CLR Write 1 to clear HP_SPM_PARITY_ERR_INT interrupt.
(WT)

Register 20.11. HP_SYSTEM_HP_CORE_TIMEOUT_INT_RAW_REG (0x01A8)



HP_SYSTEM_CORE0_AHB_TIMEOUT_INT_RAW The raw interrupt status of [HP_CORE0_AHB_TIMEOUT_INT](#). (R/WTC/SS)

HP_SYSTEM_CORE1_AHB_TIMEOUT_INT_RAW The raw interrupt status of [HP_CORE1_AHB_TIMEOUT_INT](#). (R/WTC/SS)

HP_SYSTEM_CORE0_IBUS_TIMEOUT_INT_RAW The raw interrupt status of [HP_CORE0_IBUS_TIMEOUT_INT](#). (R/WTC/SS)

HP_SYSTEM_CORE1_IBUS_TIMEOUT_INT_RAW The raw interrupt status of [HP_CORE1_IBUS_TIMEOUT_INT](#). (R/WTC/SS)

HP_SYSTEM_CORE0_DBUS_TIMEOUT_INT_RAW The raw interrupt status of
HP_CORE0_DBUS_TIMEOUT_INT. (R/WTC/SS)

HP_SYSTEM_CORE1_DBUS_TIMEOUT_INT_RAW The raw interrupt status of
HP_CORE1_DBUS_TIMEOUT_INT. (R/WTC/SS)

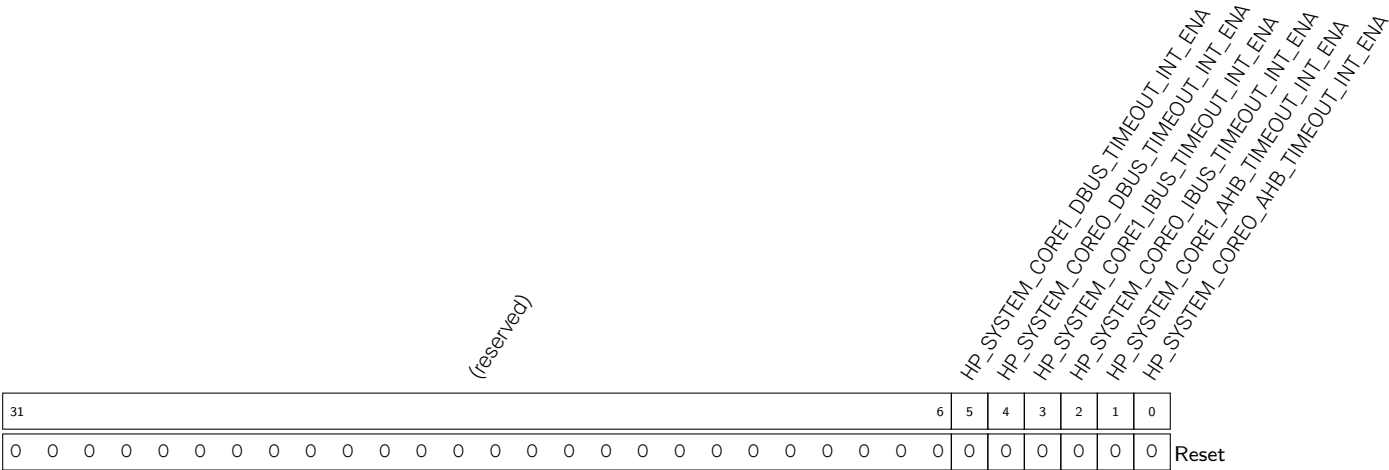
Register 20.12. HP_SYSTEM_HP_CORE_TIMEOUT_INT_ST_REG (0x01AC)

Diagram illustrating the structure of the **HP_SYSTEM_INTERRUPT_STATUS** register. The register is 32 bits wide, divided into a 26-bit reserved field (bits 31-6) and a 6-bit interrupt status field (bits 5-0). The interrupt status field contains six bits, each corresponding to a specific interrupt:

- Bit 5: **HP_SYSTEM_CORE0_AHB_TIMEOUT_INT_ST**
- Bit 4: **HP_SYSTEM_CORE1_AHB_TIMEOUT_INT_ST**
- Bit 3: **HP_SYSTEM_CORE0_IBUS_TIMEOUT_INT_ST**
- Bit 2: **HP_SYSTEM_CORE1_IBUS_TIMEOUT_INT_ST**
- Bit 1: **HP_SYSTEM_CORE0_DBUS_TIMEOUT_INT_ST**
- Bit 0: **HP_SYSTEM_CORE1_DBUS_TIMEOUT_INT_ST**

The register is reset to 0.

Register 20.13. HP_SYSTEM_HP_CORE_TIMEOUT_INT_ENA_REG (0x01B0)



- HP_SYSTEM_CORE0_AHB_TIMEOUT_INT_ENA Write 1 to enable [HP_CORE0_AHB_TIMEOUT_INT](#) interrupt. (R/W)
- HP_SYSTEM_CORE1_AHB_TIMEOUT_INT_ENA Write 1 to enable [HP_CORE1_AHB_TIMEOUT_INT](#) in-
terrupt. (R/W)
- HP_SYSTEM_CORE0_IBUS_TIMEOUT_INT_ENA Write 1 to enable [HP_CORE0_IBUS_TIMEOUT_INT](#) interrupt. (R/W)
- HP_SYSTEM_CORE1_IBUS_TIMEOUT_INT_ENA Write 1 to enable [HP_CORE1_IBUS_TIMEOUT_INT](#) interrupt. (R/W)
- HP_SYSTEM_CORE0_DBUS_TIMEOUT_INT_ENA Write 1 to enable [HP_CORE0_DBUS_TIMEOUT_INT](#) interrupt. (R/W)
- HP_SYSTEM_CORE1_DBUS_TIMEOUT_INT_ENA Write 1 to enable [HP_CORE1_DBUS_TIMEOUT_INT](#) interrupt. (R/W)

Register 20.14. HP_SYSTEM_HP_CORE_TIMEOUT_INT_CLR_REG (0x01B4)

[illegible]

HP_SYSTEM_CORE0_AHB_TIMEOUT_INT_CLR Write 1 to clear **HP_CORE0_AHB_TIMEOUT_INT** in-
terrupt. (WT)

HP_SYSTEM_CORE1_AHB_TIMEOUT_INT_CLR Write 1 to clear [HP_CORE1_AHB_TIMEOUT_INT](#) interrupt. (WT)

HP_SYSTEM_CORE0_IBUS_TIMEOUT_INT_CLR Write 1 to clear **HP_CORE0_IBUS_TIMEOUT_INT** in-
terrupt. (WT)

HP_SYSTEM_CORE1_IBUS_TIMEOUT_INT_CLR Write 1 to clear **HP_CORE1_IBUS_TIMEOUT_INT** interrupt. (WT)

HP_SYSTEM_CORE0_DBUS_TIMEOUT_INT_CLR Write 1 to clear HP_CORE0_DBUS_TIMEOUT_INT interrupt. (WT)

HP_SYSTEM_CORE1_DBUS_TIMEOUT_INT_CLR Write 1 to clear **HP_CORE1_DBUS_TIMEOUT_INT** in-
terrupt. (WT)

Register 20.15. HP_SYSTEM_HP_CACHE_CLK_CONFIG_REG (0x0020)

(reserved)																												6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	1				

HP_SYSTEM_L2_CACHE_CLK_ON Configures whether or not to enable L2 cache clock.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L1_D_CACHE_CLK_ON Configures whether or not to enable L1 dcache clock.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L1_I1_CACHE_CLK_ON Configures whether or not to enable L1 ichache 1 clock.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L1_IO_CACHE_CLK_ON Configures whether or not to enable L1 ichache 0 clock.

0: Disable

1: Enable

(R/W)

Register 20.16. HP_SYSTEM_CACHE_RESET_CONFIG_REG (0x0024)

(reserved)																												HP_SYSTEM_L1_IO_CACHE_RESET				HP_SYSTEM_L1_I1_CACHE_RESET				(reserved)				HP_SYSTEM_L1_D_CACHE_RESET						
31																												6	5	4	3	2	1	0												
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset															

HP_SYSTEM_L1_D_CACHE_RESET Configures whether or not to reset L1 dcache.
0: No effect
1: Reset
(R/W)

HP_SYSTEM_L1_I1_CACHE_RESET Configures whether or not to reset L1 ichache 1.
0: No effect
1: Reset
(R/W)

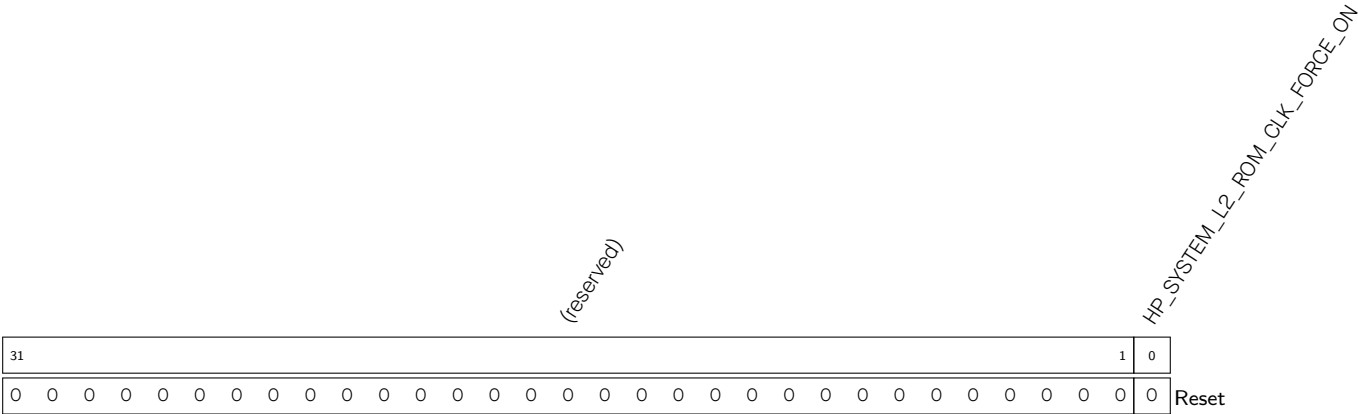
HP_SYSTEM_L1_IO_CACHE_RESET Configures whether or not to reset L1 ichache 0.
0: No effect
1: Reset
(R/W)

Register 20.17. HP_SYSTEM_HP_SPM_RAM_PWR_CTRL0_REG (0x003C)

(reserved)																															HP_SYSTEM_HP_SYSTEM_TCM_CLK_FORCE_ON															
31																															1	0														
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset											

HP_SYSTEM_HP_SPM_CLK_FORCE_ON Configures whether or not to force on HP SPM clock.
0: No effect
1: Force on
(R/W)

Register 20.18. HP_SYSTEM_L2_ROM_PWR_CTRL0_REG (0x0040)

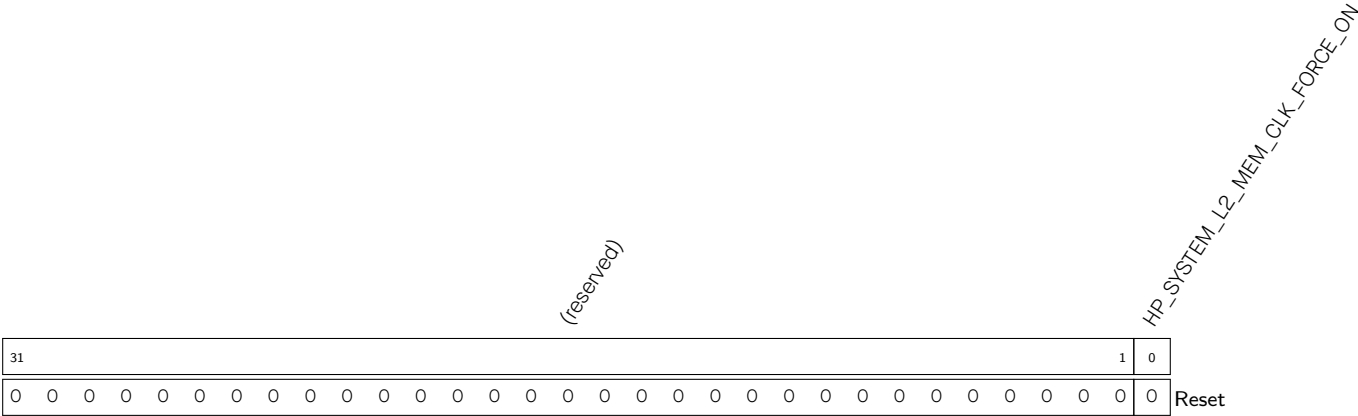


HP_SYSTEM_L2_ROM_CLK_FORCE_ON Configures whether or not to force on L2ROM clock.

- 0: No effect
- 1: Force on

(R/W)

Register 20.19. HP_SYSTEM_HP_SYSTEM_L2_MEM_RAM_PWR_CTRL0_REG (0x0060)

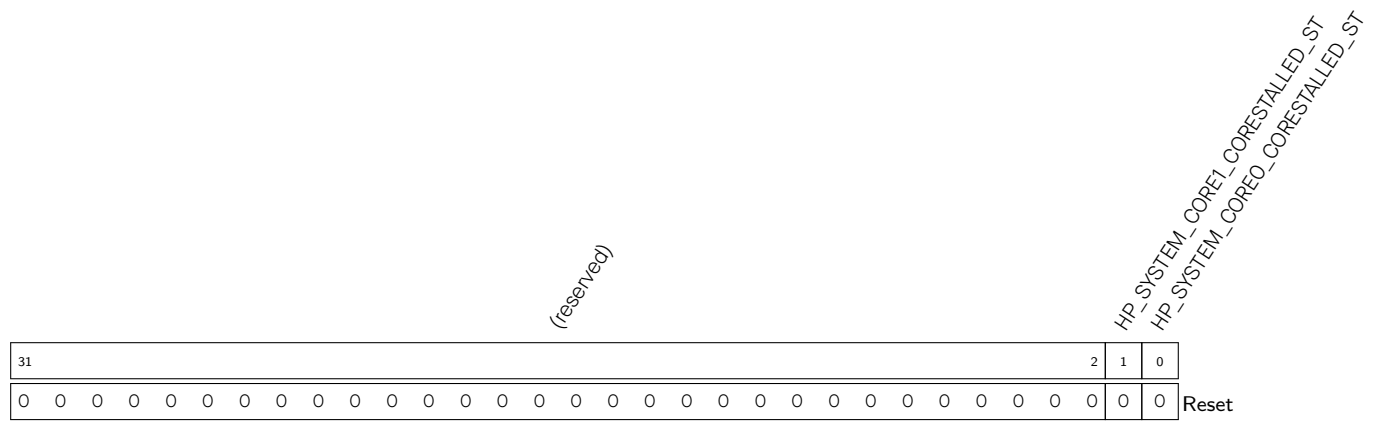


HP_SYSTEM_L2_MEM_CLK_FORCE_ON Configures whether or not to force on L2MEM clock.

- 0: No effect
- 1: Force on

(R/W)

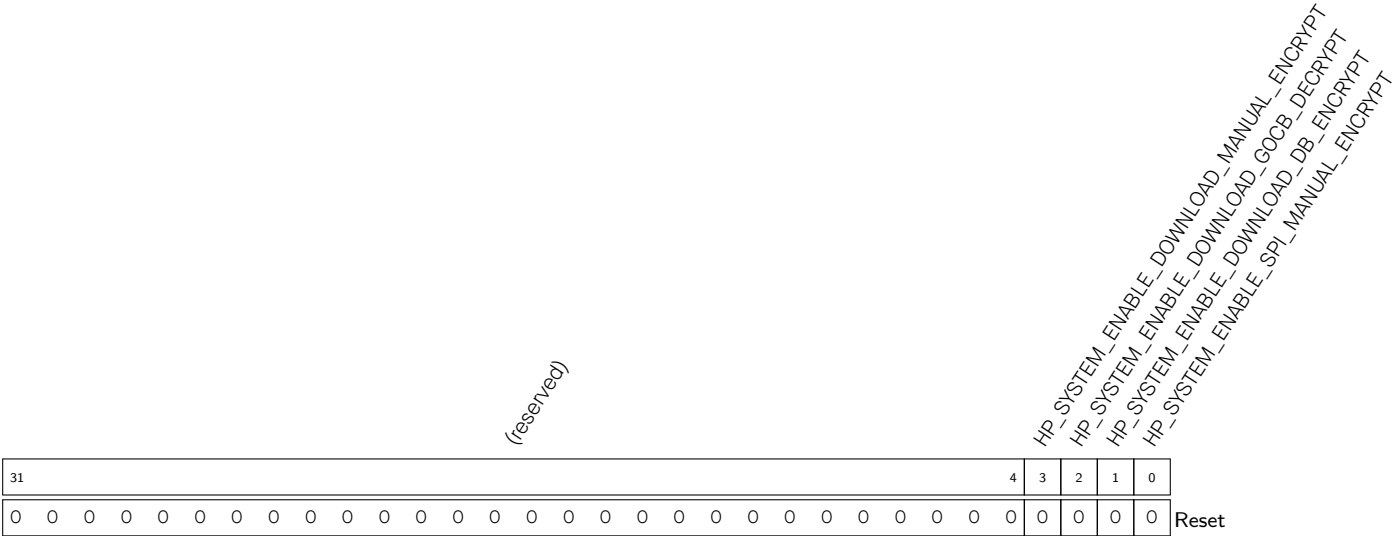
Register 20.20. HP_SYSTEM_CPU_CORESTALLED_ST_REG (0x0064)



HP_SYSTEM_CORE0_CORESTALLED_ST Represents the CPU stall status of HP CPU0. (RO)

HP_SYSTEM_CORE1_CORESTALLED_ST Represents the CPU stall status of HP CPU1. (RO)

Register 20.21. HP_SYSTEM_CRYPTO_CTRL_REG (0x0070)



HP_SYSTEM_ENABLE_SPI_MANUAL_ENCRYPT Configures whether or not to enable Manual Encryption in SPI Boot mode.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_ENABLE_DOWNLOAD_DB_ENCRYPT Configures whether or not to enable Auto Encryption in Joint Download Boot mode.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_ENABLE_DOWNLOAD_GOCB_DECRYPT Configures whether or not to enable Auto Decryption in Joint Download Boot mode.

0: Disable

1: Enable

(R/W)

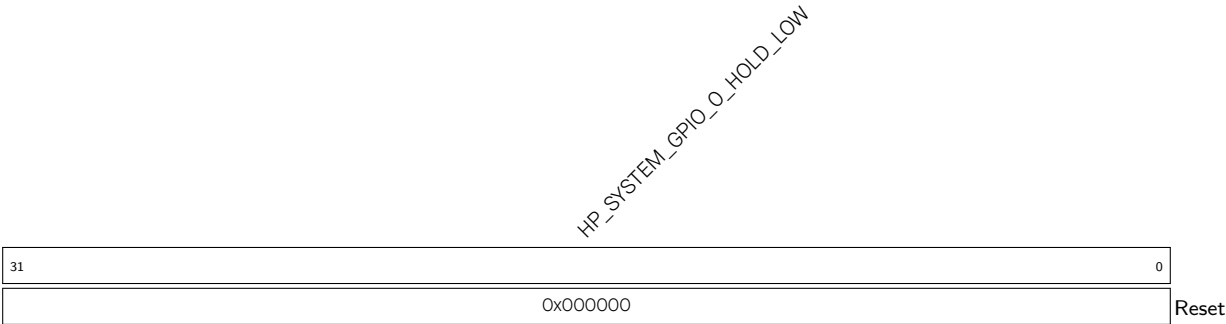
HP_SYSTEM_ENABLE_DOWNLOAD_MANUAL_ENCRYPT Configures whether or not to enable Manual Encryption in Joint Download Boot mode.

0: Disable

1: Enable

(R/W)

Register 20.22. HP_SYSTEM_GPIO_O_HOLD_CTRL0_REG (0x0074)

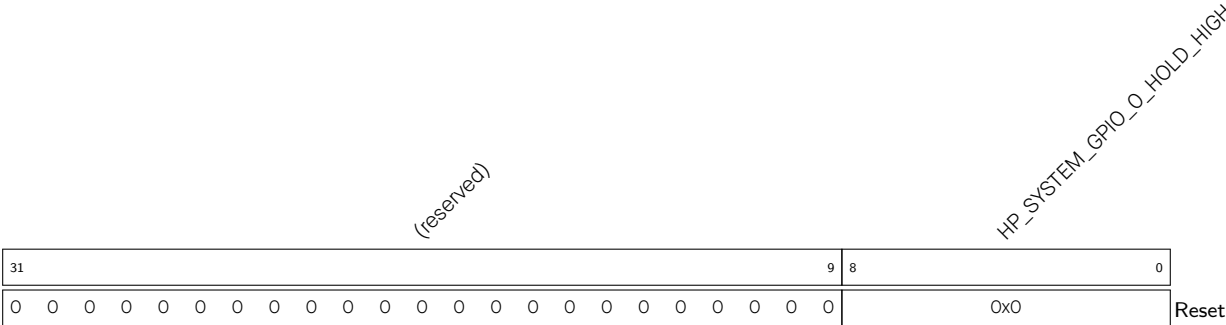


HP_SYSTEM_GPIO_O_HOLD_LOW Configures whether or not to hold HP GPIO16 ~ HP GPIO47.

- 0: No effect
- 1: Hold

(R/W)

Register 20.23. HP_SYSTEM_GPIO_O_HOLD_CTRL1_REG (0x0078)

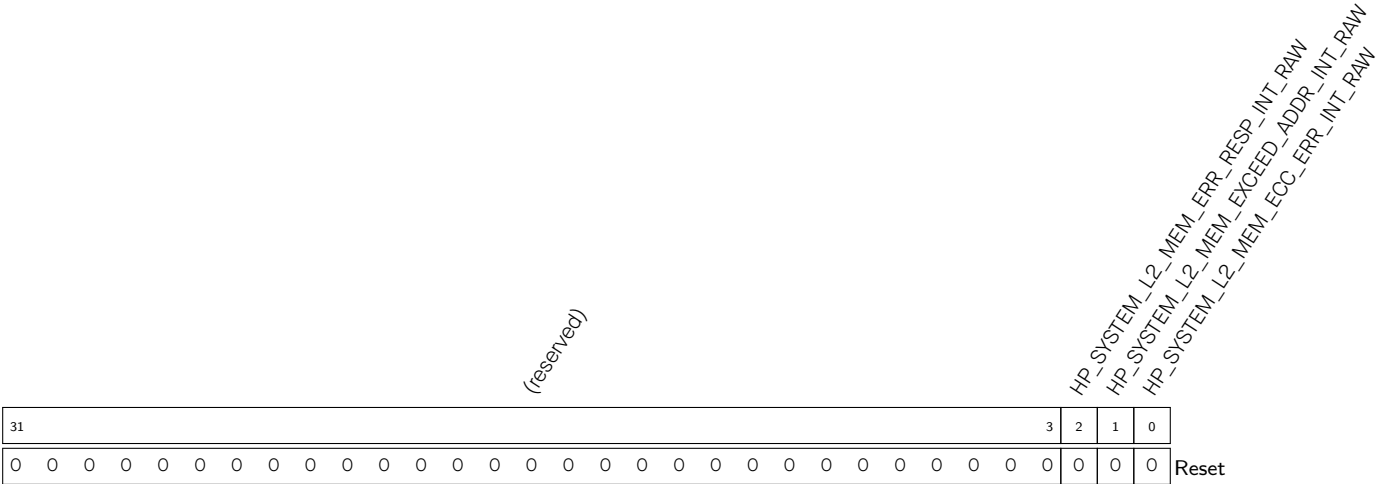


HP_SYSTEM_GPIO_O_HOLD_HIGH Configures whether or not to hold HP GPIO48 ~ HP GPIO56.

- 0: No effect
- 1: Hold

(R/W)

Register 20.24. HP_SYSTEM_L2_MEM_INT_RAW_REG (0x009C)



HP_SYSTEM_L2_MEM_ECC_ERR_INT_RAW The raw interrupt status of [L2_MEM_ECC_ERR_INT](#), triggered when a one-bit flip error is detected and corrected by ECC. (R/WTC/SS)

HP_SYSTEM_L2_MEM_EXCEED_ADDR_INT_RAW The raw interrupt status of [L2_MEM_EXCEED_ADDR_INT](#), triggered when access address exceeds limits in different modes.

Bypass mode: 0xff9fff

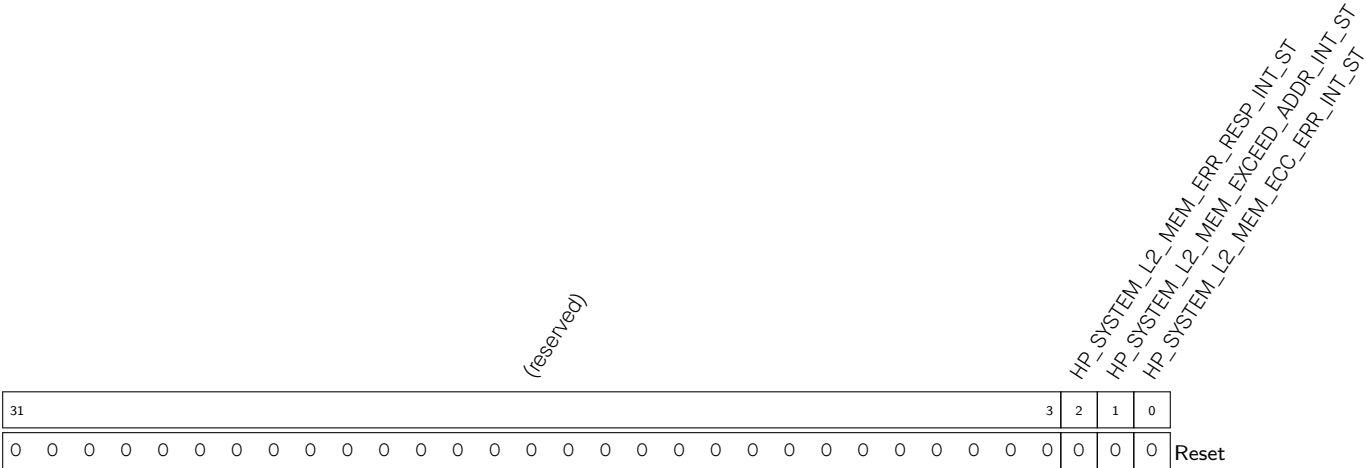
L2 cache 128 KB mode: 0xff80000

L2 cache 256 KB mode: 0xff60000

(R/WTC/SS)

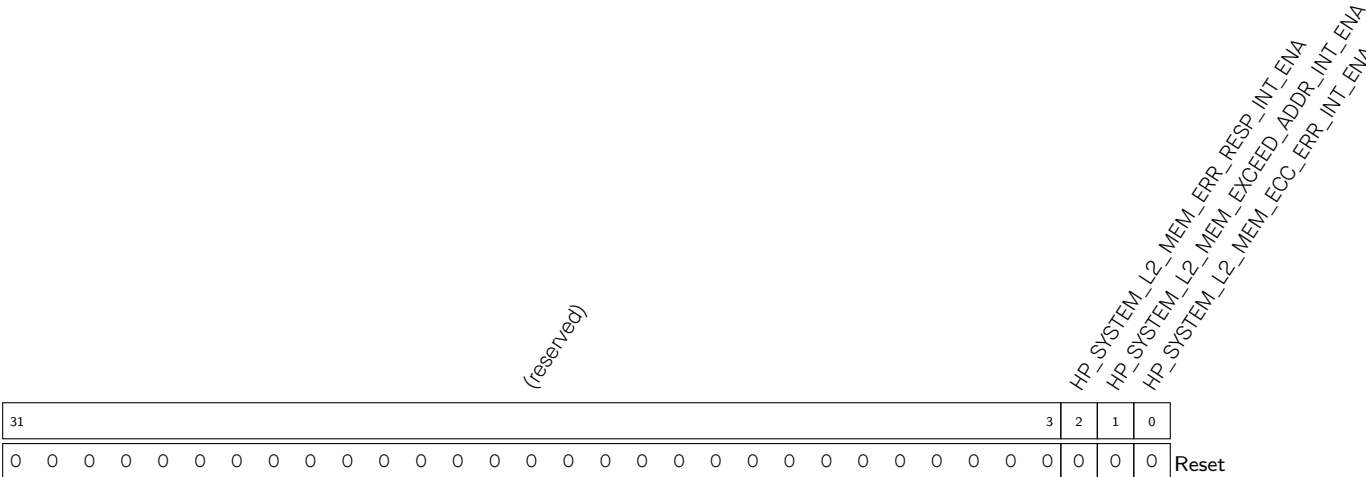
HP_SYSTEM_L2_MEM_ERR_RESP_INT_RAW The raw interrupt status of [L2_MEM_ERR_RESP_INT](#), triggered when an error response occurs. (R/WTC/SS)

Register 20.25. HP_SYSTEM_L2_MEM_INT_ST_REG (0x00A0)



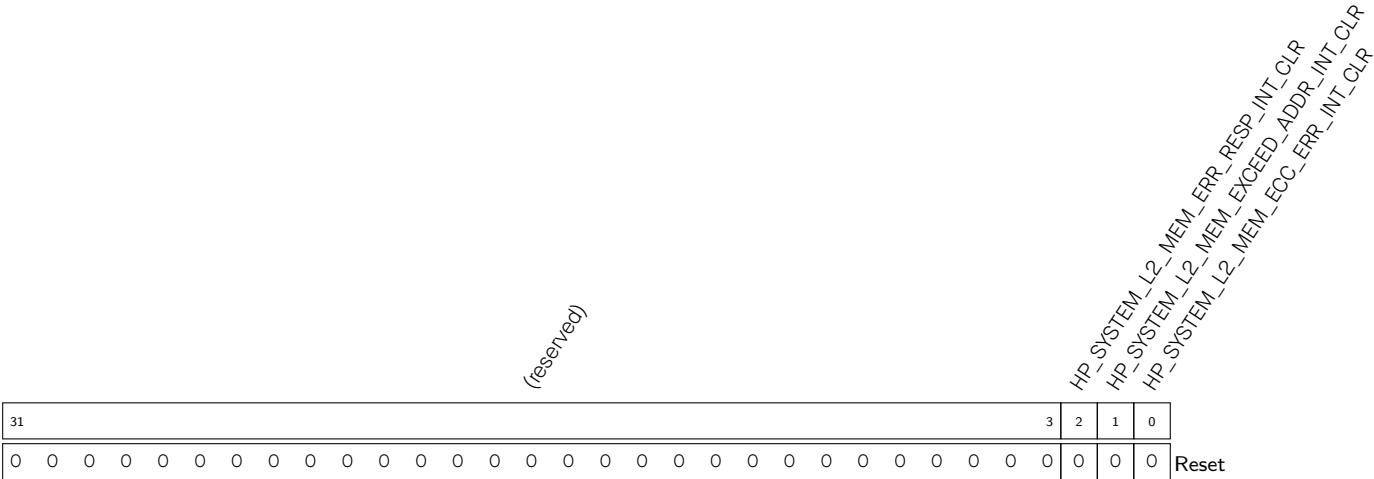
- HP_SYSTEM_L2_MEM_ECC_ERR_INT_ST The masked interrupt status of L2_MEM_ECC_ERR_INT. (RO)
- HP_SYSTEM_L2_MEM_EXCEED_ADDR_INT_ST The masked interrupt status of L2_MEM_EXCEED_ADDR_INT. (RO)
- HP_SYSTEM_L2_MEM_ERR_RESP_INT_ST The masked interrupt status of L2_MEM_ERR_RESP_INT. (RO)

Register 20.26. HP_SYSTEM_L2_MEM_INT_ENA_REG (0x00A4)



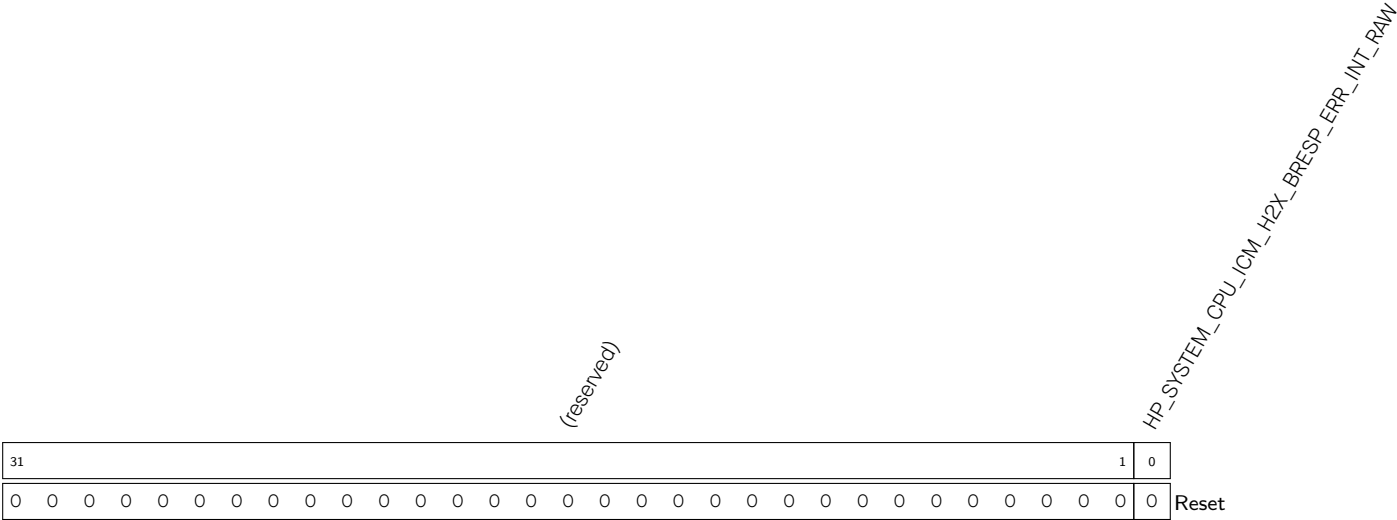
- HP_SYSTEM_L2_MEM_ECC_ERR_INT_ENA Write 1 to enable L2_MEM_ECC_ERR_INT interrupt. (R/W)
- HP_SYSTEM_L2_MEM_EXCEED_ADDR_INT_ENA Write 1 to enable L2_MEM_EXCEED_ADDR_INT interrupt. (R/W)
- HP_SYSTEM_L2_MEM_ERR_RESP_INT_ENA Write 1 to enable L2_MEM_ERR_RESP_INT interrupt. (R/W)

Register 20.27. HP_SYSTEM_L2_MEM_INT_CLR_REG (0x00A8)



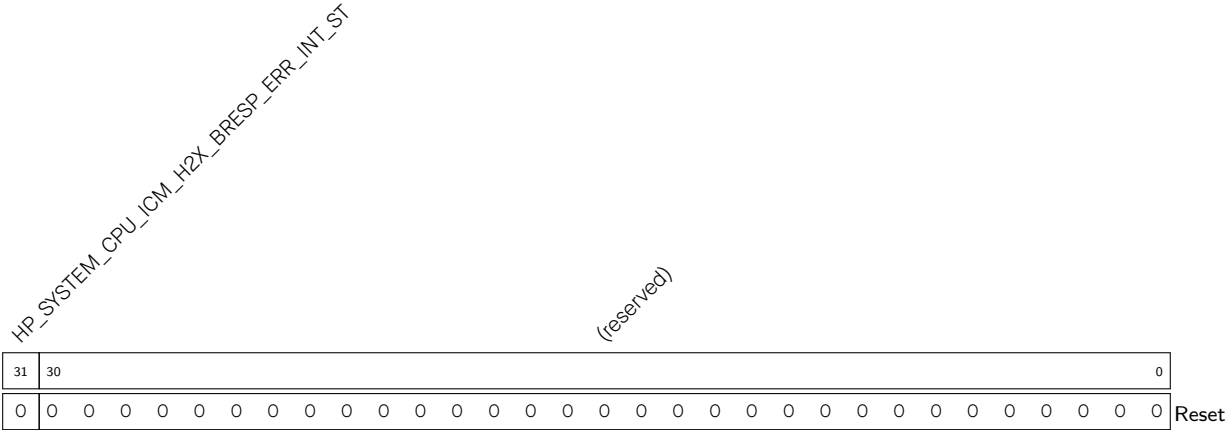
- HP_SYSTEM_L2_MEM_ECC_ERR_INT_CLR** Write 1 to clear [L2_MEM_ECC_ERR_INT](#) interrupt. (WT)
- HP_SYSTEM_L2_MEM_EXCEED_ADDR_INT_CLR** Write 1 to clear [L2_MEM_EXCEED_ADDR_INT](#) interrupt. (WT)
- HP_SYSTEM_L2_MEM_ERR_RESP_INT_CLR** Write 1 to clear [L2_MEM_ERR_RESP_INT](#) interrupt. (WT)

Register 20.28. HP_SYSTEM_AHB2AXI_BRESP_ERR_INT_RAW_REG (0x0188)



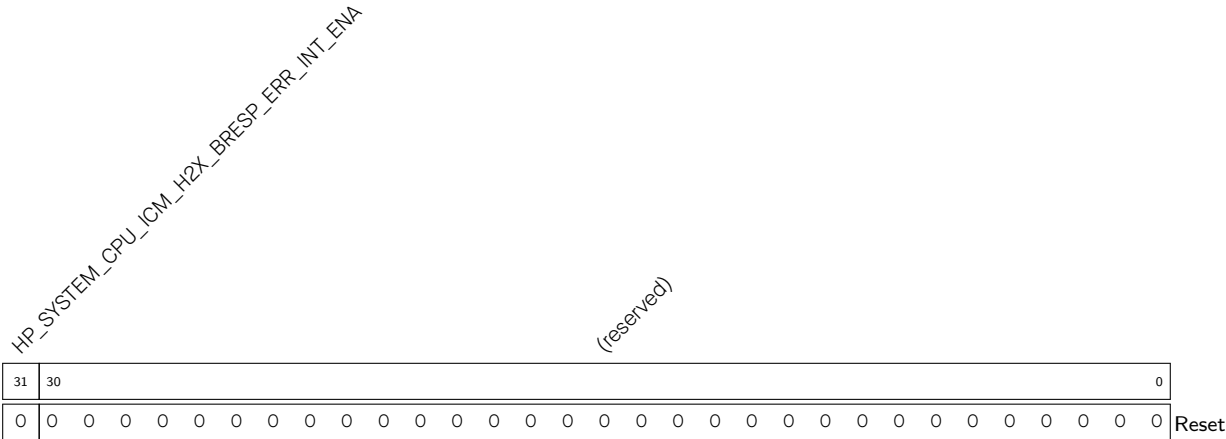
- HP_SYSTEM_CPU_ICM_H2X_BRESP_ERR_INT_RAW** The raw interrupt status of [CPU_ICM_H2X_BRESP_ERR_INT](#), triggered when a BRESP error occurs in Post Write mode in AHB2AXI. (R/WTC/SS)

Register 20.29. HP_SYSTEM_AHB2AXI_BRESP_ERR_INT_ST_REG (0x018C)



HP_SYSTEM_CPU_ICM_H2X_BRESP_ERR_INT_ST The masked interrupt status of CPU_ICM_H2X_BRESP_ERR_INT. (RO)

Register 20.30. HP_SYSTEM_AHB2AXI_BRESP_ERR_INT_ENA_REG (0x0190)



HP_SYSTEM_CPU_ICM_H2X_BRESP_ERR_INT_ENA Write 1 to enable CPU_ICM_H2X_BRESP_ERR_INT interrupt. (R/W)

HP_SYSTEM_CPU_ICM_H2X_BRESP_ERR_INT_CLR

(reserved)

Reset

CPU_ICM_H2X_BRESP_ERR_INT interrupt. (WT)

Register 20.32. HP_SYSTEM_L2_MEM_L2_RAM_ECC_REG (0x00AC)

(reserved)																												HP_SYSTEM_L2_RAM_UNIT5_ECC_EN HP_SYSTEM_L2_RAM_UNIT4_ECC_EN HP_SYSTEM_L2_RAM_UNIT3_ECC_EN HP_SYSTEM_L2_RAM_UNIT2_ECC_EN HP_SYSTEM_L2_RAM_UNIT1_ECC_EN HP_SYSTEM_L2_RAM_UNIT0_ECC_EN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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Reset

HP_SYSTEM_L2_RAM_UNIT0_ECC_EN Configures whether or not to enable the ECC check for L2MEM unit0.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L2_RAM_UNIT1_ECC_EN Configures whether or not to enable the ECC check for L2MEM unit1.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L2_RAM_UNIT2_ECC_EN Configures whether or not to enable the ECC check for L2MEM unit2.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L2_RAM_UNIT3_ECC_EN Configures whether or not to enable the ECC check for L2MEM unit3.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L2_RAM_UNIT4_ECC_EN Configures whether or not to enable the ECC check for L2MEM unit4.

0: Disable

1: Enable

(R/W)

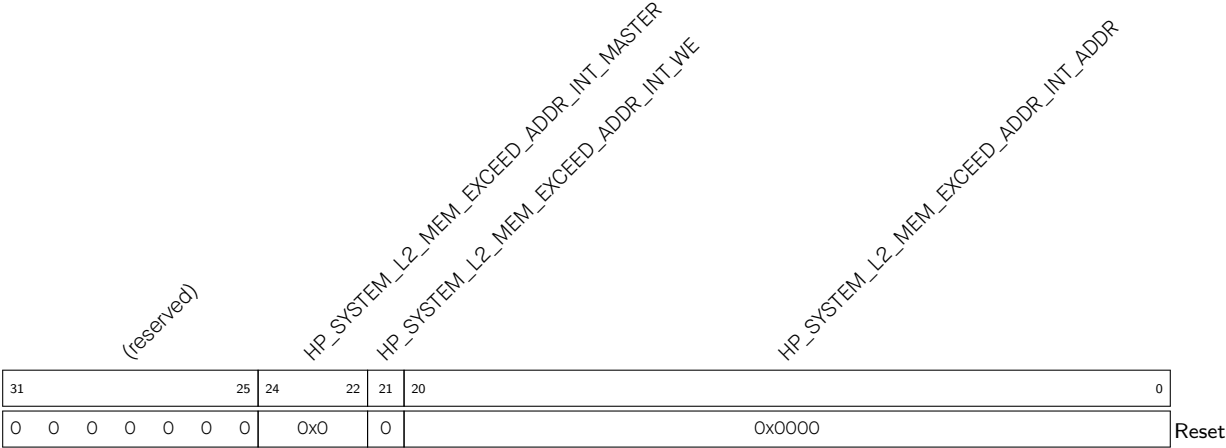
HP_SYSTEM_L2_RAM_UNIT5_ECC_EN Configures whether or not to enable the ECC check for L2MEM unit5.

0: Disable

1: Enable

(R/W)

Register 20.33. HP_SYSTEM_L2_MEM_INT_RECORD0_REG (0x00B0)



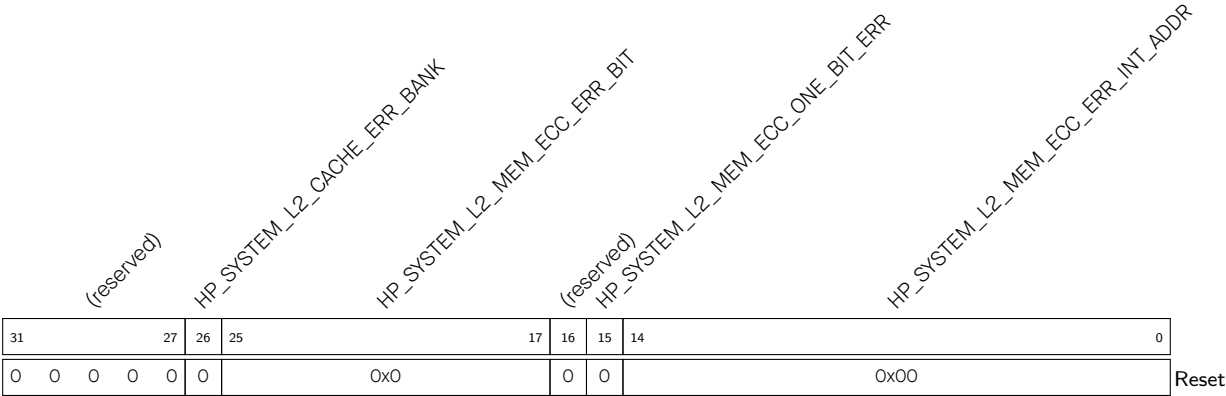
HP_SYSTEM_L2_MEM_EXCEED_ADDR_INT_ADDR Records the address when [L2_MEM_EXCEED_ADDR_INT](#) occurs. (RO)

HP_SYSTEM_L2_MEM_EXCEED_ADDR_INT_WE Records the operation (write/read) when [L2_MEM_EXCEED_ADDR_INT](#) occurs. (RO)

HP_SYSTEM_L2_MEM_EXCEED_ADDR_INT_MASTER Records source that triggered [L2_MEM_EXCEED_ADDR_INT](#).

- 1: cachebus0
 - 2: cachebus1
 - 3: dma_axi
 - 6: ahb
- (RO)

Register 20.34. HP_SYSTEM_L2_MEM_INT_RECORD1_REG (0x00B4)



- HP_SYSTEM_L2_MEM_ECC_ERR_INT_ADDR

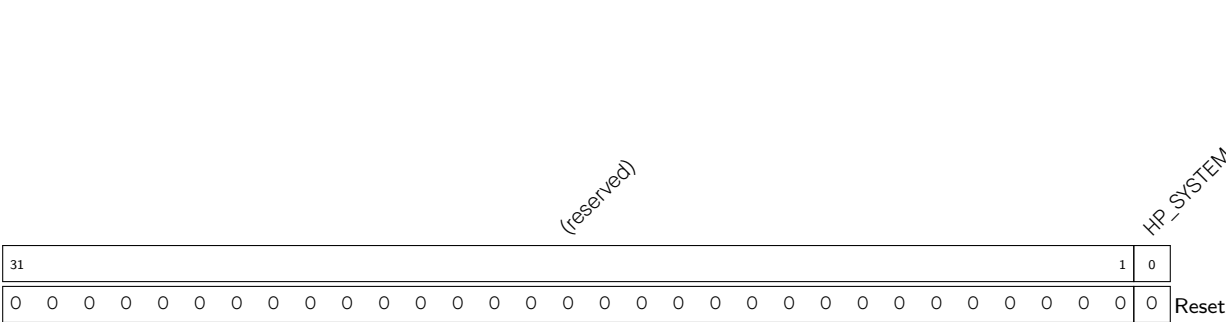
Records the address when `L2_MEM_ECC_ERR_INT` occurs. (RO)
- HP_SYSTEM_L2_MEM_ECC_ONE_BIT_ERR

Records the error when `L2_MEM_ECC_ERR_INT` occurs. (RO)
- HP_SYSTEM_L2_MEM_ECC_ERR_BIT

Records the flipped bit that triggered `L2_MEM_ECC_ERR_INT`. (RO)
- HP_SYSTEM_L2_CACHE_ERR_BANK

Records the bank number of L2 cache ECC error. (RO)

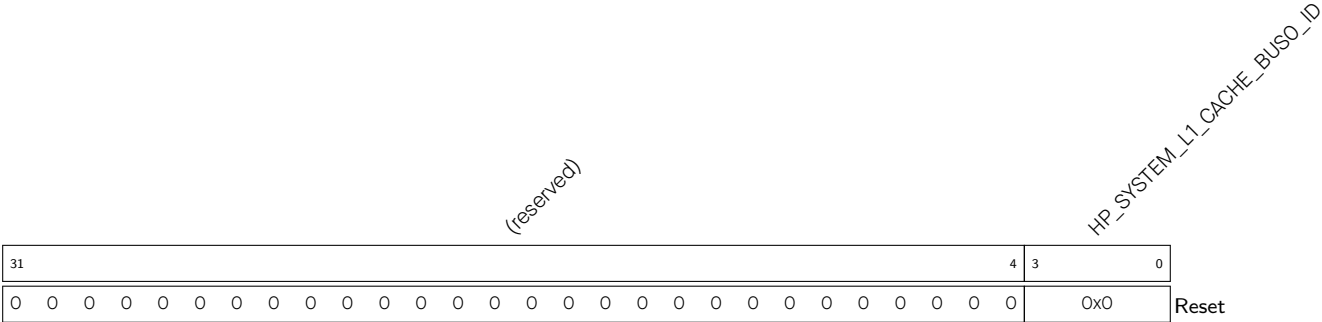
Register 20.35. HP_SYSTEM_L2_MEM_L2_CACHE_ECC_REG (0x00C4)



- HP_SYSTEM_L2_CACHE_ECC_EN

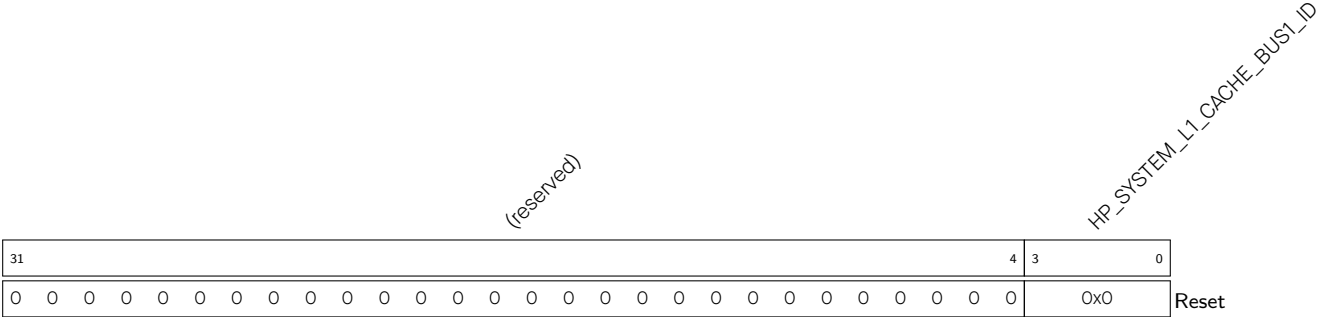
Configures whether or not to enable the ECC check for L2 cache.
0: Disable
1: Enable
(R/W)

Register 20.36. HP_SYSTEM_HP_L1CACHE_BUS0_ID_REG (0x00C8)



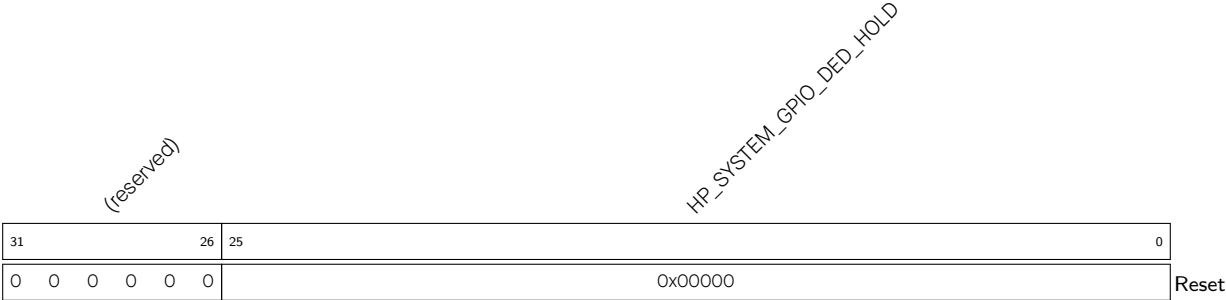
HP_SYSTEM_L1_CACHE_BUS0_ID Configures the L1 cache bus0 UID. (R/W)

Register 20.37. HP_SYSTEM_HP_L1CACHE_BUS1_ID_REG (0x00CC)



HP_SYSTEM_L1_CACHE_BUS1_ID Configures the L1 cache bus1 UID. (R/W)

Register 20.38. HP_SYSTEM_GPIO_DED_HOLD_CTRL_REG (0x00F0)



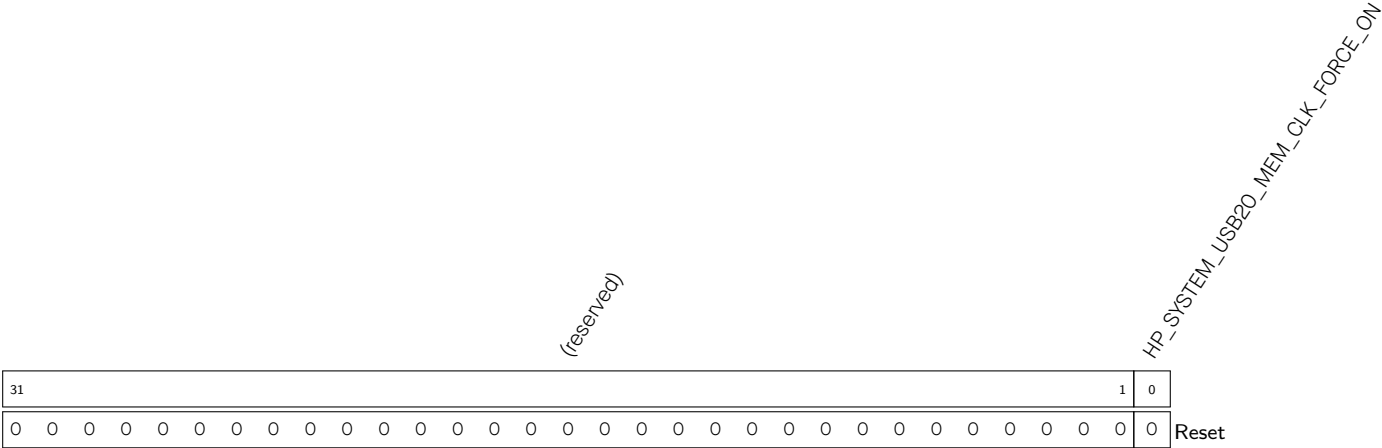
HP_SYSTEM_GPIO_DED_HOLD Configures whether or not to hold the pin status.

Lower 6 bits: flash pins

Higher 20 bits: 16 line PSRAM pins

(R/W)

Register 20.39. HP_SYSTEM_HP_USB200TG_MEM_CTRL_REG (0x00F8)

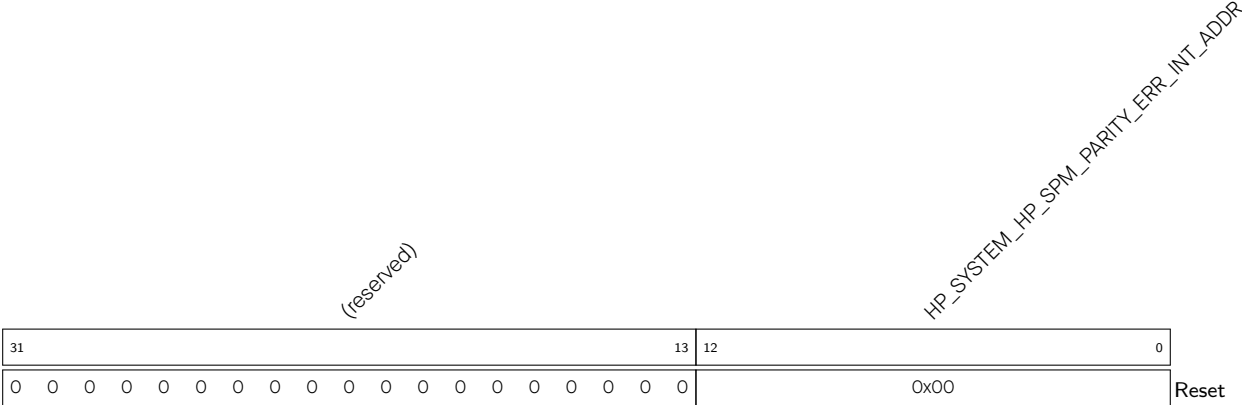


HP_SYSTEM_USB20_MEM_CLK_FORCE_ON Configures whether or not to force on the memory clock in USB OTGHS.

- 0: Disable
- 1: Enable

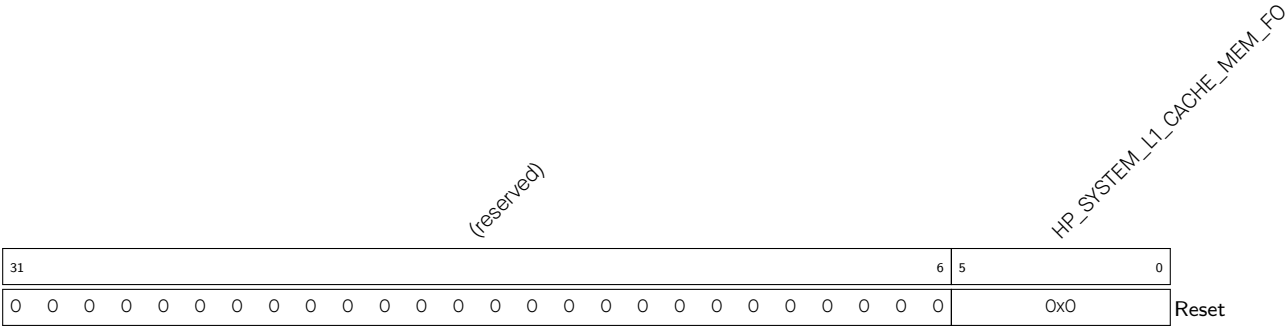
(R/W)

Register 20.40. HP_SYSTEM_HP_SPM_PARITY_INT_RECORD_REG (0x010C)



HP_SYSTEM_HP_SPM_PARITY_ERR_INT_ADDR Records the address when [HP_SPM_PARITY_ERR_INT](#) occurs. (RO)

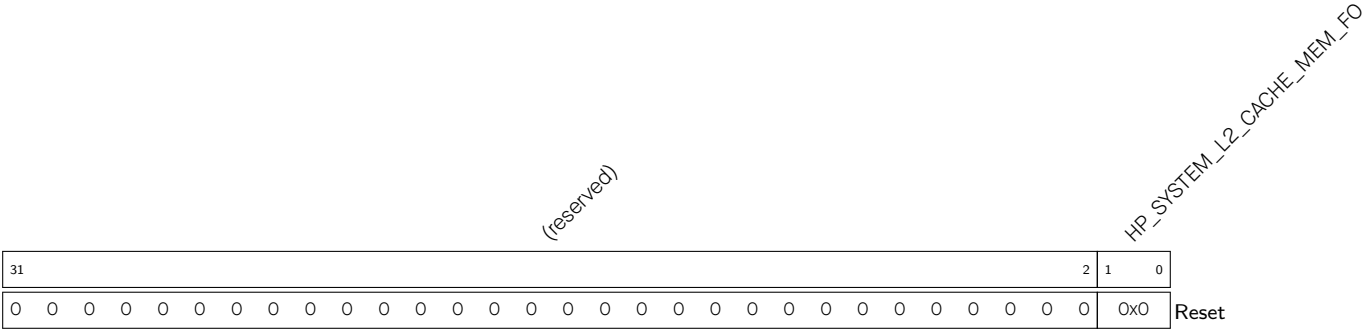
Register 20.41. HP_SYSTEM_HP_L1_CACHE_PWR_CTRL_REG (0x0110)



HP_SYSTEM_L1_CACHE_MEM_FO Configures whether or not to force L1 cache on.

0: Disable
1: Enable
(R/W)

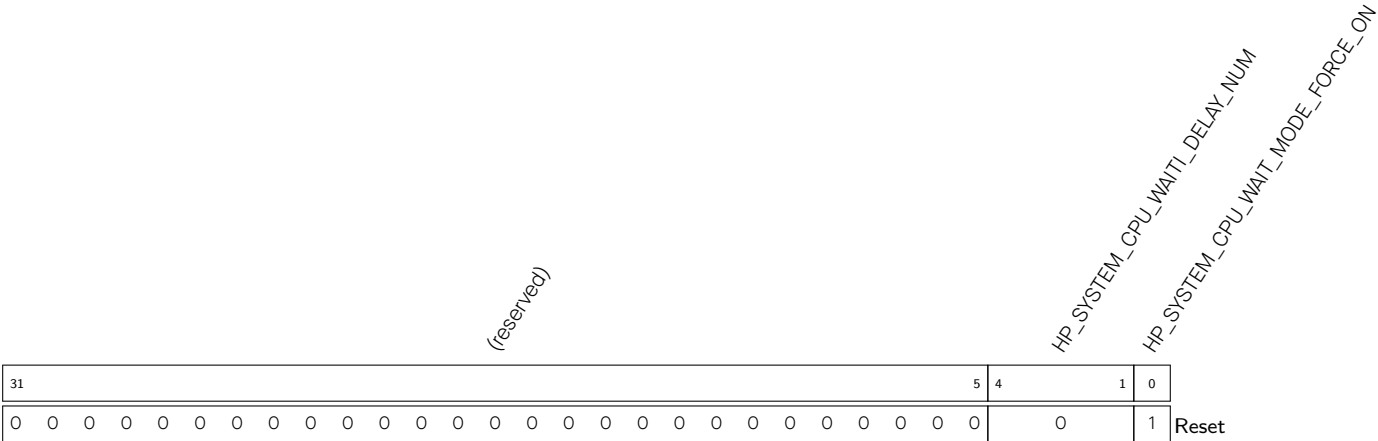
Register 20.42. HP_SYSTEM_HP_L2_CACHE_PWR_CTRL_REG (0x0114)



HP_SYSTEM_L2_CACHE_MEM_FO Configures whether or not to force L2 cache on.

0: Disable
1: Enable
(R/W)

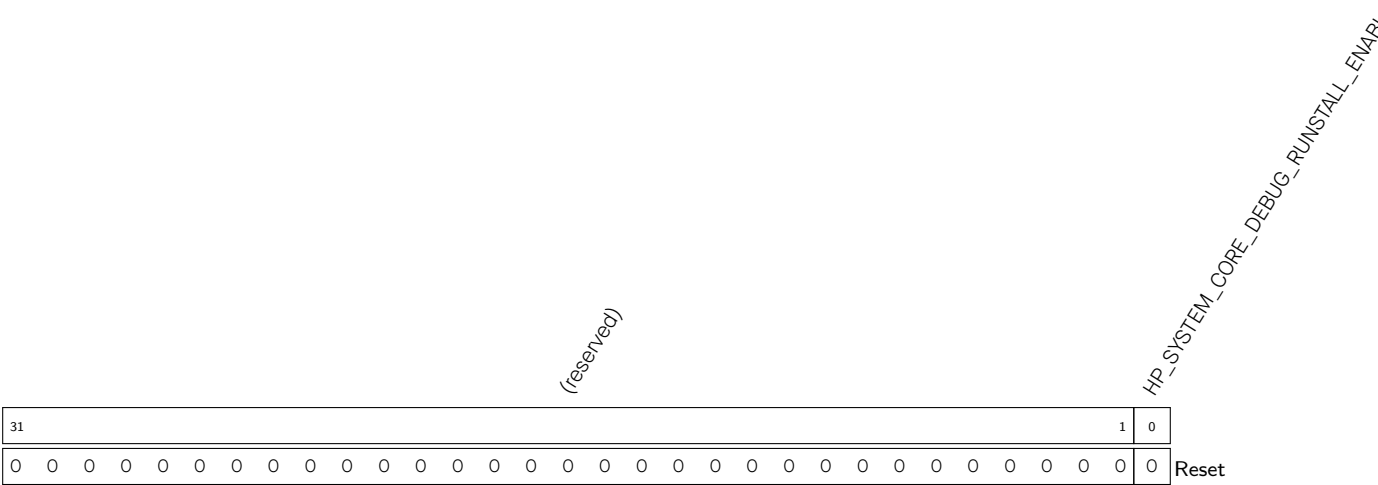
Register 20.43. HP_SYSTEM_HP_CPU_WAITI_CONF_REG (0x0118)



HP_SYSTEM_CPU_WAIT_MODE_FORCE_ON Configures whether or not to force on cpu_waiti_clk.
(R/W)

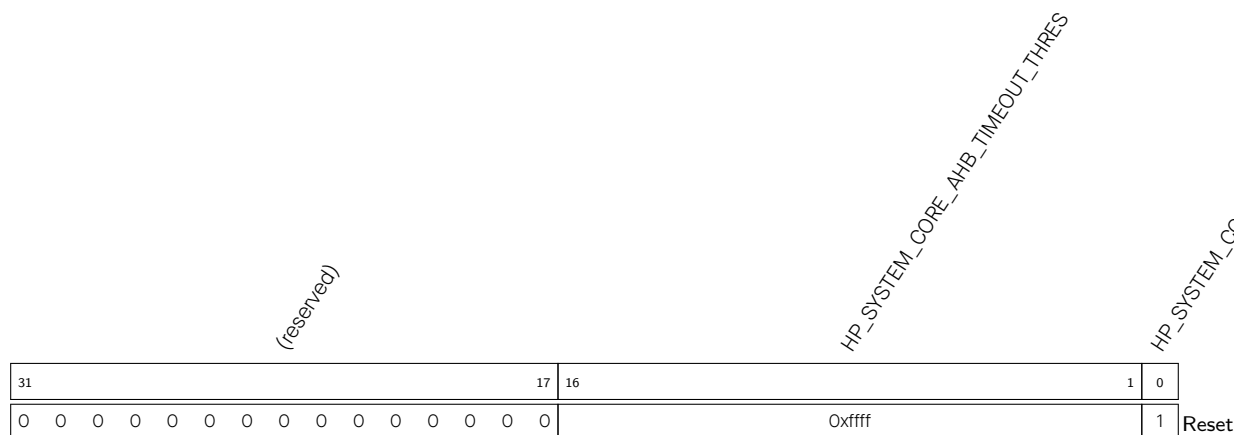
HP_SYSTEM_CPU_WAITI_DELAY_NUM Configures the delay cycles when CPU enters wait mode;
after the configured delay, cpu_waiti_clk turns off. (R/W)

Register 20.44. HP_SYSTEM_CORE_DEBUG_RUNSTALL_CONF_REG (0x011C)



HP_SYSTEM_CORE_DEBUG_RUNSTALL_ENABLE Configures whether or not to enable debug run-
stall feature between HP CPU and LP CPU.
0: Disable
1: Enable
(R/W)

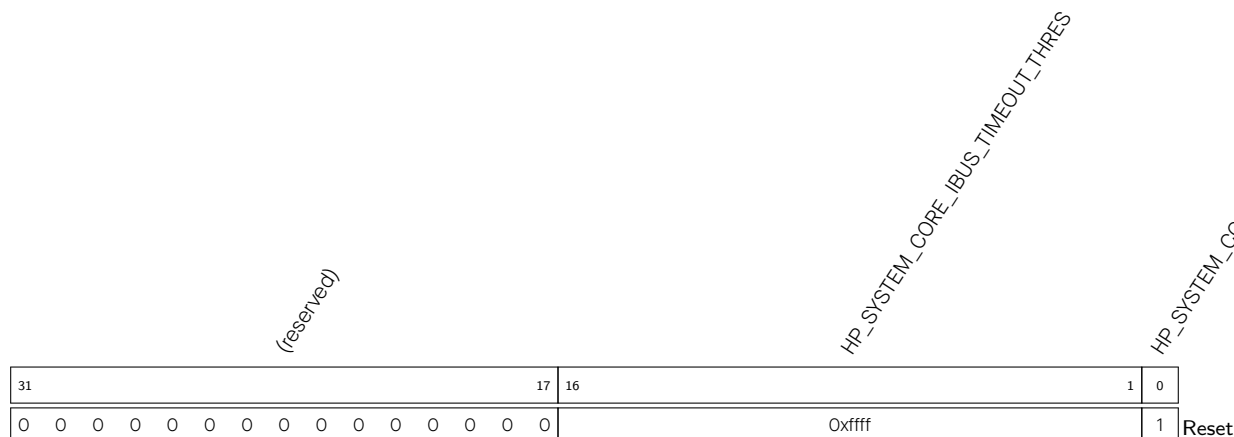
Register 20.45. HP_SYSTEM_HP_CORE_AHB_TIMEOUT_REG (0x0120)



HP_SYSTEM_CORE_AHB_TIMEOUT_EN Configures whether or not to enable timeout protection on HP CPU0/1 AHB bus. (R/W)

HP_SYSTEM_CORE_AHB_TIMEOUT_THRES Configures HP CPU0/1 AHB bus timeout threshold. (R/W)

Register 20.46. HP_SYSTEM_HP_CORE_IBUS_TIMEOUT_REG (0x0124)



HP_SYSTEM_CORE_IBUS_TIMEOUT_EN Configures whether or not to enable timeout protection on HP CPU0/1 IBUS. (R/W)

HP_SYSTEM_CORE_IBUS_TIMEOUT_THRES Configures HP CPU0/1 IBUS timeout threshold. (R/W)

Register 20.47. HP_SYSTEM_HP_CORE_DBUS_TIMEOUT_REG (0x0128)

(reserved)																HP_SYSTEM_CORE_DBUS_TIMEOUT_THRES																HP_SYSTEM_CORE_DBUS_TIMEOUT_EN																	
31																17																16																1	0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0xffff																1	Reset																

HP_SYSTEM_CORE_DBUS_TIMEOUT_EN Configures whether or not to enable timeout protection on HP CPU0/1 DBUS. (R/W)

HP_SYSTEM_CORE_DBUS_TIMEOUT_THRES Configures HP CPU0/1 DBUS timeout threshold.
(R/W)

Register 20.48. HP_SYSTEM_HP_ICM_CPU_H2X_CFG_REG (0x0138)

31	3	2	1	0	
0 0	0	1	1	Reset	

(reserved)

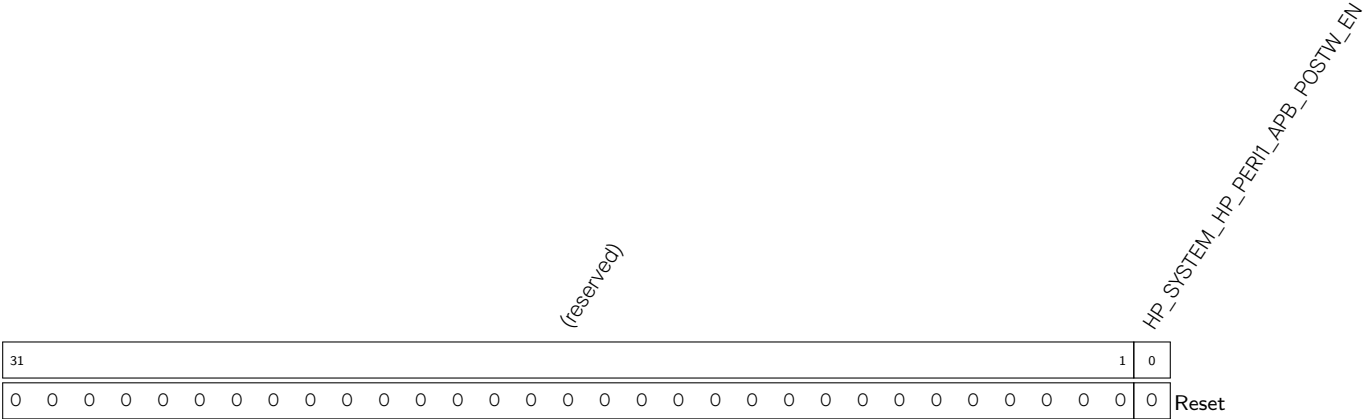
HP_SYSTEM_CPU_IOM_H2X_BRIDGE_BUSY
 HP_SYSTEM_CPU_IOM_H2X_CUT_THROUGH_EN
 HP_SYSTEM_CPU_IOM_H2X_POST_WR_EN

HP_SYSTEM_CPU_ICM_H2X_POST_WR_EN Configures whether or not to enable Post Write mode in AHB2AXI bridge for improved write performance. (R/W)

HP_SYSTEM_CPU_ICM_H2X_CUT_THROUGH_EN	Configures whether or not to enable CUT Through mode in AHB2AXI bridge for improved write performance. (R/W)
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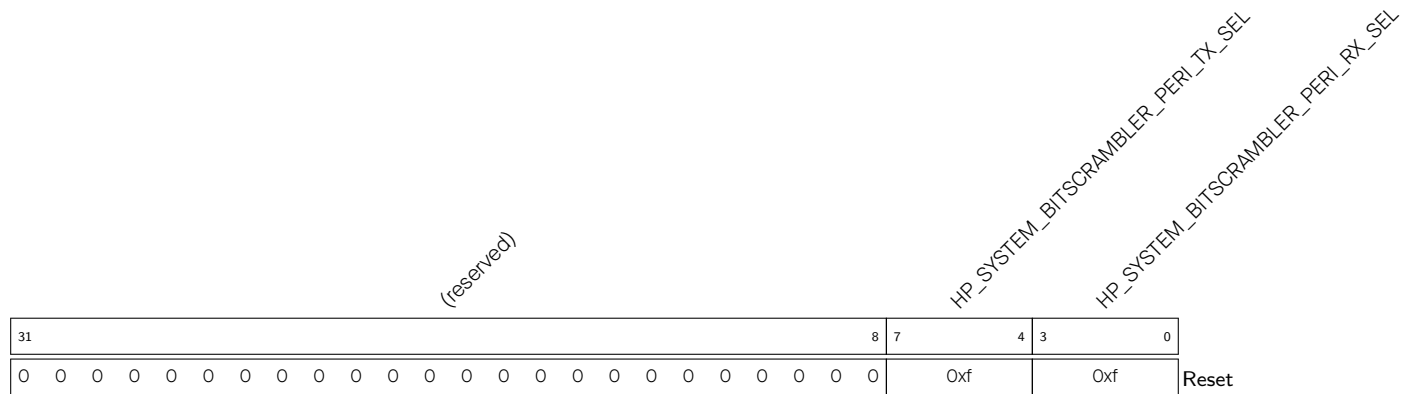
HP_SYSTEM_CPU_ICM_H2X_BRIDGE_BUSY Indicates when AHB2AXI bridge is busy. (RO)

Register 20.49. HP_SYSTEM_PERI1_APB_POSTW_EN_REG (0x013C)



HP_SYSTEM_HP_PERI1_APB_POSTW_EN Configures whether or not to enable Post Write mode for peripherals’ APB2APB bridge, which accelerates write operations but requires time to update register values. (R/W)

Register 20.50. HP_SYSTEM_BITSCRAMBLER_PERI_SEL_REG (0x0140)



HP_SYSTEM_BITSCRAMBLER_PERI_RX_SEL Configures to select the DMA-capable peripheral for BitScrambler's Rx channel.

- 0: LCD_CAM
1: GPSP12
2: GPSP13
3: PARL_IO
4: AES
5: SHA
6: ADC
7: I2S0
8: I2S1
9: I2S2
10: I3C_MST
11: UHCIO
12: RMT
others: NONE
(R/W)

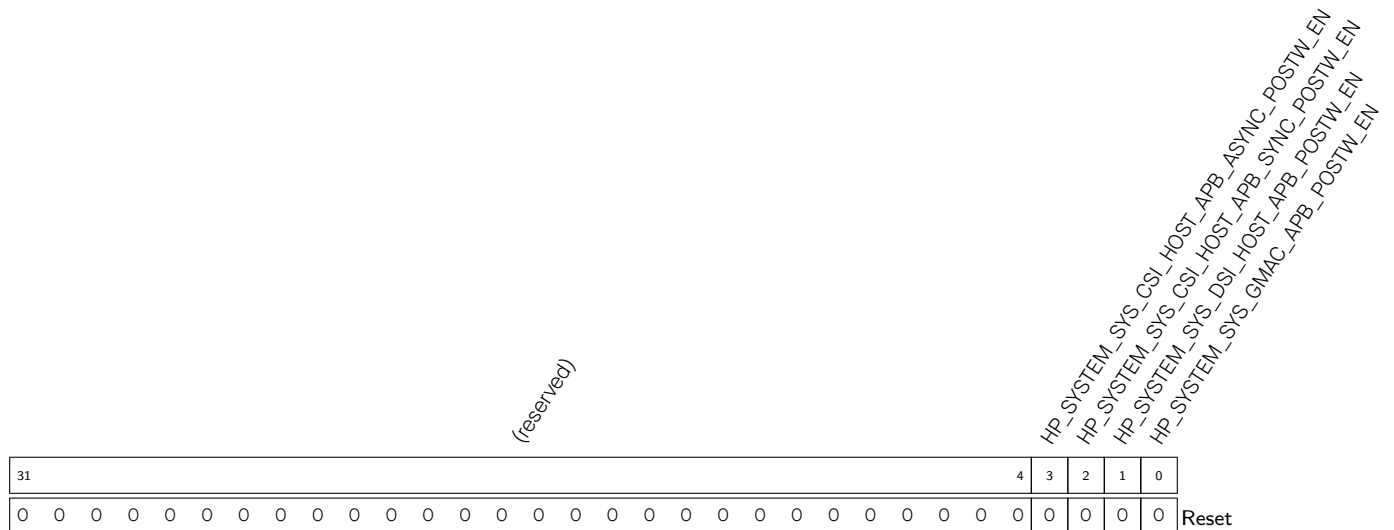
(R/W)

HP_SYSTEM_BITSCRAMBLER_PERI_TX_SEL Configures to select the DMA-capable peripheral for BitScrambler's Tx channel.

- 0: LCD_CAM
1: GPSPi2
2: GPSPi3
3: PARL_IO
4: AES
5: SHA
6: ADC
7: I2S0
8: I2S1
9: I2S2
10: I3C_MST
11: UHCIO
12: RMT
others: NONE
(R/W)

(R/W)

Register 20.51. HP_SYSTEM_APB_SYNC_POSTW_EN_REG (0x0144)



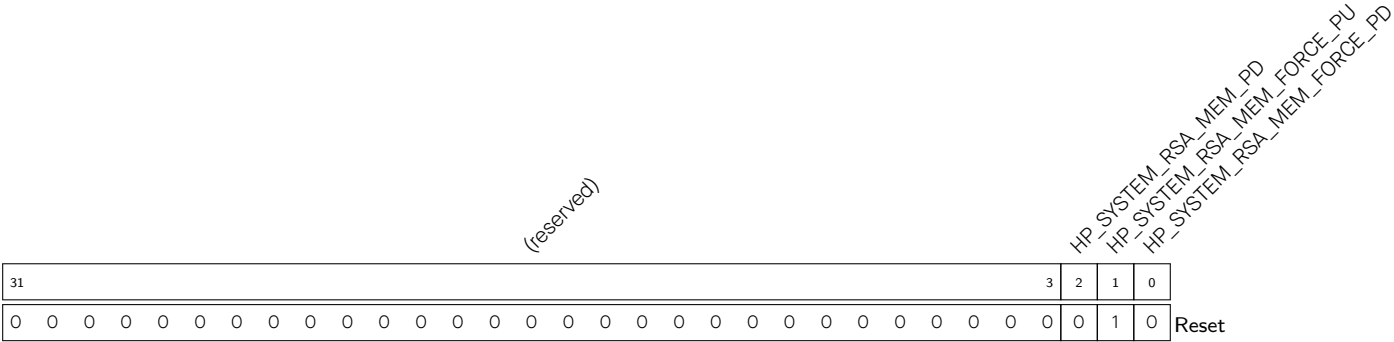
HP_SYSTEM_SYS_GMAC_APB_POSTW_EN Configures whether or not to enable Post Write mode for synchronous APB2APB bridge in GMAC, which accelerates write operations but requires time to update register values. (R/W)

HP_SYSTEM_SYS_DSI_HOST_APB_POSTW_EN Configures whether or not to enable Post Write mode for synchronous APB2APB bridge in DSI Host, which accelerates write operations but requires time to update register values. (R/W)

HP_SYSTEM_SYS_CSI_HOST_APB_SYNC_POSTW_EN Configures whether or not to enable Post Write mode for synchronous APB2APB bridge in CSI Host, which accelerates write operations but requires time to update register values. (R/W)

HP_SYSTEM_SYS_CSI_HOST_APB_ASYNC_POSTW_EN Configures whether or not to enable Post Write mode for asynchronous APB2APB bridge in CSI Host, which accelerates write operations but requires time to update register values. (R/W)

Register 20.52. HP_SYSTEM_HP_RSA_PD_CTRL_REG (0x01D0)



HP_SYSTEM_RSA_MEM_FORCE_PD Configures whether or not to force power down RSA internal memory. This setting has a lower priority than [HP_SYSTEM_RSA_MEM_FORCE_PU](#).

0: No effect
1: Power down
(R/W)

HP_SYSTEM_RSA_MEM_FORCE_PU Configures whether or not to force power up RSA internal memory.

0: No effect
1: Power up
(R/W)

HP_SYSTEM_RSA_MEM_PD Configures whether or not to power down RSA internal memory.

0: No effect
1: Power down
(R/W)

Register 20.53. HP_SYSTEM_HP_ECC_PD_CTRL_REG (0x01D4)

(reserved)																															HP_SYSTEM_ECC_MEM_PD HP_SYSTEM_ECC_MEM_FORCE_PU HP_SYSTEM_ECC_MEM_FORCE_PD				
31																															3	2	1	0	
0 0																															0	1	0	Reset	

HP_SYSTEM_ECC_MEM_FORCE_PD Configures whether or not to force power down ECC internal memory. This setting has a lower priority than **HP_SYSTEM_ECC_MEM_FORCE_PU**.

0: No effect

1: Power down

(R/W)

HP_SYSTEM_ECC_MEM_FORCE_PU	Configures whether or not to force power up ECC internal memory.
-----------------------------------	--

0: No effect

1: Power up

(R/W)

HP_SYSTEM_ECC_MEM_PD Configures whether or not to power down ECC internal memory.

0: No effect

1: Power down

(R/W)

Register 20.54. HP_SYSTEM_HP_RNG_CFG_REG (0x01D8)

HP_SYSTEM_RNG_SAMPLE_CNT										HP_SYSTEM_RNG_CHAIN_CLK_DIV_NUM										(reserved)										HP_SYSTEM_RNG_SAMPLE_ENABLE									
31	24								23	16						15													1	0									
0x0									0x0							0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset				

HP_SYSTEM_RNG_SAMPLE_ENABLE Configures whether or not to enable the RNG sample chain.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_RNG_CHAIN_CLK_DIV_NUM Configures the chain clock division number for debugging. (R/W)

HP_SYSTEM_RNG_SAMPLE_CNT Represents the RNG sample count for debugging. (RO)

Register 20.55. HP_SYSTEM_HP_UART_PD_CTRL_REG (0x01DC)

(reserved)																															HP_SYSTEM_UART_MEM_FORCE_PU HP_SYSTEM_UART_MEM_FORCE_PD		
31																															2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	Reset

HP_SYSTEM_UART_MEM_FORCE_PD Configures whether or not to force power down HP UART internal memory.

0: No effect

1: Power down

(R/W)

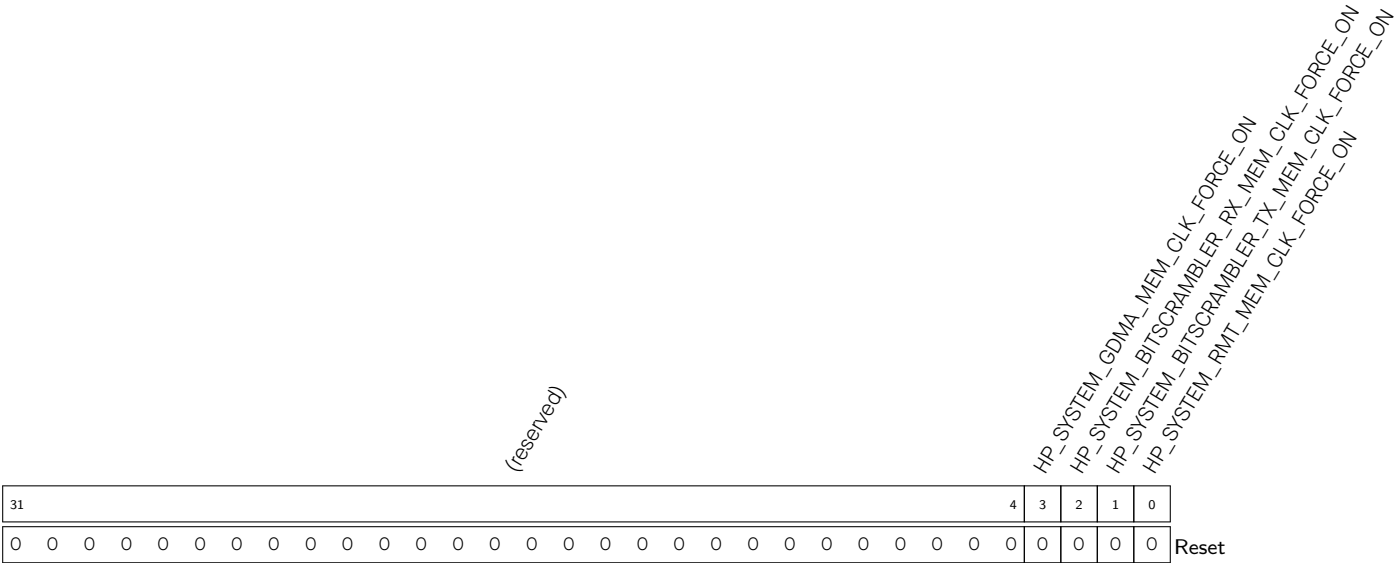
HP_SYSTEM_UART_MEM_FORCE_PU Configures whether or not to force power up HP UART internal memory.

0: No effect

1: Power up

(R/W)

Register 20.56. HP_SYSTEM_HP_PERI_MEM_CLK_FORCE_ON_REG (0x01E0)



HP_SYSTEM_RMT_MEM_CLK_FORCE_ON Configures whether or not to force enable the clock gating of the memory clock in RMT.

0: No effect

1: Force on

(R/W)

HP_SYSTEM_BITSCRAMBLER_TX_MEM_CLK_FORCE_ON Configures whether or not to force enable the clock gating of the TX memory clock in BitScrambler.

0: No effect

1: Force on

(R/W)

HP_SYSTEM_BITSCRAMBLER_RX_MEM_CLK_FORCE_ON Configures whether or not to force enable the clock gating of the RX memory clock in BitScrambler.

0: No effect

1: Force on

(R/W)

HP_SYSTEM_GDMA_MEM_CLK_FORCE_ON Configures whether or not to force enable the clock gating of the memory clock in GDMA.

0: No effect

1: Force on

(R/W)

HP_SYSTEM_SYS_DEBUG_CH_NUM

(reserved)

Reset

Register 20.58. HP_SYSTEM_GMAC_CTRL0_REG (0x014C)

(reserved)																HP_SYSTEM_SYS_GMAC_RST_CLK_RX_N		
																HP_SYSTEM_SYS_GMAC_RST_CLK_TX_N		
																HP_SYSTEM_SYS_GMAC_MEM_CLK_FORCE_ON		
																HP_SYSTEM_SYS_PHY_INTF_SEL		
																HP_SYSTEM_SYS_SBD_FLOWCTRL		
																HP_SYSTEM_SYS_PTP_PPS		
31								8	7	6	5	4			2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

HP_SYSTEM_SYS_PTP_PPS Represents pulse per second output from Ethernet MAC. (RO)

HP_SYSTEM_SYS_SBD_FLOWCTRL Configures sideband flow control signal.

0: No effect

1: Instruct Ethernet MAC to transmit Pause frames in Full-duplex mode and enable back-pressure in half-duplex mode.

(R/W)

HP_SYSTEM_SYS_PHY_INTF_SEL Configures Ethernet PHY interface.

0: MII

4: RMII

Others: Reserved

(R/W)

HP_SYSTEM_SYS_GMAC_MEM_CLK_FORCE_ON Configures whether or not to force enable the memory clock of GMAC memory.

0: No effect

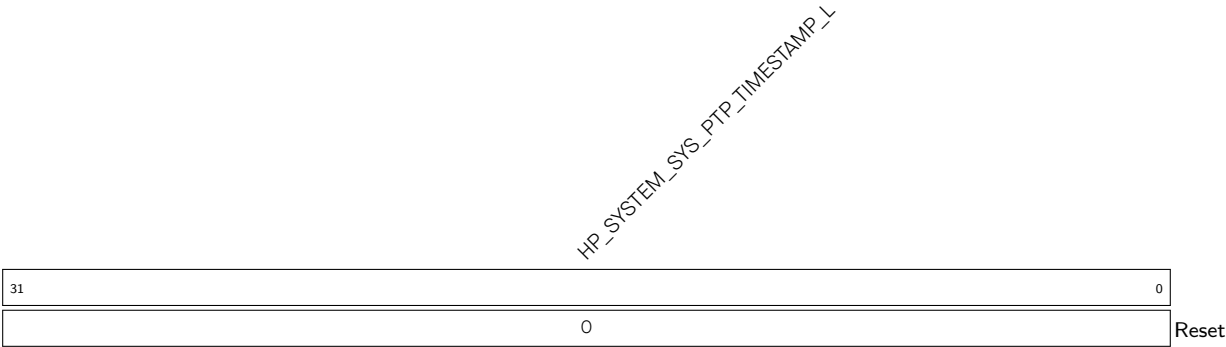
1: Force on

(R/W)

HP_SYSTEM_SYS_GMAC_RST_CLK_TX_N Represents an active-low reset signal synchronized to the Ethernet MAC Tx clock. (RO)

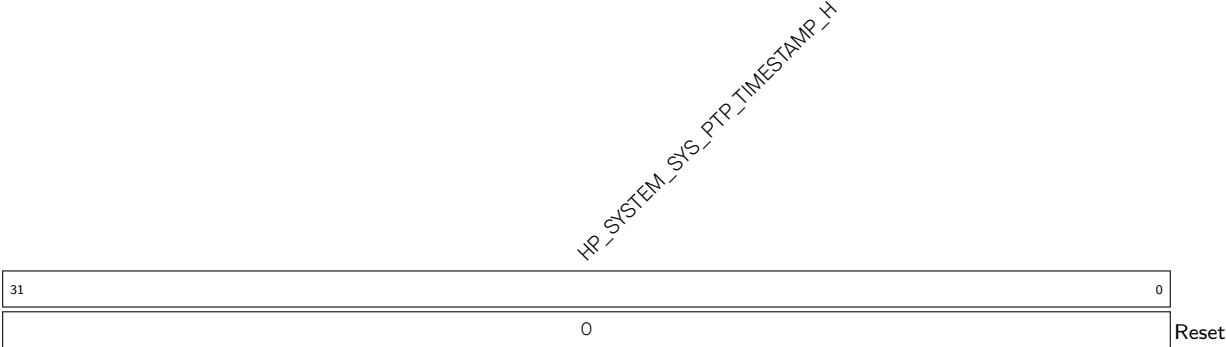
HP_SYSTEM_SYS_GMAC_RST_CLK_RX_N Represents an active-low reset signal synchronized to the Ethernet MAC Rx clock. (RO)

Register 20.59. HP_SYSTEM_GMAC_CTRL1_REG (0x0150)



HP_SYSTEM_SYS_PTP_TIMESTAMP_L Represents the lower 32 bits of timestamp output from Ethernet MAC synchronized to `clk_ptp_ref` clock. (RO)

Register 20.60. HP_SYSTEM_GMAC_CTRL2_REG (0x0154)



HP_SYSTEM_SYS_PTP_TIMESTAMP_H Represents the upper 32 bits of timestamp output from Ethernet MAC synchronized to `clk_ptp_ref` clock. (RO)

Register 20.61. HP_SYSTEM_USBOTG20_CTRL_REG (0x015C)

[illegible]

HP_SYSTEM_SYS_OTG_PHY_TEST_DONE Indicates the status of USB OTG_HS PHY BIST test.

0: Fail

1: Pass

(RO)

HP_SYSTEM_SYS_PHY_SUSPENDM Configures whether or not to drive USB OTG_HS PHY into Suspend mode. This setting is only valid when HP_SYSTEM_PHY_SUSPEND_FORCE_EN is set to 1.

0: No effect

1: In Suspend mode

(R/W)

HP_SYSTEM_SYS_PHY_SUSPEND_FORCE_EN Configures whether or not to allow forcing USB OTG HS PHY into Suspend mode by setting HP_SYSTEM_PHY_SUSPENDM.

0: No effect

1: Force in Suspend mode

(R/W)

HP_SYSTEM_SYS_PHY_RSTN Configures whether or not to reset USB OTG_HS PHY. This setting is only valid when HP_SYSTEM_PHY_RESET_FORCE_EN is set to 1.

0: Reset

1: No effect

(R/W)

HP_SYSTEM_SYS_PHY_RESET_FORCE_EN Configures whether or not to allow forcibly resetting USB OTG_HS PHY by clearing HP_SYSTEM_PHY_RSTN.

0: No effect

1: Force reset

(R/W)

Continued on the next page...

Register 20.61. HP_SYSTEM_USBOTG20_CTRL_REG (0x015C)

Continued from the previous page...

HP_SYSTEM_SYS_PHY_PLL_FORCE_EN Configures whether or not to allow forcibly enabling USB OTG_HS PHY PLL by setting HP_SYSTEM_PHY_PLL_EN.
0: No effect
1: Force enable
(R/W)

HP_SYSTEM_SYS_PHY_PLL_EN Configures whether or not to enable USB OTG_HS PHY PLL. This setting is only valid when HP_SYSTEM_PHY_PLL_FORCE_EN is set.
0: No effect
1: Enable
(R/W)

HP_SYSTEM_SYS_OTG_PHY_BISTEN Configures whether or not to drive USB OTG_HS PHY into BIST mode and perform self-test.
0: No effect
1: In BIST mode
(R/W)

Register 20.62. HP_SYSTEM_HP_SPM_ERR_RESP_CTRL_REG (0x0160)

(reserved)																															HP_SYSTEM_HP_SPM_ERR_RESP_EN		
31																															1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

HP_SYSTEM_HP_SPM_ERR_RESP_EN Configures whether or not to report parity error to the HP CPU and trigger exception.
0: Disable
1: Enable
(R/W)

Register 20.63. HP_SYSTEM_L2_MEM_REFRESH_REG (0x0164)

(reserved)																				HP_SYSTEM_L2_MEM_UNIT5_REFRESH_DONE HP_SYSTEM_L2_MEM_UNIT4_REFRESH_DONE HP_SYSTEM_L2_MEM_UNIT3_REFRESH_DONE HP_SYSTEM_L2_MEM_UNIT2_REFRESH_DONE HP_SYSTEM_L2_MEM_UNIT1_REFRESH_DONE HP_SYSTEM_L2_MEM_UNIT0_REFRESH_DONE HP_SYSTEM_L2_MEM_REFRESH_CNT_DONE HP_SYSTEM_L2_MEM_REFRESH_CNT_RESET HP_SYSTEM_L2_MEM_UNIT3_REFRESH_EN HP_SYSTEM_L2_MEM_UNIT2_REFRESH_EN HP_SYSTEM_L2_MEM_UNIT1_REFRESH_EN HP_SYSTEM_L2_MEM_UNIT0_REFRESH_EN															
31													13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset								
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0												

HP_SYSTEM_L2_MEM_UNIT0_REFRESH_EN Configures whether or not to refresh the ECC bit of L2MEM unit0.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L2_MEM_UNIT1_REFRESH_EN Configures whether or not to refresh the ECC bit of L2MEM unit1.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L2_MEM_UNIT2_REFRESH_EN Configures whether or not to refresh the ECC bit of L2MEM unit2.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L2_MEM_UNIT3_REFRESH_EN Configures whether or not to refresh the ECC bit of L2MEM unit3.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L2_MEM_UNIT4_REFRESH_EN Configures whether or not to refresh the ECC bit of L2MEM unit4.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 20.63. HP_SYSTEM_L2_MEM_REFRESH_REG (0x0164)

Continued from the previous page...

HP_SYSTEM_L2_MEM_UNIT5_REFERSH_EN Configures whether or not to refresh the ECC bit of L2MEM unit5.

0: Disable

1: Enable

(R/W)

HP_SYSTEM_L2_MEM_REFERSH_CNT_RESET Configures whether or not to reset L2MEM refresh count.

0: No effect

1: Reset

(R/W)

HP_SYSTEM_L2_MEM_UNIT0_REFRESH_DONE Represents the refresh status of L2MEM unit0 ECC bit. (RO)

HP_SYSTEM_L2_MEM_UNIT1_REFRESH_DONE Represents the refresh status of L2MEM unit1 ECC bit. (RO)

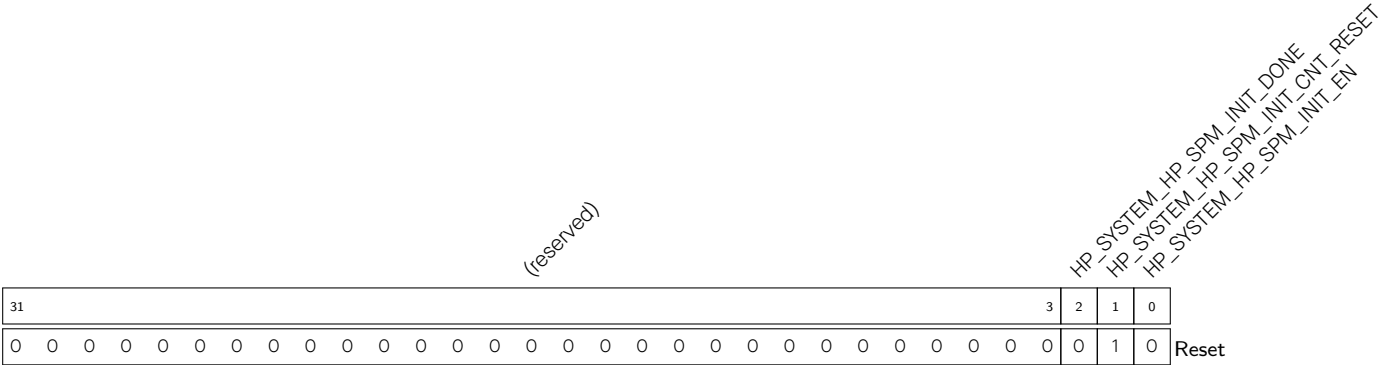
HP_SYSTEM_L2_MEM_UNIT2_REFRESH_DONE Represents the refresh status of L2MEM unit2 ECC bit. (RO)

HP_SYSTEM_L2_MEM_UNIT3_REFRESH_DONE Represents the refresh status of L2MEM unit3 ECC bit. (RO)

HP_SYSTEM_L2_MEM_UNIT4_REFRESH_DONE Represents the refresh status of L2MEM unit4 ECC bit. (RO)

HP_SYSTEM_L2_MEM_UNIT5_REFRESH_DONE Represents the refresh status of L2MEM unit5 ECC bit. (RO)

Register 20.64. HP_SYSTEM_HP_SPM_INIT_REG (0x0168)



- HP_SYSTEM_HP_SPM_INIT_EN

Configures whether or not to enable HP SPM parity bit initialization.

0: Disable

1: Enable

(R/W)
- HP_SYSTEM_HP_SPM_INIT_CNT_RESET

Configures whether or not to reset TCM initialization count.

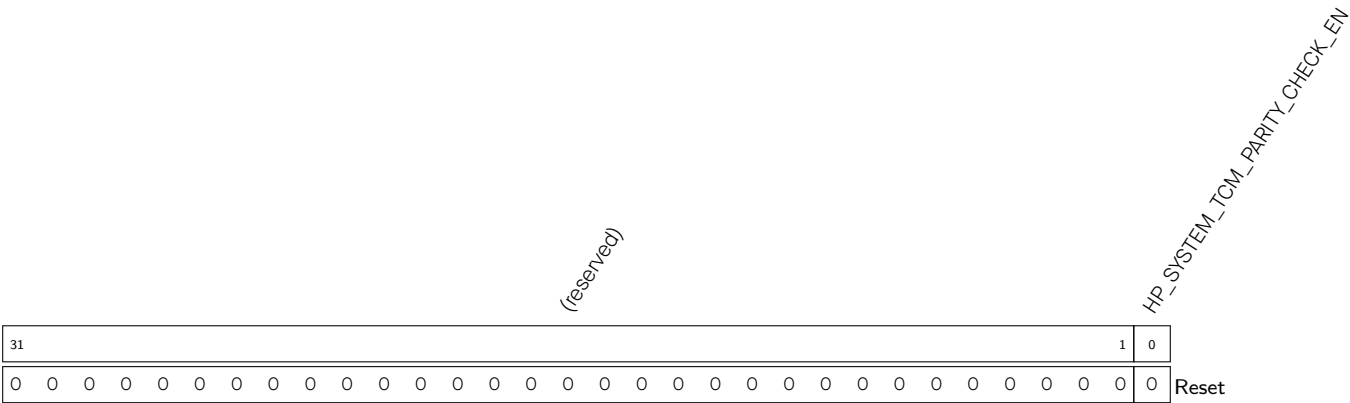
0: No effect

1: Reset

(R/W)
- HP_SYSTEM_HP_SPM_INIT_DONE

Represents the initialization status of HP SPM parity bits. (RO)

Register 20.65. HP_SYSTEM_HP_SPM_PARITY_CHECK_CTRL_REG (0x016C)



- HP_SYSTEM_HP_SPM_PARITY_CHECK_EN

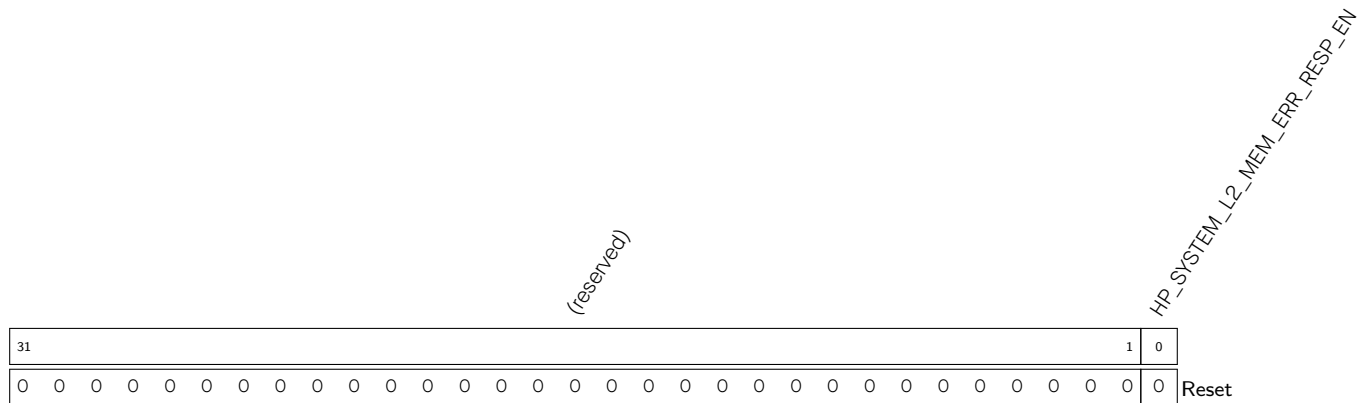
Configures whether or not to enable parity error check in HP SPM.

0: Disable

1: Enable

(R/W)

Register 20.66. HP_SYSTEM_L2_MEM_ERR_RESP_CTRL_REG (0x0198)



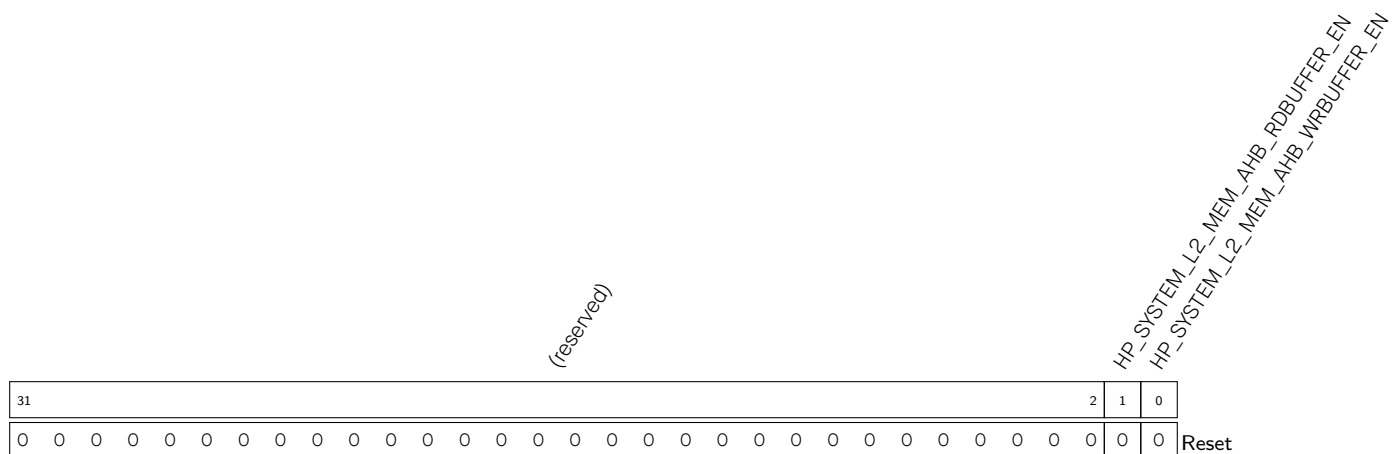
HP_SYSTEM_L2_MEM_ERR_RESP_EN Configures whether or not to report ECC error to the HP CPU and trigger exception.

0: Disable

1: Enable

(R/W)

Register 20.67. HP_SYSTEM_L2_MEM_AHB_BUFFER_CTRL_REG (0x019C)



HP_SYSTEM_L2_MEM_AHB_WRBUFFER_EN Configures whether or not to enable L2MEM AHB write buffer.

0: Disable

1: Enable

(R/W)

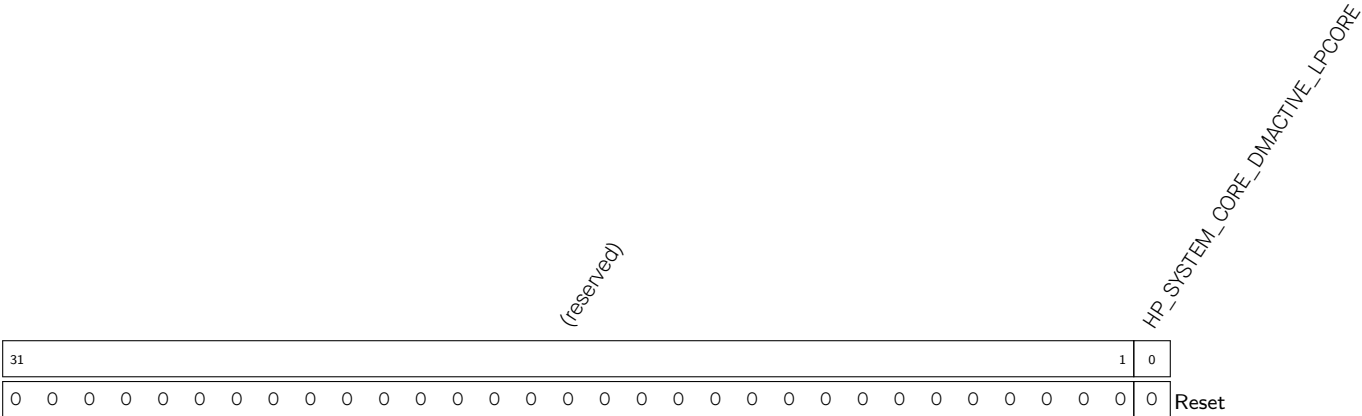
HP_SYSTEM_L2_MEM_AHB_RDBUFFER_EN Configures whether or not to enable L2MEM AHB read buffer.

0: Disable

1: Enable

(R/W)

Register 20.68. HP_SYSTEM_HP_CORE_DMACTIVE_LPCORE_REG (0x01A0)



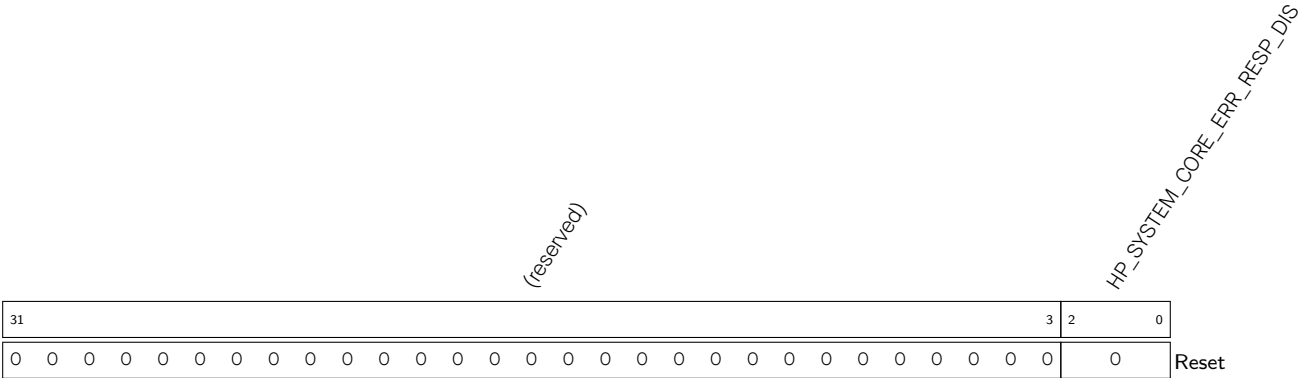
HP_SYSTEM_CORE_DMACTIVE_LPCORE Represents the dmactive value of LP CPU’s debug module.

0: Indicates that the LP CPU is not connected to the JTAG interface, or is not in debug mode.

1: Indicates that the LP CPU is connected to the JTAG interface and is in debug mode.

(RO)

Register 20.69. HP_SYSTEM_HP_CORE_ERR_RESP_DIS_REG (0x01A4)



HP_SYSTEM_CORE_ERR_RESP_DIS Configures whether or not to disable the error response on different buses.

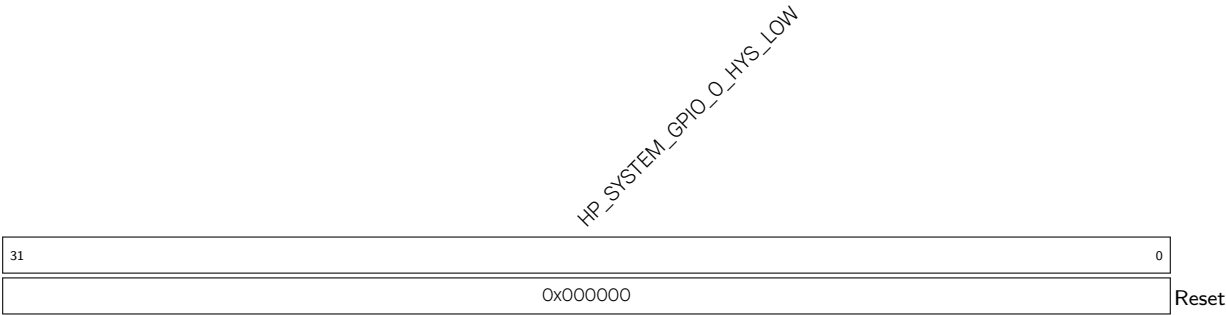
Bit 0: IBUS

Bit 1: DBUS

Bit 2: AHB bus

(R/W)

Register 20.70. HP_SYSTEM_GPIO_O_HYS_CTRL0_REG (0x01C0)



HP_SYSTEM_GPIO_O_HYS_LOW Configures whether or not to enable the hysteresis feature of HP GPIO16 ~ HP GPIO47.

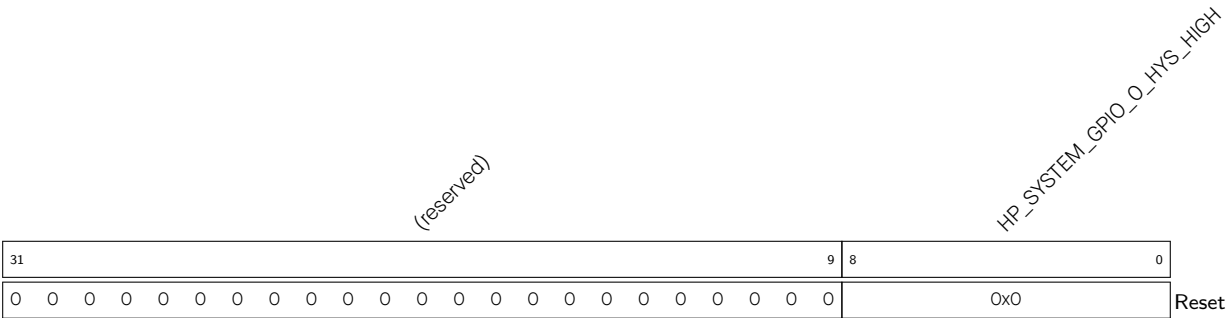
0: No effect

1: Enable

See detailed information in Section 9.10 of Chapter 9 *GPIO Matrix and IO MUX*.

(R/W)

Register 20.71. HP_SYSTEM_GPIO_O_HYS_CTRL1_REG (0x01C4)



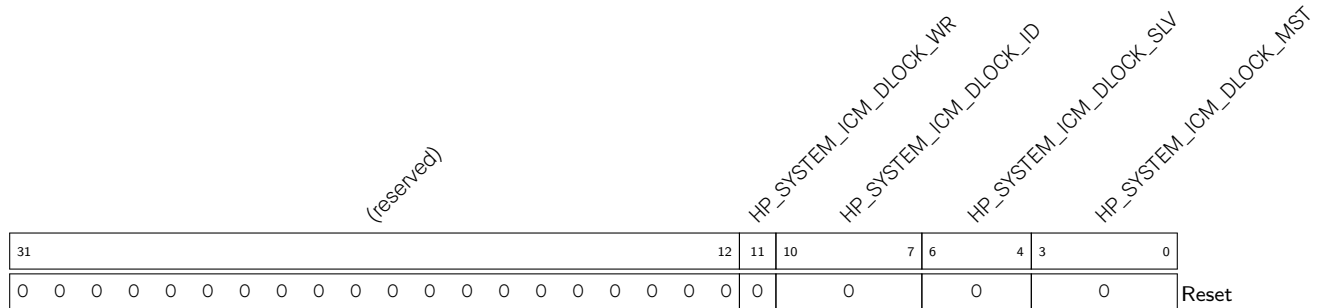
HP_SYSTEM_GPIO_O_HYS_HIGH Configures whether or not to enable the hysteresis feature of HP GPIO48 ~ HP GPIO56.

0: No effect

1: Enable

See detailed information in Section 9.10 of Chapter 9 *GPIO Matrix and IO MUX*.

(R/W)

Register 20.74. HP_SYSTEM_ICM_DLOCK_STATUS_REG (0x0008)

HP_SYSTEM_ICM_DLOCK_MST Records the lowest numbered master involved in a deadlock. (RO)

HP_SYSTEM_ICM_DLOCK_SLV Records the slave with which the deadlock has occurred. (RO)

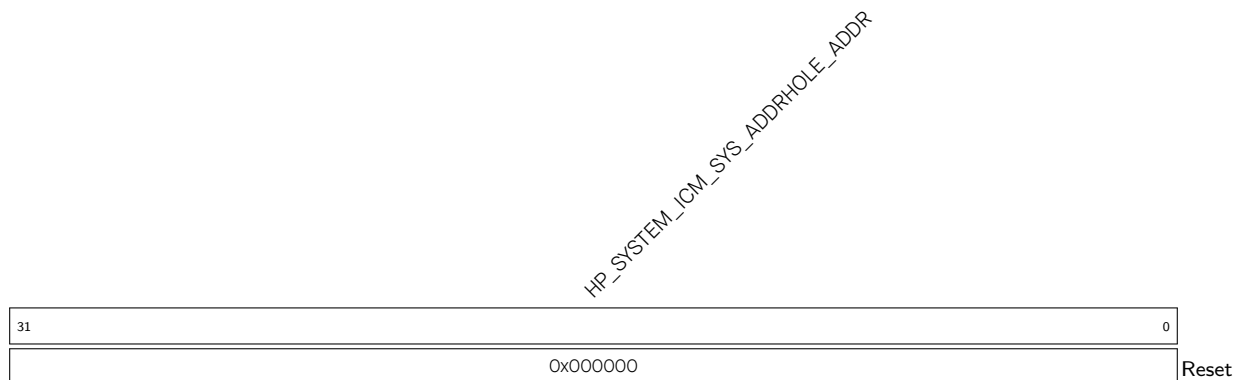
HP_SYSTEM_ICM_DLOCK_ID Records the AXI ID of the transaction that is deadlocked. (RO)

HP_SYSTEM_ICM_DLOCK_WR Indicates whether the deadlocked transaction is a write operation.

0: Not a write operation

1: Write operation

(RO)

Register 20.75. HP_SYSTEM_ICM_SYS_ADDRHOLE_ADDR_REG (0x0038)

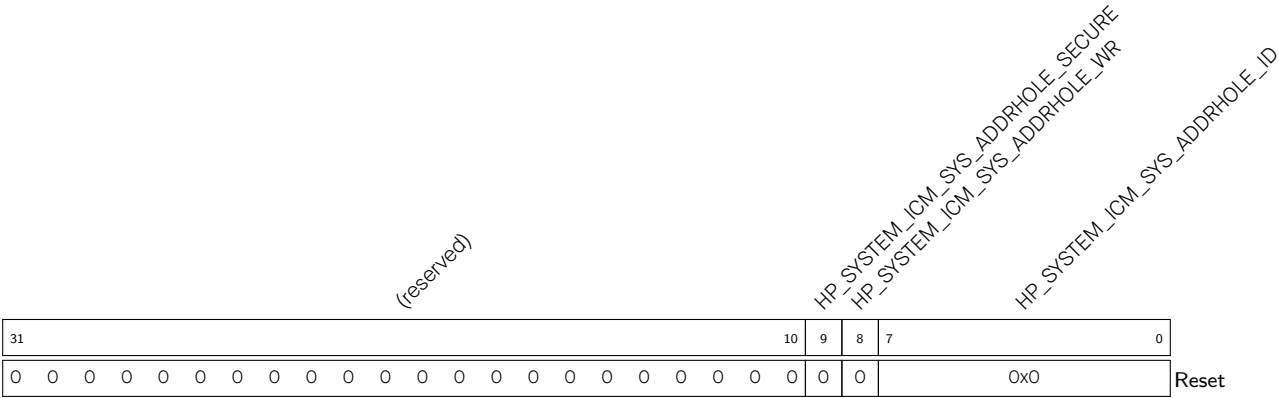
HP_SYSTEM_ICM_SYS_ADDRHOLE_ADDR Records the address when [ICM_SYS_ADDRHOLE_INT](#) occurs.

When HP_SYSTEM_ICM_SYS_ADDRHOLE_SECURE is 0, it records the unauthorized access address.

When HP_SYSTEM_ICM_SYS_ADDRHOLE_SECURE is 1, it records the illegal access address.

(RO)

Register 20.76. HP_SYSTEM_ICM_SYS_ADDRHOLE_INFO_REG (0x003C)



HP_SYSTEM_ICM_SYS_ADDRHOLE_ID Records the master ID when [ICM_SYS_ADDRHOLE_INT](#) occurs. This ID consists of a 4-bit CID and a 4-bit UID (refer to the related IP for details). The CID is used for master verification in ICM. CID values correspond to the following masters:

- 1: CACHE
- 5: GDMA MST1
- 6: GDMA MST2
- 8: AXI GDMA
- 10: DMA2D
- 11: H264 MST1
- 12: H264 MST2

(RO)

HP_SYSTEM_ICM_SYS_ADDRHOLE_WR Records the type of transfer when [ICM_SYS_ADDRHOLE_INT](#) occurs.

- 0: Read transfer
- 1: Write transfer

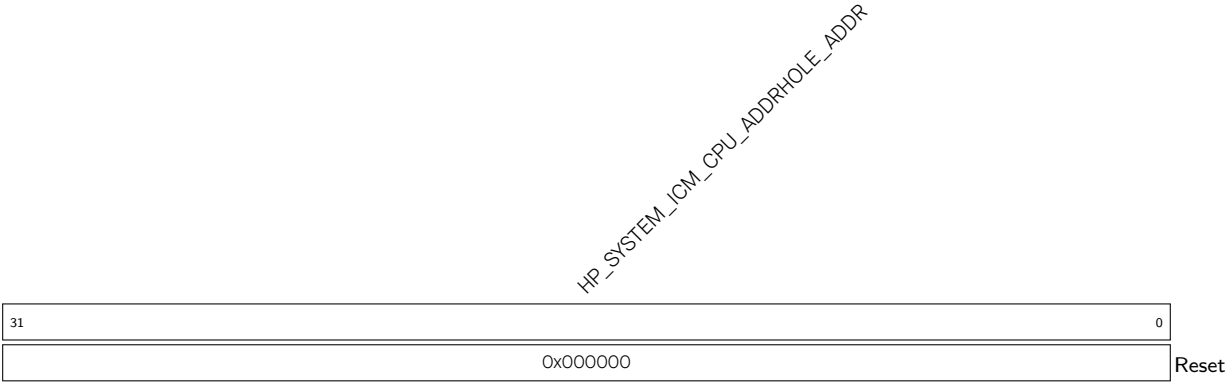
(RO)

HP_SYSTEM_ICM_SYS_ADDRHOLE_SECURE Records the type of access error when [ICM_SYS_ADDRHOLE_INT](#) occurs.

- 0: Unauthorized access
- 1: Illegal access

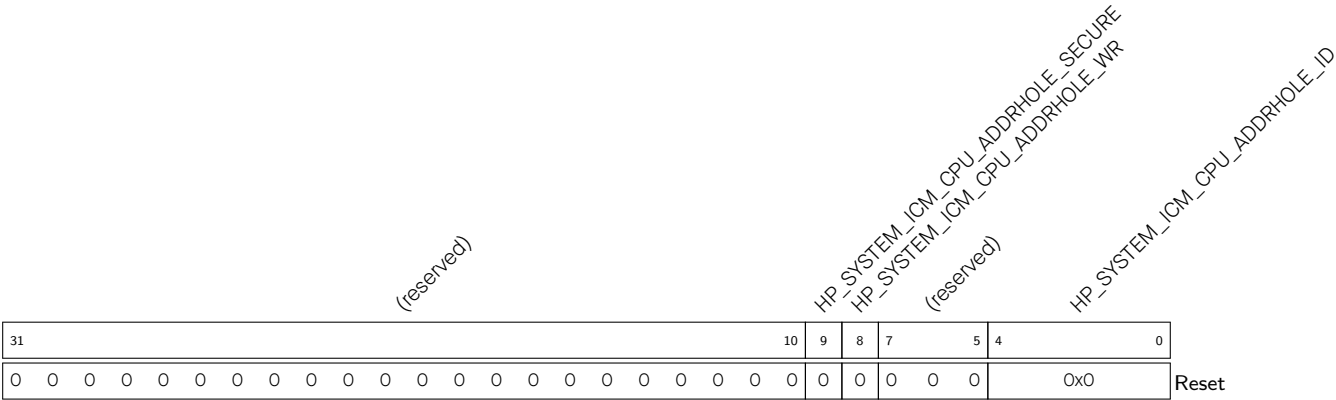
(RO)

Register 20.77. HP_SYSTEM_ICM_CPU_ADDRHOLE_ADDR_REG (0x0040)



HP_SYSTEM_ICM_CPU_ADDRHOLE_ADDR Records the address when [ICM_CPU_ADDRHOLE_INT](#) occurs.
When HP_SYSTEM_ICM_CPU_ADDRHOLE_SECURE is 0, it records the unauthorized access address.
When HP_SYSTEM_ICM_CPU_ADDRHOLE_SECURE is 1, it records the illegal access address.
(RO)

Register 20.78. HP_SYSTEM_ICM_CPU_ADDRHOLE_INFO_REG (0x0044)



HP_SYSTEM_ICM_CPU_ADDRHOLE_ID Records the master ID when [ICM_CPU_ADDRHOLE_INT](#) occurs.

- 0: HP CPU0
- 1: HP CPU1
- 2: LP CPU
- 3: USBOTGFS
- 5: GMAC
- 6: SDMMC
- 7: USBOTGHS
- 8: TRACE0
- 9: TRACE1
- 10: SPM Monitor
- 11: L2MEM Monitor
- 16-31: AHB GDMA

(RO)

HP_SYSTEM_ICM_CPU_ADDRHOLE_WR Records the type of transfer when [ICM_CPU_ADDRHOLE_INT](#) occurs.

- 0: Read transfer
- 1: Write transfer

(RO)

HP_SYSTEM_ICM_CPU_ADDRHOLE_SECURE Records the type of access error when [ICM_CPU_ADDRHOLE_INT](#) occurs.

- 0: Unauthorized access
- 1: Illegal access

(RO)

(reserved)

HP_SYSTEM_ICM_CPU_ADDRHOLE_INT_RAW
HP_SYSTEM_ICM_SYS_ADDRHOLE_INT_RAW
HP_SYSTEM_ICM_DLOCK_INT_RAW

HP_SYSTEM_ICM_CPU_ADDRHOLE_INT_RAW The raw interrupt status of ICM_CPU_ADDRHOLE_INT. (R/WTC/SS)

(reserved)

HP_SYSTEM_ICM_CPU_ADDRHOLE_INT_ST
HP_SYSTEM_ICM_SYS_ADDRHOLE_INT_ST
HP_SYSTEM_ICM_DLOCK_INT_ST

HP_SYSTEM_ICM_CPU_ADDRHOLE_INT_ST The masked interrupt status of **ICM_CPU_ADDRHOLE_INT**. (RO)

Register 20.81. HP_SYSTEM_ICM_INT_ENA_REG (0x0014)

(reserved)																																HP_SYSTEM_ICM_CPU_ADDRHOLE_INT_ENA HP_SYSTEM_ICM_SYS_ADDRHOLE_INT_ENA HP_SYSTEM_ICM_DLOCK_INT_ENA			
31																																3	2	1	0
0 0																																1	1	1	Reset

HP_SYSTEM_ICM_DLOCK_INT_ENA Write 1 to enable [DLOCK_INT](#) interrupt. (R/W)

HP_SYSTEM_ICM_SYS_ADDRHOLE_INT_ENA Write 1 to enable [ICM_SYS_ADDRHOLE_INT](#) interrupt. (R/W)

HP_SYSTEM_ICM_CPU_ADDRHOLE_INT_ENA Write 1 to enable [ICM_CPU_ADDRHOLE_INT](#) interrupt. (R/W)

Register 20.82. HP_SYSTEM_ICM_INT_CLR_REG (0x0018)

(reserved)																												HP_SYSTEM_ICM_CPU_ADDRHOLE_INT_CLR HP_SYSTEM_ICM_SYS_ADDRHOLE_INT_CLR HP_SYSTEM_ICM_DLOCK_INT_CLR				
31																												3	2	1	0	
0 0																												0	0	0	0	Reset

HP_SYSTEM_ICM_DLOCK_INT_CLR Write 1 to clear [DLOCK_INT](#) interrupt. (WT)

HP_SYSTEM_ICM_SYS_ADDRHOLE_INT_CLR Write 1 to clear [ICM_SYS_ADDRHOLE_INT](#) interrupt. (WT)

HP_SYSTEM_ICM_CPU_ADDRHOLE_INT_CLR Write 1 to clear [ICM_CPU_ADDRHOLE_INT](#) interrupt. (WT)

Register 20.83. HP_SYSTEM_ICM_SLV_ARB_PRIORITY_REG (0x0024)

(reserved)								HP_SYSTEM_ICM_CAM_PRIORITY				HP_SYSTEM_ICM_LCD_PRIORITY				HP_SYSTEM_ICM_PSRAM_MSPI_PRIORITY				(reserved)				HP_SYSTEM_ICM_L2MEM_PRIORITY				(reserved)								
31	24				23	21				20	18				17	15				14	12				11	6				5	3				2	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset				

HP_SYSTEM_ICM_L2MEM_PRIORITY Configures the arbitration priority of L2MEM for response channels among slaves connected to SYS ICM. (R/W)

HP_SYSTEM_ICM_FLASH_MSPI_PRIORITY Configures the arbitration priority of FLASH MSPI for response channels among slaves connected to SYS ICM. (R/W)

HP_SYSTEM_ICM_PSRAM_MSPI_PRIORITY Configures the arbitration priority of PSRAM MSPI for response channels among slaves connected to SYS ICM. (R/W)

HP_SYSTEM_ICM_LCD_PRIORITY Configures the arbitration priority of MIPI_LCD registers for response channels among slaves connected to SYS ICM. (R/W)

HP_SYSTEM_ICM_CAM_PRIORITY Configures the arbitration priority of MIPI_CAM registers for response channels among slaves connected to SYS ICM. (R/W)

Register 20.84. HP_SYSTEM_ICM_MST_ARQOS_REG (0x0028)

HP_SYSTEM_ICM_AXI_GDMA_INT_ARQOS				HP_SYSTEM_ICM_H264_DMA_M2_ARQOS				HP_SYSTEM_ICM_H264_DMA_M1_ARQOS				HP_SYSTEM_ICM_DW_GDMA_MST2_ARQOS				HP_SYSTEM_ICM_DW_GDMA_MST1_ARQOS				HP_SYSTEM_ICM_DMA2D_ARQOS				HP_SYSTEM_ICM_CACHE_ARQOS				HP_SYSTEM_ICM_CPU_ARQOS			
31	28	27	24	23	20	19	16	15	12	11	8	7	4	3	0																
0				0				0				0				0				0				0				Reset			

HP_SYSTEM_ICM_CPU_ARQOS Configures the ARQoS value for CPU ICM. (R/W)

HP_SYSTEM_ICM_CACHE_ARQOS Configures the ARQoS value for CACHE. (R/W)

HP_SYSTEM_ICM_DMA2D_ARQOS Configures the ARQoS value for DMA2D. (R/W)

HP_SYSTEM_ICM_DW_GDMA_MST1_ARQOS Configures the ARQoS value for DW-GDMA MST1.
(R/W)

HP_SYSTEM_ICM_DW_GDMA_MST2_ARQOS Configures the ARQoS value for DW-GDMA MST2.
(R/W)

HP_SYSTEM_ICM_H264_DMA_M1_ARQOS Configures the ARQoS value for H264 DMAMST1.
(R/W)

HP_SYSTEM_ICM_H264_DMA_M2_ARQOS Configures the ARQoS value for H264 DMAMST2.
(R/W)

HP_SYSTEM_ICM_AXI_GDMA_INT_ARQOS Configures the ARQoS value for AXI GDMA. (R/W)

Register 20.85. HP_SYSTEM_ICM_MST_AWQOS_REG (0x0030)

HP_SYSTEM_ICM_PDMA_INT_AWQOS				HP_SYSTEM_ICM_H264_DMA_M2_AWQOS				HP_SYSTEM_ICM_H264_DMA_M1_AWQOS				HP_SYSTEM_ICM_GDMA_M2_AWQOS				HP_SYSTEM_ICM_GDMA_M1_AWQOS				HP_SYSTEM_ICM_DMA2D_AWQOS				HP_SYSTEM_ICM_CACHE_AWQOS				HP_SYSTEM_ICM_CPU_AWQOS			
31	28	27	24	23	20	19	16	15	12	11	8	7	4	3	0																
0				0				0				0				0				0				0				Reset			

HP_SYSTEM_ICM_CPU_AWQOS Configures the AWQoS value for the CPU ICM. (R/W)

HP_SYSTEM_ICM_CACHE_AWQOS Configures the AWQoS value for the CACHE. (R/W)

HP_SYSTEM_ICM_DMA2D_AWQOS Configures the AWQoS value for the DMA2D. (R/W)

HP_SYSTEM_ICM_GDMA_M1_AWQOS Configures the AWQoS value for the DW-GDMA MST1. (R/W)

HP_SYSTEM_ICM_GDMA_M2_AWQOS Configures the AWQoS value for the DW-GDMA MST2.
(R/W)

HP_SYSTEM_ICM_H264_DMA_M1_AWQOS Configures the AWQoS value for the H264 DMA MST1.
(R/W)

HP_SYSTEM_ICM_H264_DMA_M2_AWQOS Configures the AWQoS value for the H264 DMA MST2.
(R/W)

HP_SYSTEM_ICM_PDMA_INT_AWQOS Configures the AWQoS value for the AXI GDMA. (R/W)

Register 20.86. HP_SYSTEM_ICM_DLOCK_TIMEOUT_REG (0x0048)

(reserved)																HP_SYSTEM_ICM_DLOCK_TIMEOUT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			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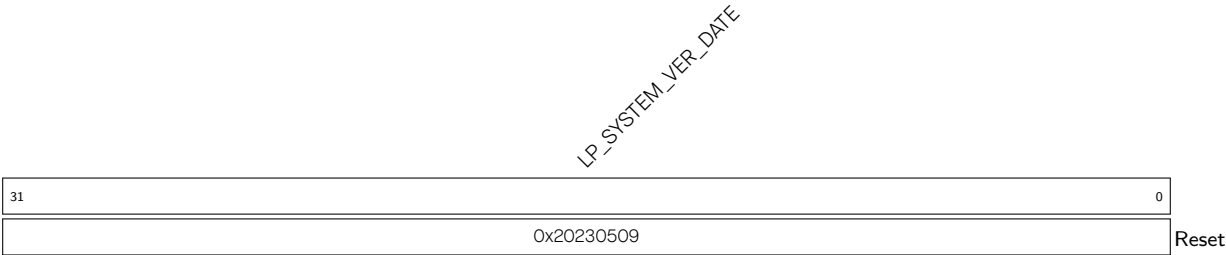
HP_SYSTEM_ICM_DLOCK_TIMEOUT Configures the deadlock timeout threshold value. Deadlock will occur if there is no response within the configured threshold.
(R/W)

20.5.3 LP Registers

The addresses in this section are relative to LP System Registers base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 20.87. LP_SYSTEM_LP_SYS_VER_DATE_REG (0x0000)



LP_SYSTEM_VER_DATE Version control register. (R/W)

Register 20.88. LP_SYSTEM_SYS_CTRL_REG (0x0008)

LP_SYSTEM_SYSTIMER_STALL_SEL				LP_SYSTEM_LP_CORE_ETM_WAKEUP_FLAG				LP_SYSTEM_LP_CORE_ETM_WAKEUP_FLAG_CLR				LP_SYSTEM_LP_FIB_SEL				LP_SYSTEM_ANA_FIB				(reserved)				LP_SYSTEM_IO_MUX_RESET_DISABLE				LP_SYSTEM_DIG_FIB				LP_SYSTEM_FORCE_DOWNLOAD_BOOT				(reserved)			
31	30	29	28	21				20	14				13	12	11	10	3				2	1	0																
0	0	0	0xff				0x7f				0	0	0	0xff				0	0	0	Reset																		

LP_SYSTEM_SYS_SW_RST Configures whether or not to trigger a software system reset. (WT)

LP_SYSTEM_FORCE_DOWNLOAD_BOOT Configures whether or not to forcibly enter Joint Download Boot mode when the pad boot mode is set to SPI Boot mode.

0: Not forcibly enter in Joint Download Boot mode

1: Forcibly enter in Joint Download Boot mode

(R/W)

LP_SYSTEM_DIG_FIB Configures software FIB.

bit-0: Unused

bit-1: Software BOD enable

bit-2: Software SWD enable

bit-3: Unused

bit-4: Software DCDC switch

bit-5: Software DCDC VSET enable

bit-6: Unused

bit-7: Unused

(R/W)

LP_SYSTEM_IO_MUX_RESET_DISABLE Configures whether or not to disable the HP IOMUX reset.

0: Enable IOMUX reset

1: Disable IOMUX reset

(R/W)

Continued on the next page...

Register 20.88. LP_SYSTEM_SYS_CTRL_REG (0x0008)

Continued from the previous page...

LP_SYSTEM_ANA_FIB Represents analog FIB state.

bit-0: Unused

bit-1: Analog BOD enable

bit-2: Analog SWD enable

bit-3: Unused

bit-4: Unused

bit-5: Unused

bit-6: Unused

bit-7: Unused

(RO)

LP_SYSTEM_LP_FIB_SEL Configures FIB control source select.

bit-0: Unused

bit-1: BOD enable control select. 0: Controlled by software, 1: Controlled by analog

bit-2: SWD enable control select. 0: Controlled by software, 1: Controlled by analog

bit-3: Unused

bit-4: Unused

bit-5: Unused

bit-6: Unused

bit-7: Unused

(R/W)

LP_SYSTEM_LP_CORE_ETM_WAKEUP_FLAG_CLR Write 1 to clear
LP_SYSTEM_LP_CORE_ETM_WAKEUP_FLAG. (WT)

LP_SYSTEM_LP_CORE_ETM_WAKEUP_FLAG Represents ETM task of ULP wakeup. (R/WTC/SS)

LP_SYSTEM_SYSTIMER_STALL_SEL Configures whether the LP systimer_stall signal is from HP CPU0 or HP CPU1.

0: From HP CPU0

1: From HP CPU1

(R/W)

Register 20.89. LP_SYSTEM_LP_CLK_CTRL_REG (0x000C)

(reserved)																															LP_SYSTEM_CLK_EN	
31																															1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	Reset

LP_SYSTEM_CLK_EN Configures whether the register clock is always on.

0: The clock is enabled only during register access

1: The clock is always on

(R/W)

Register 20.90. LP_SYSTEM_LP_RST_CTRL_REG (0x0010)

(reserved)																															LP_SYSTEM_SYS_RST_BYPASS		LP_SYSTEM_ANA_RST_BYPASS	
31																														2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	Reset		

LP_SYSTEM_ANA_RST_BYPASS Configures whether or not to bypass this analog reset sources: brown-out detector, super WDT and power glitch detector.

0: Not bypass

1: Bypass

(R/W)

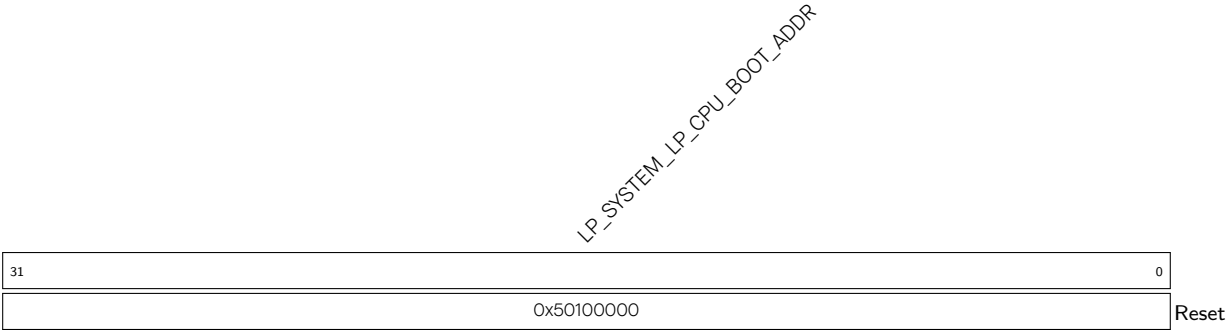
LP_SYSTEM_SYS_RST_BYPASS Configures whether or not to bypass this digital reset sources: software, HP watchdog timer, LP watchdog time and eFuse.

0: Not bypass

1: Bypass

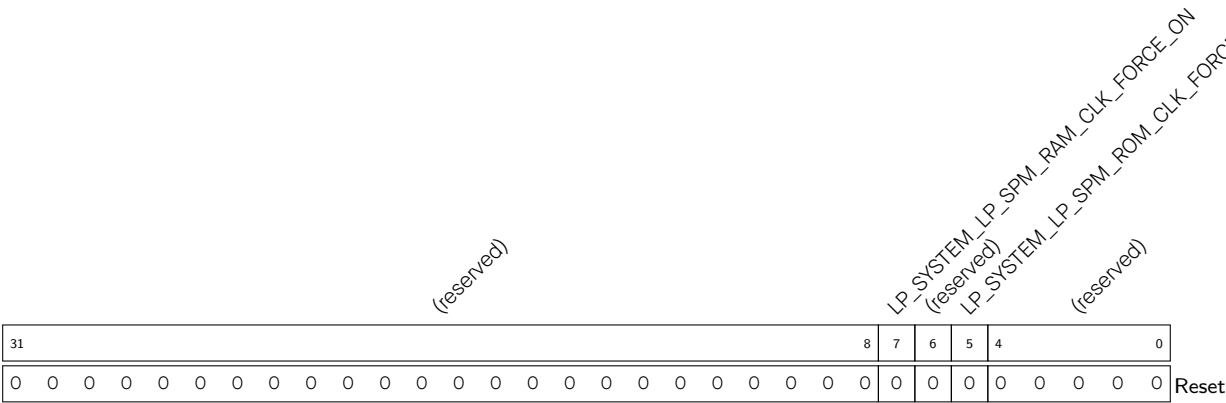
(R/W)

Register 20.91. LP_SYSTEM_LP_CORE_BOOT_ADDR_REG (0x0018)



LP_SYSTEM_LP_CPU_BOOT_ADDR Configures the LP CPU boot address. (R/W)

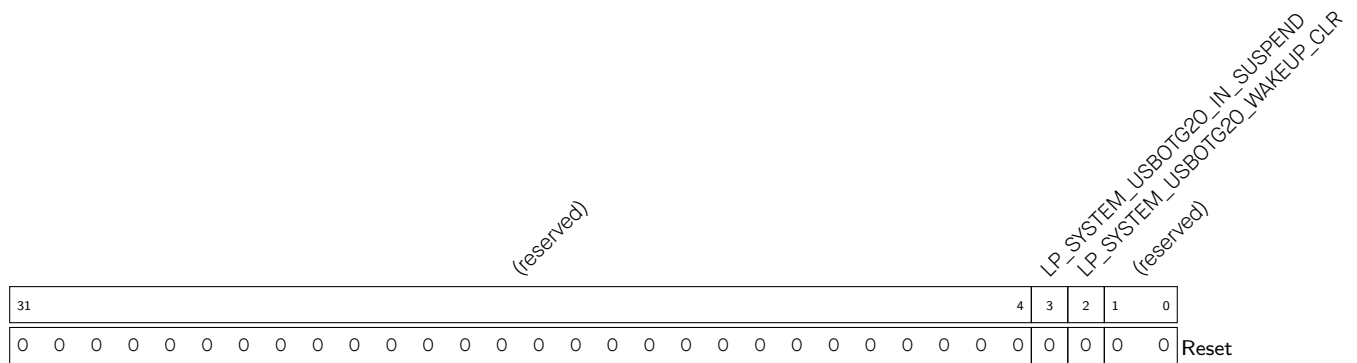
Register 20.92. LP_SYSTEM_LP_SPM_PWR_CTRL_REG (0x0024)



LP_SYSTEM_LP_SPM_ROM_CLK_FORCE_ON Configures whether or not to force on LP SPM_ROM clock when LP CPU clock is on.
0: Not force on
1: Force on
(R/W)

LP_SYSTEM_LP_SPM_RAM_CLK_FORCE_ON Configures whether or not to force on LP SPM_RAM clock.
0: Not force on
1: Force on
(R/W)

Register 20.93. LP_SYSTEM_USB_CTRL_REG (0x0100)



LP_SYSTEM_USBOTG20_WAKEUP_CLR Write 1 to clear wakeup signals from USB OTG2.0 to PMU.
(WT)

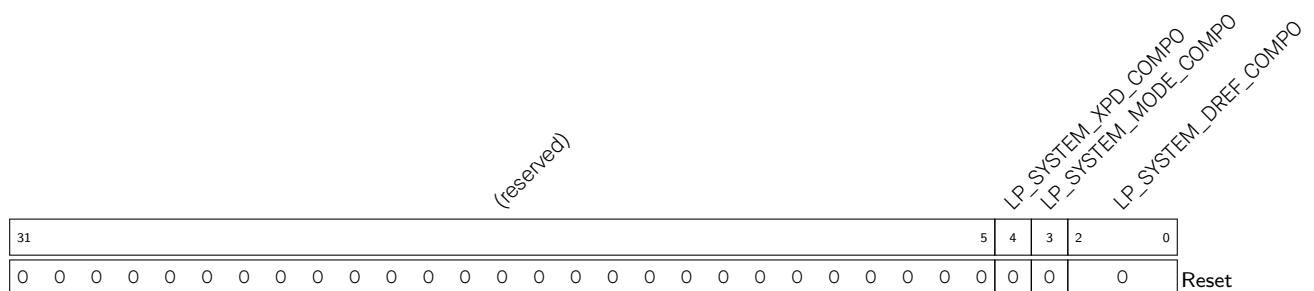
LP_SYSTEM_USBOTG20_IN_SUSPEND Configures whether or not to enable wakeup signals from USB OTG2.0.

0: Disable

1: Enabled

(R/W)

Register 20.94. LP_SYSTEM_PAD_COMPO_REG (0x0148)



LP_SYSTEM_DREF_COMPO Configures the internal reference voltage of pad comparator0. (R/W)

LP_SYSTEM_MODE_COMP0 Configures the comparison mode for pad comparator0.

0: Comparing the main voltage with the external reference voltage

1: Comparing the main voltage with the internal reference voltage

(R/W)

LP_SYSTEM_XPD_COMPO Configures whether or not to enable pad comparator0.

0: Disable

1: Enable

(R/W)

Register 20.95. LP_SYSTEM_PAD_COMP1_REG (0x014C)

(reserved)																										LP_SYSTEM_XPD_COMP1			LP_SYSTEM_MODE_COMP1			LP_SYSTEM_DREF_COMP1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
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LP_SYSTEM_DREF_COMP1 Configures the internal reference voltage of pad comparator1. (R/W)

LP_SYSTEM_MODE_COMP1 Configures the comparison mode for pad comparator1.

0: Comparing the main voltage with the external reference voltage

1: Comparing the main voltage with the internal reference voltage

(R/W)

LP_SYSTEM_XPD_COMP1 Configures whether or not to enable pad comparator1.

0: Disable

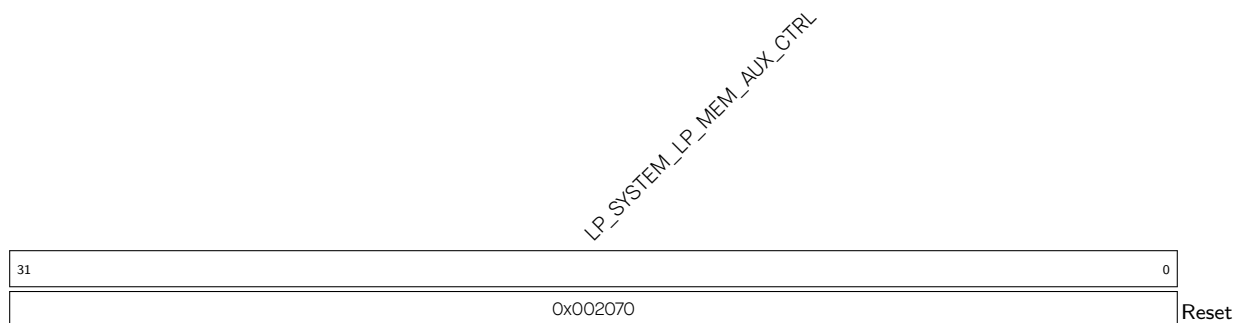
1: Enable

(R/W)

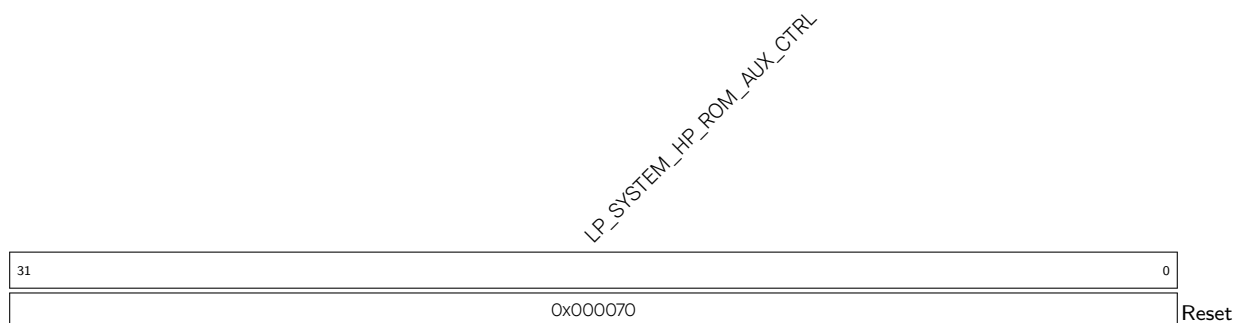
Register 20.96. LP_SYSTEM_HP_MEM_AUX_CTRL_REG (0x0180)

LP_SYSTEM_HP_MEM_AUX_CTRL																															
31																															0
0x002070																															
Reset																															

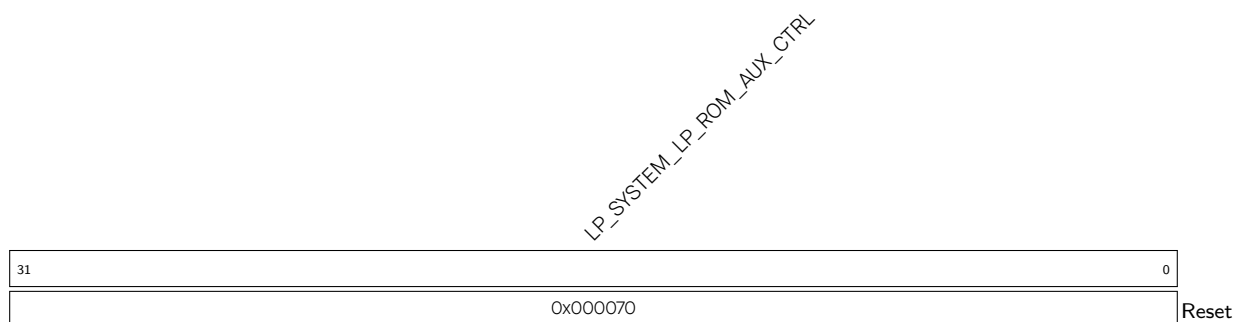
LP_SYSTEM_HP_MEM_AUX_CTRL Configures HP system memory aux control value. (R/W)

Register 20.97. LP_SYSTEM_LP_MEM_AUX_CTRL_REG (0x0184)

LP_SYSTEM_LP_MEM_AUX_CTRL Configures LP system memory aux control value. (R/W)

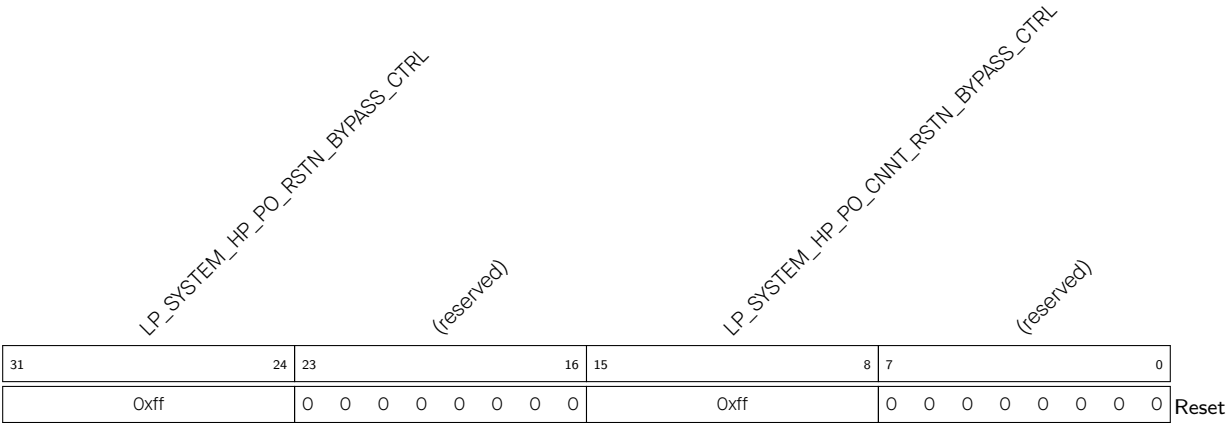
Register 20.98. LP_SYSTEM_HP_ROM_AUX_CTRL_REG (0x0188)

LP_SYSTEM_HP_ROM_AUX_CTRL Configures L2ROM aux control value. (R/W)

Register 20.99. LP_SYSTEM_LP_ROM_AUX_CTRL_REG (0x018C)

LP_SYSTEM_LP_ROM_AUX_CTRL Configures LPRom aux control value. (R/W)

Register 20.100. LP_SYSTEM_HP_POR_RST_BYPASS_CTRL_REG (0x01A0)



LP_SYSTEM_HP_PO_CNNT_RSTN_BYPASS_CTRL Configures whether or not to bypass a certain reset source for the HP_CNNT power domain during the Power-On reset.

0: No effect

1: Bypass

See section 20.2.2.5 for detailed information about respective fields. (R/W)

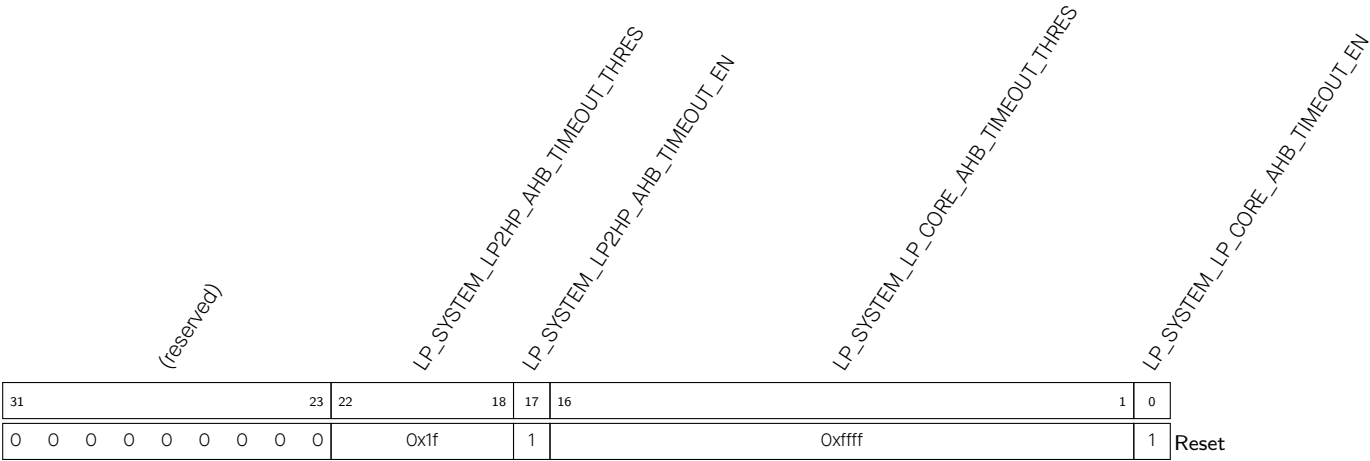
LP_SYSTEM_HP_PO_RSTN_BYPASS_CTRL configures whether or not to bypass the following reset sources for the HP power domain during the Power-On reset.

0: No effect

1: Bypass

See section 20.2.2.5 for detailed information about respective fields. (R/W)

Register 20.101. LP_SYSTEM_LP_CORE_AHB_TIMEOUT_REG (0x01B0)

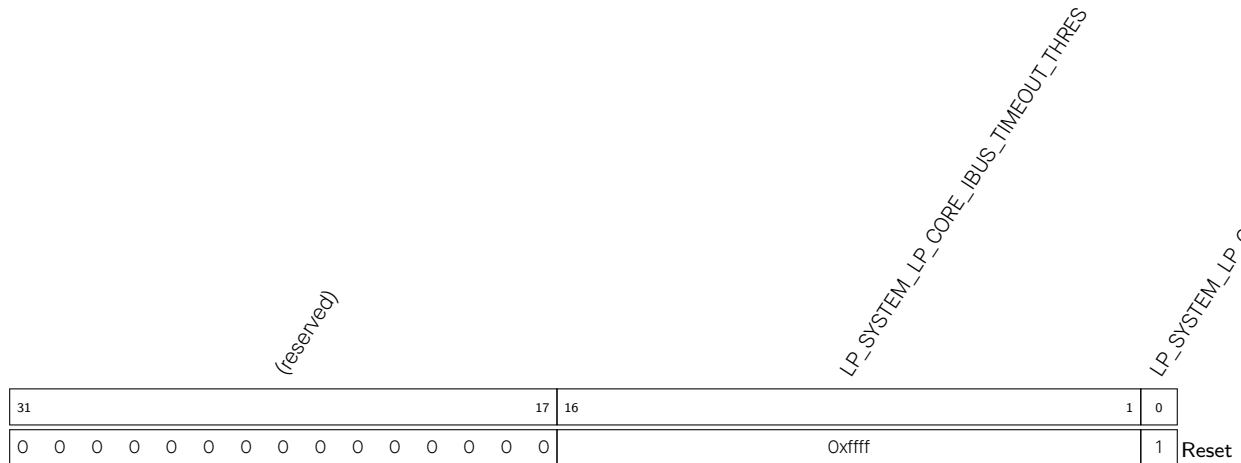


LP_SYSTEM_LP_CORE_AHB_TIMEOUT_EN Configures whether or not to enable timeout protection on LP AHB bus.
0: Disable
1: Enable
(R/W)

LP_SYSTEM_LP_CORE_AHB_TIMEOUT_THRES Configures [LP_CORE_AHB_TIMEOUT_INT](#) threshold. (R/W)

LP_SYSTEM_LP2HP_AHB_TIMEOUT_EN Configures whether or not to enable timeout protection on LP2HP AHB bus.
0: Disable
1: Enable
(R/W)

LP_SYSTEM_LP2HP_AHB_TIMEOUT_THRES Configures [LP_CORE_AHB_TIMEOUT_INT](#) threshold. (R/W)

Register 20.102. LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_REG (0x01B4)

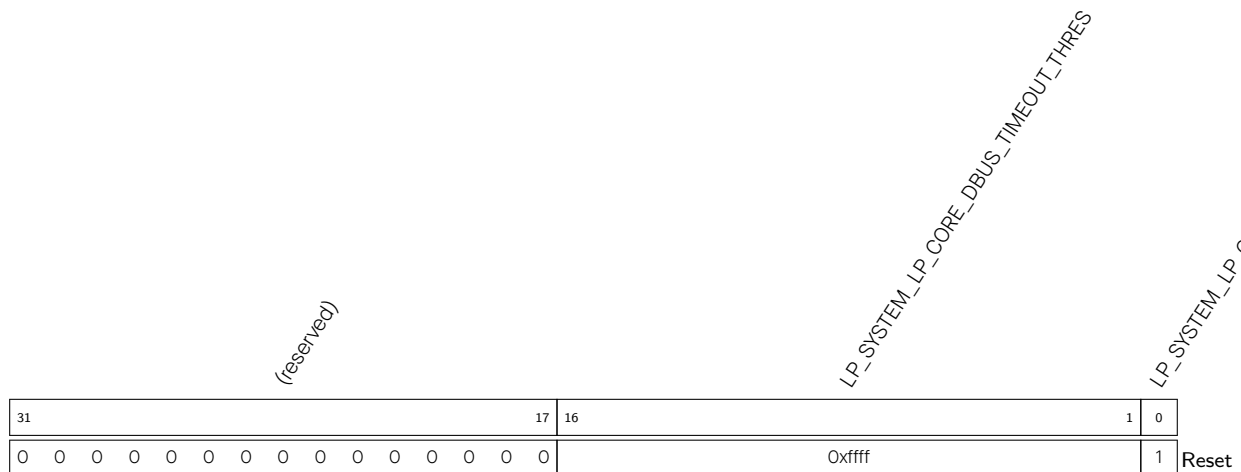
LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_EN Configures whether or not to enable timeout protection on LP CPU IBUS.

0: Disable

1: Enable

(R/W)

LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_THRES Configures [LP_CORE_IBUS_TIMEOUT_INT](#) threshold. (R/W)

Register 20.103. LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_REG (0x01B8)

LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_EN Configures whether or not to enable timeout protection on LP CPU DBUS.

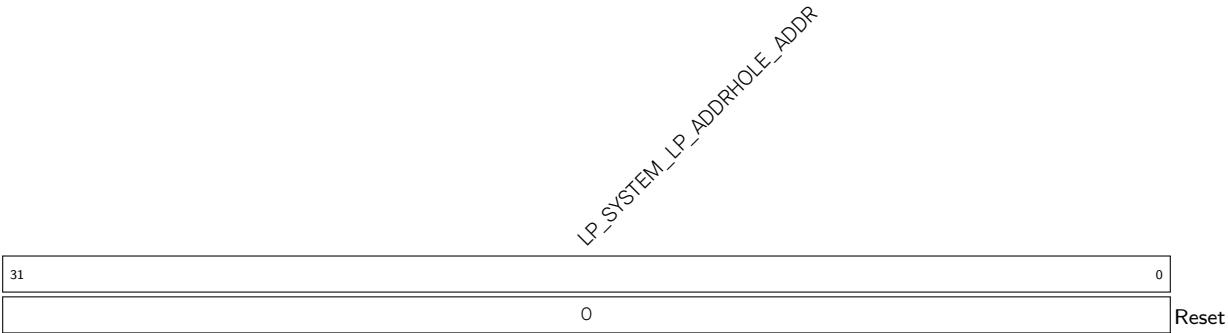
0: Disable

1: Enable

(R/W)

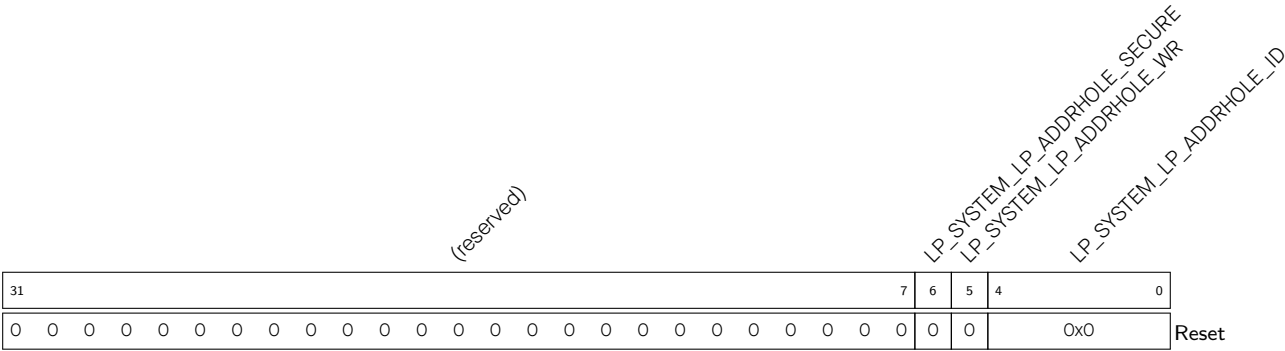
LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_THRES Configures [LP_CORE_DBUS_TIMEOUT_INT](#) threshold. (R/W)

Register 20.104. LP_SYSTEM_LP_ADDRHOLE_ADDR_REG (0x0168)



LP_SYSTEM_LP_ADDRHOLE_ADDR Represents the address of transfer when LP_ADDRHOLE_INT occurs. (RO)

Register 20.105. LP_SYSTEM_LP_ADDRHOLE_INFO_REG (0x016C)



LP_SYSTEM_LP_ADDRHOLE_ID Represents the master ID when LP_ADDRHOLE_INT occurs.

- 0: LP CPU0
 - 1: LP CPU1
 - 2: LP CPU
 - 3: USB OTG11
 - 4: Reserved
 - 5: GMAC
 - 6: SDMMC
 - 7: USBOTG20
 - 8: TRACE0
 - 9: TRACE1
 - 10: SPM monitor
 - 11: L2MEM monitor
 - 16~31: AHB PDMA
- (RO)

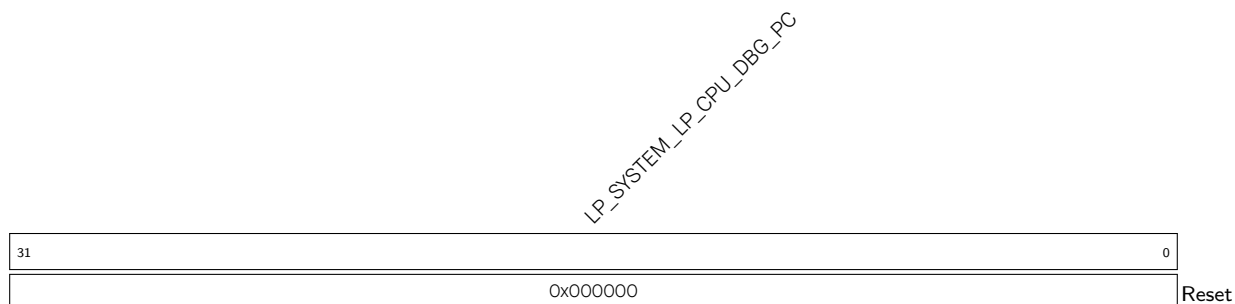
LP_SYSTEM_LP_ADDRHOLE_WR Represents the direction of transfer when LP_ADDRHOLE_INT occurs.

- 1: Write transfer
 - 0: Read transfer
- (RO)

LP_SYSTEM_LP_ADDRHOLE_SECURE Represents the address hole type when LP_ADDRHOLE_INT occurs.

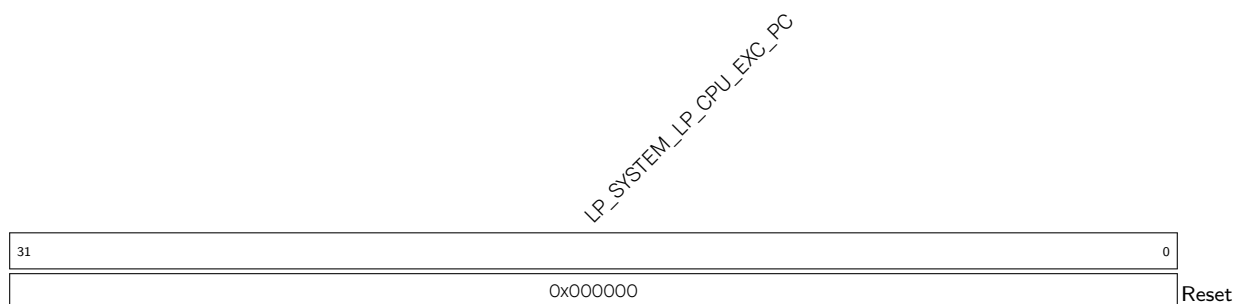
- 1: Illegal address access
 - 0: Access without permission
- (RO)

Register 20.106. LP_SYSTEM_LP_CPU_DBG_PC_REG (0x0190)



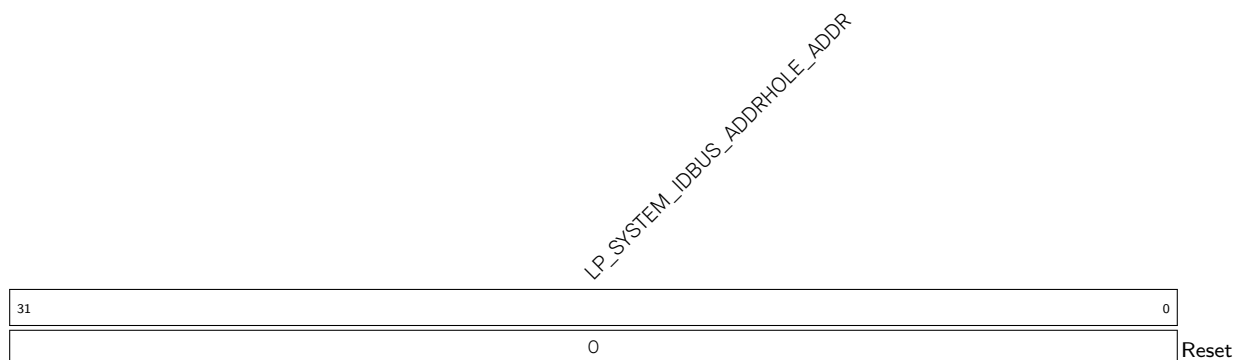
LP_SYSTEM_LP_CPU_DBG_PC Represents LP CPU current PC address. (RO)

Register 20.107. LP_SYSTEM_LP_CPU_EXC_PC_REG (0x0194)



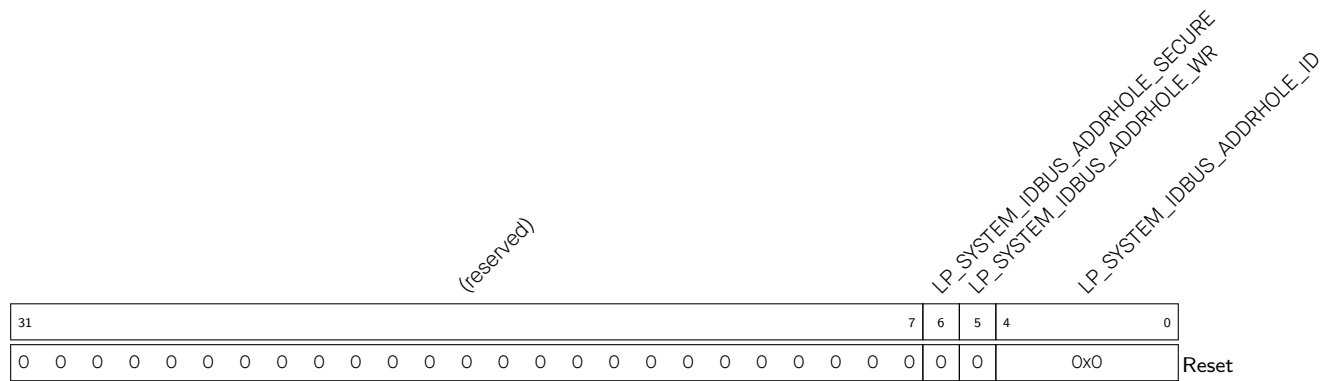
LP_SYSTEM_LP_CPU_EXC_PC Represents LP CPU exception PC address. (RO)

Register 20.108. LP_SYSTEM_IDBUS_ADDRHOLE_ADDR_REG (0x0198)



LP_SYSTEM_IDBUS_ADDRHOLE_ADDR Represents the address of transfer when [ID-BUS_ADDRHOLE_INT](#) occurs. (RO)

Register 20.109. LP_SYSTEM_IDBUS_ADDRHOLE_INFO_REG (0x019C)

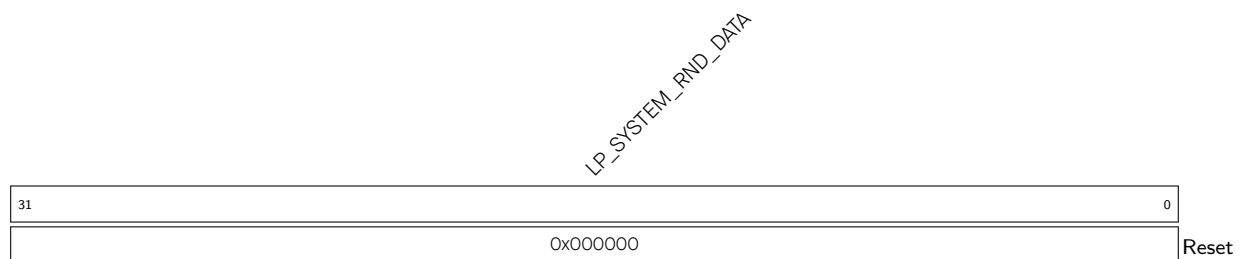


LP_SYSTEM_IDBUS_ADDRHOLE_ID Represents the LP CPU's ID value (00010) when ID-BUS_ADDRHOLE_INT occurs. (RO)

LP_SYSTEM_IDBUS_ADDRHOLE_WR Represents the direction of transfer when IDBUS_ADDRHOLE_INT occurs. (RO)

LP_SYSTEM_IDBUS_ADDRHOLE_SECURE Represents the address hole type when **IDBUS_ADDRHOLE_INT** occurs. (RO)

Register 20.110. LP_SYSTEM_RNG_DATA_REG (0x01A4)



LP_SYSTEM_RND_DATA Represents the RNG-generated random numbers. (RO)

Register 20.111. LP_SYSTEM_INT_RAW_REG (0x0170)

(reserved)																								LP_SYSTEM_SLOW_CLK_TICK_INT_RAW LP_SYSTEM_ETM_TASK_ULP_INT_RAW LP_SYSTEM_LP_CORE_ULP_INT_RAW LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_INT_RAW LP_SYSTEM_LP_CORE_AHB_TIMEOUT_INT_RAW LP_SYSTEM_IDBUS_ADDRHOLE_INT_RAW LP_SYSTEM_LP_ADDRHOLE_INT_RAW							
31																							7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0								

LP_SYSTEM_LP_ADDRHOLE_INT_RAW The raw interrupt status of [LP_ADDRHOLE_INT](#). (R/SS/WTC)

LP_SYSTEM_IDBUS_ADDRHOLE_INT_RAW The raw interrupt status of [IDBUS_ADDRHOLE_INT](#). (R/SS/WTC)

LP_SYSTEM_LP_CORE_AHB_TIMEOUT_INT_RAW The raw interrupt status of [LP_CORE_AHB_TIMEOUT_INT](#). (R/SS/WTC)

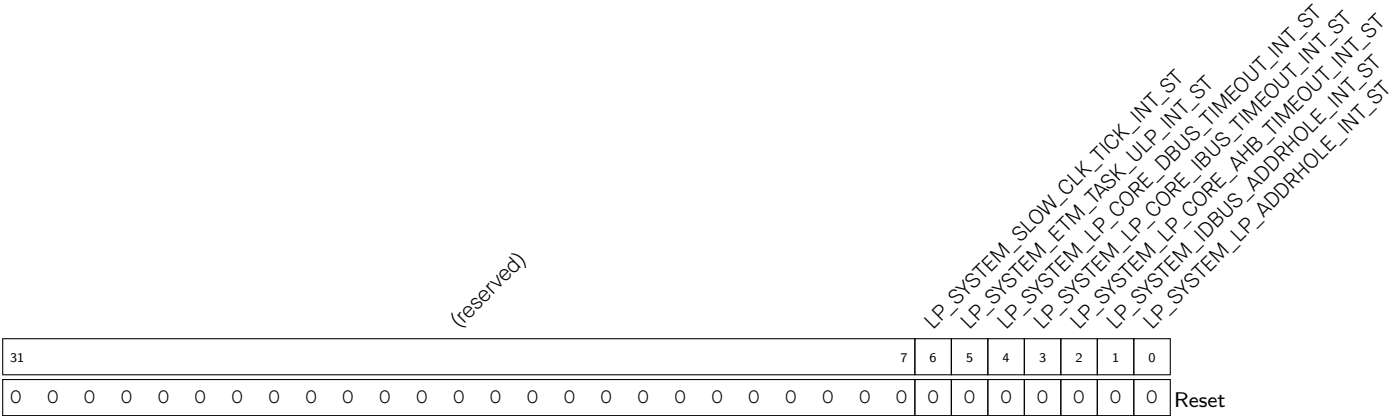
LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_INT_RAW The raw interrupt status of [LP_CORE_IBUS_TIMEOUT_INT](#). (R/SS/WTC)

LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_INT_RAW The raw interrupt status of [LP_CORE_DBUS_TIMEOUT_INT](#). (R/SS/WTC)

LP_SYSTEM_ETM_TASK_ULP_INT_RAW The raw interrupt status of [ETM_TASK_ULP_INT](#). (R/SS/WTC)

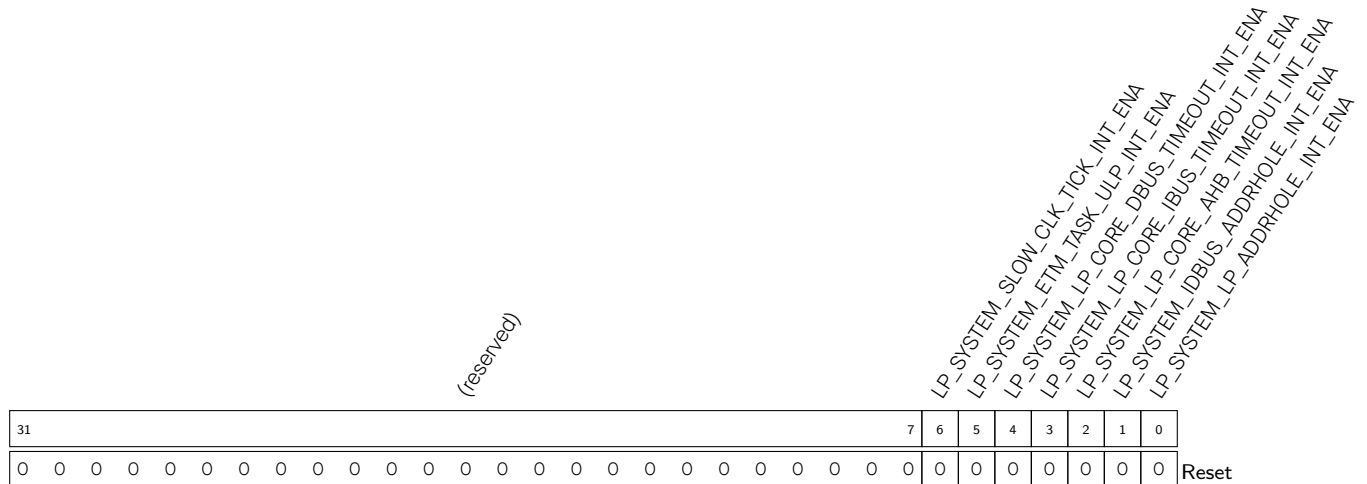
LP_SYSTEM_SLOW_CLK_TICK_INT_RAW The raw interrupt status of [SLOW_CLK_TICK_INT](#). (R/SS/WTC)

Register 20.112. LP_SYSTEM_INT_ST_REG (0x0174)



- LP_SYSTEM_LP_ADDRHOLE_INT_ST The masked interrupt status of LP_ADDRHOLE_INT. (RO)
- LP_SYSTEM_IDBUS_ADDRHOLE_INT_ST The masked interrupt status of IDBUS_ADDRHOLE_INT. (RO)
- LP_SYSTEM_LP_CORE_AHB_TIMEOUT_INT_ST The masked interrupt status of LP_CORE_AHB_TIMEOUT_INT. (RO)
- LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_INT_ST The masked interrupt status of LP_CORE_IBUS_TIMEOUT_INT. (RO)
- LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_INT_ST The masked interrupt status of LP_CORE_DBUS_TIMEOUT_INT. (RO)
- LP_SYSTEM_ETM_TASK_ULP_INT_ST The masked interrupt status of ETM_TASK_ULP_INT. (RO)
- LP_SYSTEM_SLOW_CLK_TICK_INT_ST The masked interrupt status of SLOW_CLK_TICK_INT. (RO)

Register 20.113. LP_SYSTEM_INT_ENA_REG (0x0178)



LP_SYSTEM_LP_ADDRHOLE_INT_ENA Write 1 to enable [LP_ADDRHOLE_INT](#) interrupt. (R/W)

LP_SYSTEM_IDBUS_ADDRHOLE_INT_ENA Write 1 to enable [IDBUS_ADDRHOLE_INT](#) interrupt. (R/W)

LP_SYSTEM_LP_CORE_AHB_TIMEOUT_INT_ENA Write 1 to enable LP_CORE_AHB_timeout interrupt. (R/W)

LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_INT_ENA Write 1 to enable LP_CORE_IBUS_timeout interrupt. (R/W)

LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_INT_ENA Write 1 to enable LP_CORE_DBUS_timeout interrupt. (R/W)

LP_SYSTEM_ETM_TASK_ULP_INT_ENA Write 1 to enable [ETM_TASK_ULP_INT](#) interrupt. (R/W)

LP_SYSTEM_SLOW_CLK_TICK_INT_ENA Write 1 to enable [SLOW_CLK_TICK_INT](#) interrupt. (R/W)

Register 20.114. LP_SYSTEM_INT_CLR_REG (0x017C)

(reserved)																												LP_SYSTEM_SLOW_CLK_TICK_INT_CLR LP_SYSTEM_ETM_TASK_ULP_INT_CLR LP_SYSTEM_LP_CORE_ULP_INT_CLR LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_INT_CLR LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_INT_CLR LP_SYSTEM_IDBUS_ADDRHOLE_INT_CLR LP_SYSTEM_LP_ADDRHOLE_INT_CLR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
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LP_SYSTEM_LP_ADDRHOLE_INT_CLR Write 1 to clear [LP_ADDRHOLE_INT](#) interrupt. (WT)

LP_SYSTEM_IDBUS_ADDRHOLE_INT_CLR Write 1 to clear [IDBUS_ADDRHOLE_INT](#) interrupt. (WT)

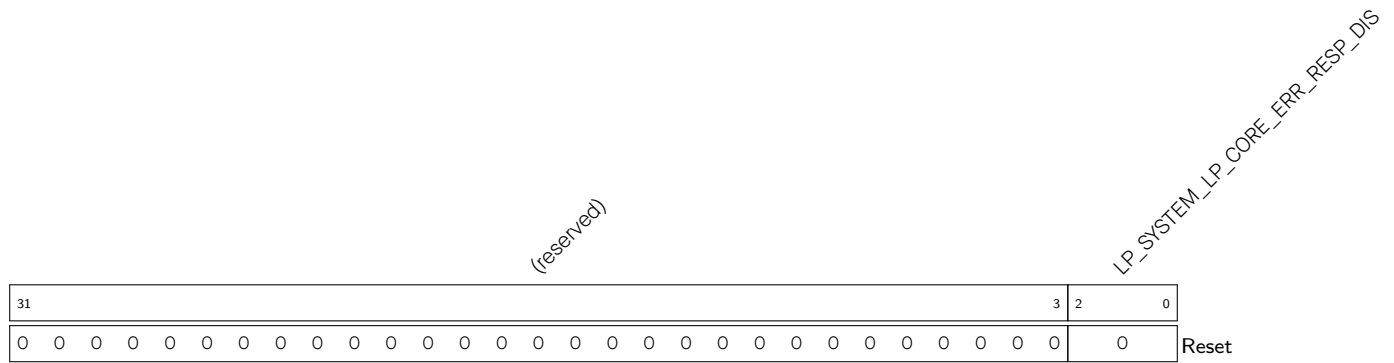
LP_SYSTEM_LP_CORE_AHB_TIMEOUT_INT_CLR Write 1 to clear LP_CORE_AHB_timeout interrupt. (WT)

LP_SYSTEM_LP_CORE_IBUS_TIMEOUT_INT_CLR Write 1 to clear LP_CORE_IBUS_timeout interrupt. (WT)

LP_SYSTEM_LP_CORE_DBUS_TIMEOUT_INT_CLR Write 1 to clear LP_CORE_DBUS_timeout interrupt. (WT)

LP_SYSTEM_ETM_TASK_ULP_INT_CLR Write 1 to clear [ETM_TASK_ULP_INT](#) interrupt. (WT)

LP_SYSTEM_SLOW_CLK_TICK_INT_CLR Write 1 to clear [SLOW_CLK_TICK_INT](#) interrupt. (WT)

Register 20.115. LP_SYSTEM_LP_CORE_ERR_RESP_DIS_REG (0x01BC)

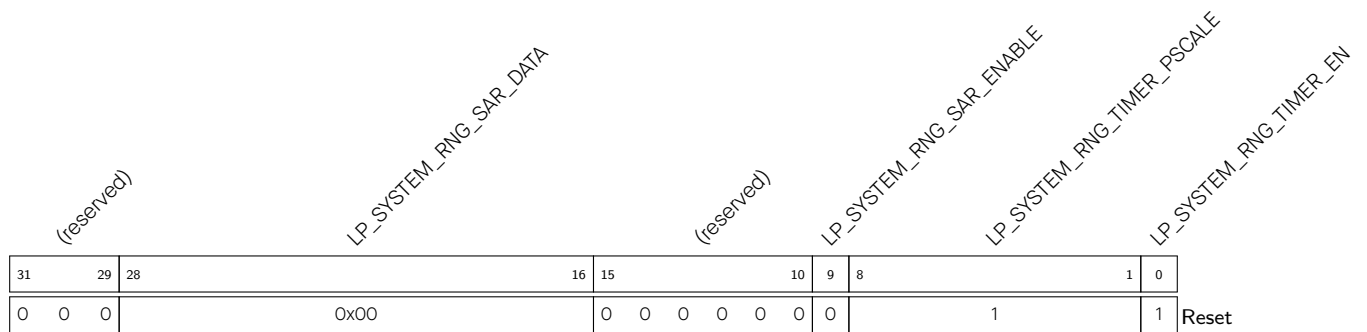
LP_SYSTEM_LP_CORE_ERR_RESP_DIS Configures whether or not to disable error response for respective LP CPU buses.

bit-0: IBUS

bit-1: DBUS

bit-2: AHB bus

(R/W)

Register 20.116. LP_SYSTEM_RNG_CFG_REG (0x01C0)

LP_SYSTEM_RNG_TIMER_EN Configures whether or not to enable RNG timer.

0: Disable

1: Enable

(R/W)

LP_SYSTEM_RNG_TIMER_PSCALE Configures the clock division factor for the RNG timer. (R/W)

LP_SYSTEM_RNG_SAR_ENABLE Configures whether or not to enable RNG_SARADC.

0: Disable

1: Enable

(R/W)

LP_SYSTEM_RNG_SAR_DATA Represents the analog SAR3_CAP value, which is used for debugging RNG_SARADC sample count. (RO)